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Methods and compositions for detecting transfusion-transmitted pathogens

Abstract

Probe sets capable of detecting pathogen nucleic acids in a sample are described. The probe set can be provided on a solid support, such as a microarray. Methods of detecting pathogen nucleic acids in a sample using the probe set are also provided. In some examples, the probes and methods are capable of detecting one or more RNA viruses, one or more DNA viruses, one or more bacterial nucleic acids, and/or one or more protozoan nucleic acids in a sample.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is the U.S. National Stage of International Application No. PCT/US2020/016262, filed Jan. 31, 2020, which was published in English under PCT Article 21 (2), which in turn claims the benefit of U.S. Provisional Application No. 62/799,482, filed Jan. 31, 2019, which is incorporated herein by reference in its entirety.

FIELD

(1) This disclosure relates to compositions and methods for detecting pathogens in a sample, particularly probes and microarrays and methods of their use.

BACKGROUND

(2) Each year millions of blood donations are collected globally and millions of blood components are transfused to patients. Though screening of these blood units using serologic and nucleic acid testing (NAT) has greatly reduced the risk of some transfusion-transmitted infections (TTIs), the vast majority of bloodborne agents are not screened (Alter et al., *Semin. Hematol.* 44:32-41, 2007; Glynn et al., *Transfusion* 53:438-454, 2013; Leveton et al., *Transfusion* 36:919-927, 1996). The U.S. Food and Drug Administration-licensed methods for infectious disease screening of donor blood include: 1) nucleic acid testing (NAT) for Hepatitis B virus (HBV), Hepatitis C virus (HCV), HIV-1 and -2, *Babesia*, West Nile virus (WNV) and Zika virus (ZIKV); and 2) immunoassays for HBV, HCV, HIV-1 and -2, cytomegalovirus (CMV), human T-cell lymphotropic virus I and II (HTLV), *Treponema pallidum* (syphilis), and *Trypanosoma cruzi* (Chagas). HTLV, syphilis, and Chagas antibody testing fail to detect these pathogens during a window period and Chagas is screened only once on samples from first-time blood donors (Duncan et al., *Exp. Rev. Mol. Diagn.* 16:83-95, 2016).

(3) The American Association of Blood Banks Transfusion-Transmitted Diseases Committee produced a list of over 30 pathogens of concern for transmission via blood that included bacteria, parasites, prions and viruses (Stramer et al., *Transfusion* 49:1S-29S, 2009). Only prions cannot be detected by currently available technology. Nearly all the other agents currently require individual qPCR or serologic testing and it is logistically impractical and cost prohibitive to test all known and potential agents individually (Stramer *ISBT Science Series* 9:30-37, 2014; Atrey et al., *Transfusion* 51:1855-1871, 2011).

(4) Multiplex PCR-based devices for testing blood-borne pathogens are limited. FDA-approved blood donor screening assays that use transcription-mediated amplification for multiplex detection of HBV, HCV, and HIV 1 and 2 include the cobas TaqScreen MPX Test (Roche Molecular Systems, Inc.) and the Procleix Ultrio Plus (Gen-Probe, Inc.) (Duncan et al., *Exp. Rev. Mol. Diagn.* 16:83-95, 2016).

SUMMARY

(5) A multiplex assay capable of detecting many, most, or all known pathogens of concern in a single small blood sample with high sensitivity and specificity could significantly increase the safety of the blood supply. Further, to counter emerging pathogens, the platform should be adaptable for rapid addition and validation of probes to detect new agents. Microarray-based technology offers the advantage of multiplex detection in a miniaturized format with high adaptability.

(6) Disclosed herein are probe sets that include probes with at least 90% identity (such as at least 90%, at least 95%, at least 98%, or at least 99% identity) with the nucleic acid sequences of SEQ ID NOs: 1-1300, 1391-1570, and 1691-1769, or subsets thereof. In some examples, the probe set includes probes with the nucleic acid sequences of SEQ ID NOs: 1-1300, 1391-1570, and 1691-1769, or a subset thereof. In some embodiments, the probe set includes one or more probes (such as 30 or more probes) for one or more RNA viruses, such as one or more of Chikungunya virus, Dengue virus type 1, Dengue virus type 2, Dengue virus type 3, Dengue virus type 4, Hepatitis A virus, Hepatitis C virus type 1, Hepatitis C virus type 2, Hepatitis C virus type 3, Hepatitis E virus, Human immunodeficiency virus type 1, Human immunodeficiency virus type 2, Human T-lymphotropic virus type I, Human T-lymphotropic virus type II, West Nile virus, and Zika virus.

(7) In other embodiments, the probe set includes one or more probes for one or more DNA viruses, such as one or more of cytomegalovirus (CMV, also known as HHV-5), Epstein Barr virus (EBV, also known as HHV-4, for example subtype B95-8 and/or AG876)), human herpes virus 8 (HHV-8), Hepatitis B virus (such as one or more of Hepatitis B virus subtype adw, subtype ayw, subtype adr, and subtype ayr), human parvovirus B19, and human papillomavirus (HPV, such as one or more of type 6, 11, 16, and 18). In some embodiments, the probe set includes probes with at least 90% identity (such as at least 90%, at least 95%, at least 98%, or at least 99% identity) with the nucleic acid sequences of SEQ ID NOs: 1770-2647, or a subset thereof. In some examples, the probe set includes probes with the nucleic acid sequences of SEQ ID NOs: 1770-2647, or a subset thereof.

(8) Further disclosed are probe sets that include one or more probes for one or more bacterial or protozoan pathogens, such as one or more of *Treponema pallidum*, *Ehrlichia chaffeensis*, *Ehrlichia ewingii*, *Ehrlichia muris*, *Borrelia burgdorferi*, *Coxiella burnetii*, *Trypanosoma brucei*, *Trypanosoma cruzi*, *Leishmania major*, *Babesia microti*, *Plasmodium falciparum*, and *Plasmodium vivax*. In some embodiments, the probe set includes probes with at least 90% identity (such as at least 90%, at least 95%, at least 98%, or at least 99% identity) with the nucleic acid sequences of SEQ ID NOs: 2648-3207, or a subset thereof. In some examples, the probe set includes probes with the nucleic acid sequences of SEQ ID NOs: 2648-3207, or a subset thereof.

(9) In some embodiments, the disclosed probe sets include at least one negative control probe and/or at least one positive control probe. In some examples the negative control probe is a probe with at least 90% identity (such as at least 90%, at least 95%, at least 98%, or at least 99% identity) with the nucleic acid sequences of any one of SEQ ID NOs: 1571-1690. In other examples the control probe is a probe with at least 90% identity (such as at least 90%, at least 95%, at least 98%, or at least 99% identity) with the nucleic acid sequences of any one of SEQ ID NOs: 3208-3628.

(10) In one non-limiting embodiment, the probe set is a set of probes including each of SEQ ID NOs: 1-1769. In another non-limiting embodiment, the probe set is a set of probes including each of SEQ ID NOs: 1770-2647 and 3208-3628, each of SEQ ID NOs: 2648-3628, or each of SEQ ID NOs: 1770-3628. In a further non-limiting embodiment, the microarray includes probes including

each of SEQ ID NOs: 1-3628.

(11) Also disclosed are microarrays that include a probe set described herein, for example, wherein the probes are covalently linked to a solid support. In one non-limiting example, the microarray includes probes with at least 90% identity (such as at least 90%, at least 95%, at least 98%, at least 99%, or at least 100% identity) with the nucleic acid sequences of SEQ ID NOs: 1-1300, 1391-1570, and 1691-1769, or subsets thereof. In another non-limiting example, the microarray includes probes with at least 90% identity (such as at least 90%, at least 95%, at least 98%, at least 99%, or at least 100% identity) with the nucleic acid sequences of SEQ ID NOs: 1770-2647, or subsets thereof. In a further non-limiting example, the microarray includes probes with at least 90% identity (such as at least 90%, at least 95%, at least 98%, at least 99%, or at least 100% identity) with the nucleic acid sequences of SEQ ID NOs: 2648-3207, or subsets thereof. In yet another non-limiting embodiment, the microarray includes probes with at least 90% identity (such as at least 90%, at least 95%, at least 98%, at least 99%, or at least 100% identity) with the nucleic acid sequences of SEQ ID NOs: 1770-3207, or subsets thereof. The microarray may further include negative and/or positive control probes. In one non-limiting embodiment, the microarray includes probes including each of SEQ ID NOs: 1-1769. In other non-limiting embodiments, the microarray includes probes including each of SEQ ID NOs: 1770-2647 and 3208-3628, each of SEQ ID NOs: 2648-3628, each of SEQ ID NOs: 1770-3628. In a further non-limiting embodiment, the microarray includes probes including each of SEQ ID NOs: 1-3628.

(12) Disclosed herein are methods of detecting one or more pathogen nucleic acids in a sample. In some examples, the methods include detecting nucleic acids from one or more RNA viruses, such as one or more of Chikungunya virus, Dengue virus type 1, Dengue virus type 2, Dengue virus type 3, Dengue virus type 4, Hepatitis A virus, Hepatitis C virus type 1, Hepatitis C virus type 2, Hepatitis C virus type 3, Hepatitis E virus, Human immunodeficiency virus type 1, Human immunodeficiency virus type 2, Human T-lymphotropic virus type I, Human T-lymphotropic virus type II, West Nile virus, and Zika virus in a sample. In other examples, the methods include detecting nucleic acids from one or more DNA viruses, such as one or more of cytomegalovirus, Epstein Barr virus, human herpes virus 8, Hepatitis B virus, human parvovirus B19, and human papillomavirus.

(13) Also disclosed are methods of detecting one or more bacterial and/or protozoan nucleic acids in a sample. In some examples, the methods include detecting nucleic acids from one or more of *Treponema pallidum*, *Ehrlichia chaffeensis*, *Ehrlichia ewingii*, *Ehrlichia muris*, *Borrelia burgdorferi*, *Coxiella burnetii*, *Trypanosoma brucei*, *Trypanosoma cruzi*, *Leishmania major*, *Babesia microti*, *Plasmodium falciparum*, and *Plasmodium vivax*.

(14) In some examples, the methods include contacting a sample with a disclosed probe set or microarray under conditions sufficient to allow hybridization of pathogen nucleic acids present in the sample to the probes of the probe set or microarray and measuring hybridization of the sample to one or more of the probes, thereby detecting one or more nucleic acids in the sample. The sample may be a blood, serum, or plasma sample, or nucleic acids (such as RNA or cDNA) isolated from the sample. In particular examples, the sample is a blood donation sample or nucleic acids isolated from a blood donation sample. In particular examples, nucleic acids (such as RNA or cDNA) from the sample are labeled prior to contacting the probe set or microarray with the nucleic acids. In one example, the method includes preparing cDNA from the sample and labeling the cDNA. In some examples, the method does not include amplifying RNA from the sample prior to preparing the cDNA.

(15) The foregoing and other features of the disclosure will become more apparent from the following detailed description, which proceeds with reference to the accompanying figures.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIGS. 1A-1C are a series of panels showing pathogen chip design (FIG. 1A), sample preparation work flow (FIG. 1B), and analysis strategy (FIG. 1C) for pathogen detection microarrays.

(2) FIGS. 2A-2C are a series of graphs showing amplification method and Pathogen Chip assay performance assessed using positive control viral RNAs. FIG. 2A shows SPIA amplification vs. standard (STD) method. cDNA concentration after amplification for four representative viral RNAs is shown. Starting RNA concentration was <10 ng/ μ l each. FIG. 2B shows Pathogen chip assay performance 1. Bars are the mean of Cy3 signal for the Chikungunya and West Nile probes hybridized to test samples positive for CHIKV and WNV and a negative plasma sample. Only probes specific to target showed a specific hybridization signal. No signal was detected for negative plasma. FIG. 2C shows Pathogen chip assay performance 2. Detection responses of four representative samples (Dengue-4) were measured over a dilution series from 10,000 to 10 genomic copies per sample. Bars are the mean of Cy3 signals for all probes to the indicated viruses hybridized to test samples.

SEQUENCE LISTING

(3) Any nucleic acid and amino acid sequences listed herein or in the accompanying sequence listing are shown using standard letter abbreviations for nucleotide bases and amino acids, as defined in 37 C.F.R. § 1.822. In at least some cases, only one strand of each nucleic acid sequence is shown, but the complementary strand is understood as included by any reference to the displayed strand.

(4) The Sequence Listing is submitted as an ASCII text file in the form of the file named Sequence_Listing.txt, which was created on Jul. 22, 2021, and is 676,876 bytes, which is incorporated by reference herein.

(5) In the accompanying sequence listing: SEQ ID NOs: 1-110 are Hepatitis C virus genotype 1 probes SEQ ID NOs: 111-210 are Hepatitis C virus genotype 2 probes SEQ ID NOs: 211-310 are Hepatitis C virus genotype 3 probes SEQ ID NOs: 311-400 are Human Immunodeficiency virus 1 probes SEQ ID NOs: 401-510 are Human Immunodeficiency virus 2 probes SEQ ID NOs: 511-570 are Human T-lymphotropic virus I probes SEQ ID NOs: 571-660 are Human T-lymphotropic virus II probes SEQ ID NOs: 661-760 are West Nile virus NY99 probes SEQ ID NOs: 761-870 are West Nile virus 956 probes SEQ ID NOs: 871-900 are Chikungunya virus probes SEQ ID NOs: 901-1000 are Dengue virus 1 probes SEQ ID NOs: 1001-1100 are Dengue virus 2 probes SEQ ID NOs: 1101-1199 are Dengue virus 3 probes SEQ ID NOs: 1200-1300 are Dengue virus 4 probes SEQ ID NOs: 1301-1390 are GB virus C/Hepatitis G virus probes SEQ ID NOs: 1391-1500 are Hepatitis A virus probes SEQ ID NOs: 1501-1570 are Hepatitis E virus probes SEQ ID NOs: 1571-1580 are White clover cryptic virus 1 probes SEQ ID NOs: 1581-1620 are Broad bean wilt virus 1 probes SEQ ID NOs: 1621-1690 are Lettuce necrotic yellows virus probes SEQ ID NOs: 1691-1700 are Zika virus isolate Brazil-ZKV2015 probes SEQ ID NOs: 1701-1710 are Zika virus strain PRVABC59 probes SEQ ID NOs: 1711-1720 are Zika virus isolate Z1106033 probes SEQ ID NOs: 1721-1730 are Zika virus isolate SSABR1 probes SEQ ID NOs: 1731-1769 are Zika virus strain ZikaSPH2015 probes SEQ ID NOs: 1770-1852 are Cytomegalovirus probes SEQ ID NOs: 1853-1917 are Epstein Barr virus B95-8 probes SEQ ID NOs: 1918-2023 are Epstein Barr virus AG876 probes SEQ ID NOs: 2024-2108 are Human herpesvirus 8 probes SEQ ID NOs: 2109-2192 are Human papillomavirus subtype 6b probes SEQ ID NOs: 2193-2271 are Human papillomavirus subtype 11 probes SEQ ID NOs: 2272-2342 are Human papillomavirus subtype 16 probes SEQ ID NOs: 2343-2419 are Human papillomavirus subtype 18 probes SEQ ID NOs: 2420-2470 are Hepatitis B virus subtype adw probes SEQ ID NOs: 2471-2520 are Hepatitis B virus subtype ayw probes SEQ ID NOs: 2521-2556 are Hepatitis B virus subtype adr probes SEQ ID NOs: 2557-2602 are Hepatitis B virus subtype ayr probes SEQ ID NOs: 2603-2647 are Human parvovirus B19

probes SEQ ID NOs: 2648-2751 are *Treponema pallidum* probes SEQ ID NOs: 2752-2852 are *Ehrlichia chaffeensis* probes SEQ ID NOs: 2853-2861 are *Ehrlichia ewingii* probes SEQ ID NOs: 2862-2922 are *Ehrlichia muris* probes SEQ ID NOs: 2923-3001 are *Borrelia burgdorferi* probes SEQ ID NOs: 3002-3085 are *Coxiella burnetii* probes SEQ ID NOs: 3086-3097 are *Trypanosoma brucei* probes SEQ ID NO: 3098 is a *Trypanosoma cruzi* probe SEQ ID NOs: 3099-3113 are *Leishmania major* probes SEQ ID NOs: 3114-3154 are *Babesia microti* probes SEQ ID NOs: 3155-3185 are *Plasmodium falciparum* probes SEQ ID NOs: 3186-3207 are *Plasmodium vivax* probes SEQ ID NOs: 3208-3301 are human ACTB probes SEQ ID NOs: 3302-3385 are human ARL1 probes SEQ ID NOs: 3386-3519 are human CCDN1 probes SEQ ID NOs: 3520-3557 are *Aedes albopictus* dengue virus 2 probes SEQ ID NOs: 3558-3598 are Maize streak virus probes SEQ ID NOs: 3599-3628 are Tomato pseudo-curly top virus probes

DETAILED DESCRIPTION

(6) Disclosed herein are customized sets of probes, including microarray-based pathogen chips, for simultaneous detection of nucleic acids from RNA viruses, DNA viruses, and/or bacteria or protozoan pathogens in blood samples (such as human plasma) that are designed to have the flexibility to expand to detect emerging agents in a relatively short time frame. The presence of multiple probes per target represents an advantage in comparison to traditional NAT or EIA assays since the pathogen(s) can be detected even in the case of failure of one of the probes due to mutation (Petrik *Vox Sanguinis* 80:1-11, 2001). The flexibility and high-throughput capability of microarrays hold great potential for pathogen detection and identification, but historically have had limitations in detecting the presence of the low viral levels (Chen et al., *J. Vis. Exp.* 50:e2536, 2011; Wang et al., *Proc. Natl. Acad. Sci. USA* 99:15687-15692, 2002; Eckburg et al., *Clin. Infect. Dis.* 43:e71-e76, 2006). Disclosed herein are probe sets and microarray assays that include: 1) a platform design that simultaneously detects and distinguishes multiple pathogens and closely related strains or subtypes; and 2) a combination of amplification and labeling protocols to detect multiple targets present at low levels in a sample.

I. Terms

(7) Unless otherwise noted, technical terms are used according to conventional usage. Definitions of common terms in molecular biology may be found in *Lewin's Genes X*, ed. Krebs et al., Jones and Bartlett Publishers, 2009 (ISBN 0763766321); Kendrew et al. (eds.), *The Encyclopedia of Molecular Biology*, published by Blackwell Publishers, 1994 (ISBN 0632021829); Robert A. Meyers (ed.), *Molecular Biology and Biotechnology: a Comprehensive Desk Reference*, published by Wiley, John & Sons, Inc., 1995 (ISBN 0471186341); and George P. Rédei, *Encyclopedic Dictionary of Genetics, Genomics, Proteomics and Informatics*, 3rd Edition, Springer, 2008 (ISBN: 1402067534), and other similar references.

(8) Unless otherwise explained, all technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this disclosure belongs. The singular terms “a,” “an,” and “the” include plural referents unless the context clearly indicates otherwise. Similarly, the word “or” is intended to include “and” unless the context clearly indicates otherwise. Hence “comprising A or B” means including A, or B, or A and B. It is further to be understood that all base sizes or amino acid sizes, and all molecular weight or molecular mass values, given for nucleic acids or polypeptides are approximate, and are provided for description.

(9) Although methods and materials similar or equivalent to those described herein can be used in the practice or testing of the present disclosure, suitable methods and materials are described below. All publications, patent applications, patents, and other references mentioned herein are incorporated by reference in their entirety. In case of conflict, the present specification, including explanations of terms, will control. In addition, the materials, methods, and examples are illustrative only and not intended to be limiting.

(10) In order to facilitate review of the various embodiments of the disclosure, the following explanations of specific terms are provided:

- (11) **Array or microarray:** An arrangement of nucleic acids (such as DNA or RNA) or proteins (such as antibodies) in assigned locations on a matrix or substrate. In some examples, the nucleic acid molecules or proteins are attached covalently to the matrix or substrate.
- (12) ***Babesia*:** A tick-borne protozoan parasite that infects vertebrate red blood cells. In humans, *Babesia* species may cause asymptomatic infection or babesiosis, characterized by flu-like symptoms. Most cases of transmission between humans are attributed to a tick vector; however, it may also be transmitted through blood transfusion or organ donation. The most common pathogenic species in humans are *Babesia divergens* and *Babesia microti*. *Babesia* sequences are publicly available, and include GenBank Accession Nos. ASM107745v2 (*Babesia divergens*) and ASM69194v2 and ASM165006v1 (*Babesia microti*), which are incorporated by reference in their entirety as present in GenBank on Jan. 30, 2020.
- (13) ***Borrelia*:** A genus of tick-borne spirochete bacteria that cause Lyme disease. The major species of *Borrelia* that cause Lyme disease include *Borrelia burgdorferi*, *Borrelia afzelii*, *Borrelia garinii*, and *Borrelia mayonii*. *Borrelia* has been identified in blood stored for donation, though there is currently no evidence of Lyme disease linked to blood transfusion. *Borrelia* sequences are publicly available, and include GenBank Accession Nos. ASM868v2 (*Borrelia burgdorferi*), ASM30473v1 (*Borrelia afzelii*), ASM192254v1 (*Borrelia garinii*), and ASM194566v1 (*Borrelia mayonii*), which are incorporated by reference in their entirety as present in GenBank on Jan. 30, 2020).
- (14) **Chikungunya virus (CHIKV):** A positive-sense single-stranded RNA virus of the alphavirus genus in the family Togaviridae. This virus is primarily transmitted by *Aedes* mosquitoes, particularly *A. albopictus* and *A. aegypti*. The symptoms of CHIKV infection include rash, high fever and joint pain. CHIKV was first isolated in Tanzania in 1952 and re-emerged in Kenya in 2004. The evolution and spread of this virus into new geographic areas, and the disease severity resulting from CHIKV infection, present a serious public health concern. CHIKV sequences are publicly available, and include GenBank Accession No. NC_004162 (gi|27754751)), which is incorporated by reference in its entirety as present in GenBank on Jan. 30, 2019.
- (15) ***Coxiella burnetii*:** A Gram-negative bacteria that causes Q fever. Symptoms are typically flu-like and may be mild or severe, and a small percentage of infected individuals develop chronic Q fever. The bacteria infects livestock (such as cows, sheep, and goats) and is transmitted to humans by contact with feces, urine, milk, or other products from an infected animal, typically by breathing dust contaminated with the bacteria. *Coxiella burnetii* sequences are publicly available, and includes GenBank Accession No. ASM776v2, which is incorporated by reference in its entirety as present in GenBank on Jan. 30, 2020.
- (16) **Cytomegalovirus (CMV):** Also known as human herpesvirus 5. A common virus that infects up to 50% of adults by the age of 40. Most people show no symptoms of infection or only mild symptoms; however, babies born with congenital CMV infection may have long-term health problems. CMV is transmitted by body fluids, including blood transfusions. CMV sequences are publicly available, and include GenBank Accession No. NC_006273, which is incorporated by reference in its entirety as present in GenBank on Jan. 30, 2020.
- (17) **Dengue virus (DEN):** An RNA virus of the family Flaviviridae, genus *Flavivirus*. There are four serotypes of dengue virus, referred to as DEN1, DEN2, DEN3 and DEN4. All four serotypes can cause the full spectrum of dengue disease. Infection with one serotype can produce lifelong immunity to that serotype. However, severe complications can occur upon subsequent infection by a different serotype. Dengue virus is primarily transmitted by *Aedes* mosquitoes, particularly *A. aegypti*. Symptoms of dengue virus infection include fever, headache, muscle and joint pain and a skin rash similar to measles. In a small percentage of cases, the infection develops into a life-threatening dengue hemorrhagic fever, typically resulting in bleeding, low platelet levels and blood plasma leakage, or into dengue shock syndrome, characterized by dangerously low blood pressure. DEN sequences are publicly available, and include GenBank Accession Nos. NC_001477 (gi|9626685) (DEN1), NC_001474 (gi|158976983) (DEN2), NC_001475 (gi|163644368) (DEN3),

and NC_002640 (gi|12084822) (DEN4), which are incorporated by reference in their entirety as present in GenBank on Jan. 30, 2019.

(18) Epstein-Barr virus (EBV): Also known as human herpesvirus 4. EBV is a common virus that is spread primarily through saliva, though it can also be spread by sexual contact, blood transfusion, and organ transplantation. EBV causes infectious mononucleosis, characterized by fatigue, fever, swollen lymph nodes, and sore throat; however, EBV infection may also be asymptomatic. EBV sequences are publicly available, and include GenBank Accession Nos. AJ278309 (EBV strain B95-8), DQ279927 (EBV strain AG876), and NC_009334, all of which are incorporated by reference in their entirety as present in GenBank on Jan. 30, 2020.

(19) *Ehrlichia*: A genus of tick-borne bacteria that causes ehrlichiosis. In some cases, *Ehrlichia* has been transmitted through blood transfusion or organ transplantation. Symptoms can include rash, fever, headache, muscle aches, nausea, vomiting, and diarrhea. Severe, late stage illness can include neural damage, respiratory failure, and organ failure. Disease causing species include *Ehrlichia chaffeensis*, *Ehrlichia ewingii*, and *Ehrlichia muris*. *Ehrlichia* sequences are publicly available, and include GenBank Accession Nos. NC_007799 (*E. chaffeensis*) and NC_023063 (*E. muris*), which are incorporated by reference in their entirety as present in GenBank on Jan. 30, 2020.

(20) Hepatitis A virus (HAV): A single-stranded RNA virus in the order Picornavirales, family Picornaviridae. The virus is transmitted through fecal-oral and blood routes. HAV causes symptoms such as nausea, vomiting, diarrhea, jaundice, fever, and abdominal pain and typically lasts about 8 weeks. Acute liver failure may occur in some cases. HAV sequences are publicly available, and include GenBank Accession No. NC_001489 (gi|9626732), which is incorporated by reference in its entirety as present in GenBank on Jan. 30, 2019.

(21) Hepatitis B virus (HBV): A DNA virus of the Hepadnaviridae family. HBV is transmitted through blood or bodily fluids and new infections are frequently asymptomatic in healthy adults.

(22) Immunosuppressed adults and children less than 5 years of age more commonly exhibit symptoms, including flu-like symptoms and jaundice. HBV sequences are publicly available and include GenBank Accession Nos. AY518556 (subtype adw), NC_003977 (subtype ayw), AY123041 (subtype adr), and X04615 (subtype ayr), all of which are incorporated by reference in their entirety as present in GenBank on Jan. 30, 2020.

(23) Hepatitis C virus (HCV): A single-stranded positive sense RNA virus of the family Flaviviridae. HCV is transmitted primarily through blood and acute infection typically causes mild or no symptoms. However, chronic infection frequently leads to liver disease, including cirrhosis, liver failure, and/or hepatocellular carcinoma. HCV type 1 sequences are publicly available, and include GenBank Accession No. NC_004102 (gi|22129792). HCV type 2 sequences are also publicly available, and include GenBank Accession No. NC_009823 (gi|157781212). HCV type 3 sequences are also publicly available, and include GenBank Accession No. NC_009824 (gi|157781216). Each of these sequences are incorporated by reference in their entirety as present in GenBank on Jan. 30, 2019.

(24) Hepatitis E virus (HEV): A single-stranded positive sense RNA virus that is currently classified in the Hepeviridae family, genus *Orthohepevirus*. HEV causes liver inflammation, and is typically an acute and self-limiting infection. However, it can cause chronic hepatitis in individuals with weakened immune systems, particularly organ transplant recipients. HEV sequences are publicly available, and include GenBank Accession No. NC_001434 (gi|9626440), which is incorporated by reference in its entirety as present in GenBank on Jan. 30, 2019.

(25) Human Immunodeficiency virus (HIV): A single-stranded positive-sense RNA virus (retrovirus) that causes HIV infection and acquired immunodeficiency syndrome (AIDS). HIV is transmitted by blood or sexual contact. HIV type 1 sequences are publicly available, and include GenBank Accession No. NC_001802 (gi|9629357). HIV type 2 sequences are also publicly available and include GenBank Accession No. NC_001722 (gi|9628880). Each sequence is

incorporated by reference in their entirety as present in GenBank on Jan. 30, 2019

(26) Human Herpesvirus 8 (HHV-8): Also known as Kaposi sarcoma-associated herpesvirus. HHV-8 is associated with Kaposi sarcoma and other cancers, including some lymphomas. It is transmitted through bodily fluids, including blood, saliva, and sexual contact. HHV-8 sequences are publicly available and include GenBank Accession No. NC_009333, which is incorporated by reference in its entirety as present in GenBank on Jan. 30, 2020.

(27) Human parvovirus: A single-stranded DNA virus of the Parvoviridae family. Parvovirus B19 is the only parvovirus known to infect humans. B19 primarily causes disease in children, and causes what is sometimes called “fifth disease,” a mild rash. Parvovirus B19 can be transmitted via respiratory secretions or through blood or blood products. Human parvovirus B19 sequences are publicly available and include GenBank Accession No. NC_000883, which is incorporated by reference in its entirety as present in GenBank on Jan. 30, 2020.

(28) Human papillomavirus (HPV): A DNA virus of the family Papillomaviridae. HPV is a common sexually transmitted virus that can cause warts and cancers, including cervical cancer and head and neck cancer, in some individuals. HPV DNA can be detected in the blood in some cases; however, it is not clear whether it can be transmitted by blood transfusion. There are over 100 known types of HPV to date. HPV sequences are publicly available, and include GenBank Accession Nos. HG793809 (type 6), HE574701 (type 11), NC_001526 (type 16), and NC_001357 (type 18), each of which are incorporated by reference in their entirety as present in GenBank on Jan. 30, 2020.

(29) Human T-lymphotropic virus (HTLV): A group of positive-sense RNA retroviruses that are implicated in cancer (for example, T-cell lymphomas) and myelopathy. HTLV type I sequences are publicly available, and include GenBank Accession Nos. AF033817 and NC_001436 (gi|9626453). HTLV type II sequences are also publicly available and include GenBank Accession No. NC_001488 (gi|9626726). Each sequence is incorporated by reference in their entirety as present in GenBank on Jan. 30, 2019

(30) *Leishmania major*: A trypanosomatid parasite transmitted by sand flies. *L. major* causes cutaneous leishmaniasis. *L. major* sequences are publicly available, and include GenBank Accession No. ASM272v2, incorporated by reference in its entirety as present in GenBank on Jan. 30, 2020.

(31) *Plasmodium*: A genus of mosquito-transmitted protozoan parasites that causes malaria in humans. The two major malaria causing *Plasmodium* species in humans are *P. falciparum* and *P. vivax*. *P. falciparum* is also associated with Burkitt's lymphoma. *Plasmodium* can be transmitted by blood transfusion, causing transfusion-transmitted malaria. *Plasmodium* sequences are publicly available and include GenBank Accession No. ASM276v2 (*P. falciparum*) and ASM241v2 (*P. vivax*), which are incorporated by reference in their entirety as present in GenBank on Jan. 30, 2020.

(32) Probe: A probe typically comprises an isolated nucleic acid (for example, at least 10 or more nucleotides in length, such as 10-60, 15-50, 20-40, 20-50, 25-50, or 30-60 nucleotides in length). In some examples, a probe includes a detectable label, while in other examples a probe does not include a detectable label.

(33) Sample (or biological sample): A biological specimen containing nucleic acids (for example, DNA, RNA, and/or mRNA), proteins, or combinations thereof, obtained from a subject. Examples include, but are not limited to, peripheral blood, serum, plasma, urine, saliva, tissue biopsy, fine needle aspirate, surgical specimen, and autopsy material. In some examples, a sample includes blood, serum, or plasma.

(34) Subject: A living multi-cellular vertebrate organism, a category that includes human and non-human mammals. In one example, a subject is a blood donor.

(35) *Treponema*: A genus of spirochete bacteria. The major pathogenic species in humans is *Treponema pallidum*, of which subspecies *T. pallidum pallidum* causes syphilis. The bacteria is

transmitted primarily by sexual contact. Nucleic acid sequences for *T. pallidum pallidum* are publicly available and include GenBank Accession Nos. NC_016844 and NC_00919, which are incorporated by reference in their entirety as present in GenBank on Jan. 30, 2020.

(36) *Trypanosoma*: A genus of protozoan parasites transmitted by blood-feeding insects. *T. brucei* is transmitted by infected tsetse flies and causes sleeping sickness (trypanosomiasis) in humans. There are two types of trypanosomiasis: East African trypanosomiasis, caused by *Trypanosoma brucei rhodesiense* and West African trypanosomiasis, caused by *Trypanosoma brucei gambiense*. *Trypanosoma brucei brucei* infects primarily cattle, and does not normally infect humans. *T. cruzi* causes Chagas disease and is transmitted by triatomine bugs. *Trypanosoma* sequences are publicly available and include GenBank Accession Nos. ASM21029v1 (*T. brucei gambiense*), and ASM20906v1 (*T. cruzi*), each of which are incorporated by reference in their entirety as present in GenBank on Jan. 30, 2020.

(37) West Nile virus (WNV): A member of the virus family Flaviviridae and the genus *Flavivirus*. WNV was first isolated from a woman in the West Nile district of Uganda in 1937. The virus was later identified in birds in the Nile delta region in 1953. Human infections attributable to WNV have been reported in many countries for over 50 years. In 1999, a WNV circulating in Israel and Tunisia was imported into New York, producing a large and dramatic outbreak that spread throughout the continental United States in the following years. Human infection is most often the result of bites from infected mosquitoes, but may also be transmitted through contact with other infected animals, their blood or other tissues. Infection with WNV is asymptomatic in about 80% of infected people, but about 20% develop West Nile fever. Symptoms include fever, headache, fatigue, body aches, nausea, vomiting, swollen lymph glands and in some cases, a skin rash. Approximately 1 in 150 of infected individuals develop severe, neuroinvasive disease, such as encephalitis, meningitis, or poliomyelitis. WNV sequences are publicly available, and include GenBank Accession Nos. NC_009942 (gi|158516887) (NY99, lineage 1) and NC_001563 (gi|11528013) (956, lineage 2), which are incorporated by reference in their entirety as present in GenBank on Jan. 30, 2019.

(38) Zika virus (ZKV or ZIKV): A member of the virus family Flaviviridae and the genus *Flavivirus*. ZIKV is spread by the daytime-active mosquitoes *Aedes aegypti* and *A. albopictus*. This virus was first isolated from a Rhesus macaque from the Zika Forest of Uganda in 1947. Since the 1950s, ZIKV has been known to occur within a narrow equatorial belt from Africa to Asia. The virus spread eastward across the Pacific Ocean in 2013-2014, resulting in ZIKV outbreaks in Oceania to French Polynesia, New Caledonia, the Cook islands, and Easter Island. In 2015, ZKV spread to Mexico, Central America, the Caribbean and South America, where ZKV has reached pandemic levels. Infection by ZIKV generally causes either no symptoms or mild symptoms, including mild headache, maculopapular rash, fever, malaise, conjunctivitis and joint pain. However, ZKV infection has been linked to the birth of microcephalic infants following maternal infection. Reports have also indicated that ZIKV has the potential for human blood-borne and sexual transmission. ZIKV sequences are publicly available, and include GenBank Accession Nos. KU497555 (gi|985578255) (isolate Brazil-ZK2015), KU501215 (gi|984874581) (strain PRVABC59), KU312312 (gi|973447404) (isolate Z1106033), KU707826 (gi|992324757) (isolate SSABR1), and KU321639 (strain ZikaSPH2015), which are incorporated by reference in their entirety as present in GenBank on Jan. 30, 2019.

II. Probes and Microarrays

(39) Disclosed herein is a nucleic acid probe set capable of detecting nucleic acid molecules from one or more RNA viruses, including Chikungunya virus (CHIKV), Dengue virus types 1, 2, 3, or 4, (DEN1, DEN2, DEN3, DEN4), Hepatitis A virus (HAV), Hepatitis C virus (HCV) types 1, 2, or 3, Hepatitis E virus (HEV), Human Immunodeficiency virus (HIV) types 1 or 2, Human T-lymphotropic virus (HTLV) types I or II, West Nile virus (WNV), and Zika virus (ZKV). In some embodiments, the probe set includes 30 or more probes for one or more of the viruses (such as 30

or more, 50 or more, 60 or more, 70 or more, 80 or more, 90 or more, 100 or more, 110 or more, or 120 or more), for example 30-120 probes, 50-100 probes, or 70-110 probes for one or more of CHIKV, DEN1, DEN2, DEN3, DEN4, HAV, HCV type 1, HCV type 2, HCV type 3, HEV, HIV type 1, HIV type 2, HTLV type 1, HTLV type 2, WNV, and ZIKV.

(40) In some embodiments, the probe set includes nucleic acid probes that are at least 80%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical to the nucleic acid sequences of SEQ ID NOs: 1-1300, 1391-1570, and 1691-1769. In other embodiments, the probe set includes a subset of the probes of SEQ ID NOs: 1-1300, 1391-1570, and 1691-1769, such as at least 10%, at least 20%, at least 30%, at least 40%, at least 50%, at least 60%, at least 70%, at least 80%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or at least 99.9% of the probes of SEQ ID NOs: 1-1300, 1391-1570, and 1691-1769. In some examples, the subset includes at least one probe for each of CHIKV, DEN1, DEN2, DEN3, DEN4, HAV, HCV type 1, HCV type 2, HCV type 3, HEV, HIV type 1, HIV type 2, HTLV type 1, HTLV type 2, WNV, and ZKV, such as at least 1, at least 2, at least 5, at least 10, at least 20, at least 30, or more probes for each virus. In some examples, the subset includes at least 40 probes (such as at least 50, at least 60, at least 70, at least 80, at least 90, at least 100, or at least 110 probes) for one or more of CHIKV, DEN1, DEN2, DEN3, DEN4, HAV, HCV type 1, HCV type 2, HCV type 3, HEV, HIV type 1, HIV type 2, HTLV type 1, HTLV type 2, WNV, and ZKV.

(41) In one non-limiting example, the probe set includes or consists of each of the probes of SEQ ID NOs: 1-1769. In another non-limiting example, the probe set includes or consists of each of the probes of SEQ ID NOs: 1-1300, 1391-1570, and 1691-1769. In another example, the probe set includes or consists of each of the probes of SEQ ID NOs: 1-1300 and 1391-1769. In other embodiments, the probe set includes a subset of the probes of SEQ ID NOs: 1-1300, 1391-1570, and 1691-1769 or a subset of the probes of SEQ ID NOs: 1-1300 and 1391-1769.

(42) Also disclosed herein is a nucleic acid probe set capable of detecting nucleic acid molecules from one or more DNA viruses, including one or more of cytomegalovirus, Epstein Barr virus (e.g., one or more of EBV subtype B95-8 and EBV subtype AG876), human herpes virus 8, Hepatitis B virus (e.g., one or more of HBV subtypes adw, ayw, adr, and ayr), human parvovirus B19, and human papillomavirus (e.g., one or more of HPV types 6, 11, 16, and 18). In some embodiments, the probe set includes 10 or more probes for one or more of the viruses (such as 15 or more, 20 or more, 30 or more, 50 or more, 60 or more, 70 or more, 80 or more, 90 or more, 100 or more, 110 or more, or 120 or more), for example 10-50 probes, 30-120 probes, 50-100 probes, or 70-110 probes for one or more of cytomegalovirus, Epstein Barr virus, human herpes virus 8, Hepatitis B virus, human parvovirus B19, and human papillomavirus.

(43) In some embodiments, the probe set includes nucleic acid probes that are at least 80%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical to the nucleic acid sequences of SEQ ID NOs: 1770-2647. In other embodiments, the probe set includes a subset of the probes of SEQ ID NOs: 1770-2647, such as at least 10%, at least 20%, at least 30%, at least 40%, at least 50%, at least 60%, at least 70%, at least 80%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or at least 99.9% of the probes of SEQ ID NOs: 1770-2647. In some examples, the subset includes at least one probe for each of CMV, EBV subtype B95-8, EBV subtype AG876, human herpes virus 8, Hepatitis B virus subtype adw, Hepatitis B virus subtype ayw, Hepatitis B virus subtype adr, Hepatitis B virus subtype ayr, human parvovirus B19, HPV type 6, HPV type 11, HPV type 16, and HPV type 18, such as at least 1, at least 2, at least 5, at least 10, at least 20, at least 30, or more probes for each virus. In some examples, the subset includes at least 10 probes (such as at least 20, at least 30, at least 40, at least 50, at least 60, at least 70, at least 80, at least 90, at least 100, or at least 110 probes) for one or more of CMV, EBV subtype B95-8, EBV subtype AG876, human herpes virus 8, Hepatitis B virus subtype adw, Hepatitis B virus subtype ayw, Hepatitis B virus subtype adr, Hepatitis B virus subtype ayr, human parvovirus B19, HPV type 6, HPV type 11, HPV type 16, and HPV type 18. In

one non-limiting example, the probe set includes or consists of each of the probes of SEQ ID NOs: 1770-2647.

(44) Further disclosed herein is a nucleic acid probe set capable of detecting nucleic acid molecules from one or more bacterial and/or protozoan pathogens, including one or more of *Treponema pallidum*, *Ehrlichia chaffeensis*, *Ehrlichia ewingii*, *Ehrlichia muris*, *Borrelia burgdorferi*, *Coxiella burnetii*, *Trypanosoma brucei*, *Trypanosoma cruzi*, *Leishmania major*, *Babesia microti*, *Plasmodium falciparum*, and *Plasmodium vivax*. In some embodiments, the probe set includes 10 or more probes for one or more of the viruses (such as 10 or more, 20 or more, 30 or more, 50 or more, 60 or more, 70 or more, 80 or more, 90 or more, 100 or more, 110 or more, or 120 or more), for example 30-120 probes, 50-100 probes, or 70-110 probes for one or more of *Treponema pallidum*, *Ehrlichia chaffeensis*, *Ehrlichia ewingii*, *Ehrlichia muris*, *Borrelia burgdorferi*, *Coxiella burnetii*, *Trypanosoma brucei*, *Leishmania major*, *Babesia microti*, *Plasmodium falciparum*, and *Plasmodium vivax*.

(45) In some embodiments, the probe set includes nucleic acid probes that are at least 80%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical to the nucleic acid sequences of SEQ ID NOs: 2648-3207. In other embodiments, the probe set includes a subset of the probes of SEQ ID NOs: 2648-3207, such as at least 10%, at least 20%, at least 30%, at least 40%, at least 50%, at least 60%, at least 70%, at least 80%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or at least 99.9% of the probes of SEQ ID NOs: 2648-3207. In some examples, the subset includes at least one probe for each of *Treponema pallidum*, *Ehrlichia chaffeensis*, *Ehrlichia ewingii*, *Ehrlichia muris*, *Borrelia burgdorferi*, *Coxiella burnetii*, *Trypanosoma brucei*, *Trypanosoma cruzi*, *Leishmania major*, *Babesia microti*, *Plasmodium falciparum*, and *Plasmodium vivax*, such as at least 1, at least 2, at least 5, at least 10, at least 20, at least 30, or more probes for each pathogen. In some examples, the subset includes at least 10 probes (such as at least 20, at least 30, at least 50, at least 60, at least 70, at least 80, at least 90, at least 100, or at least 110 probes) for one or more of *Treponema pallidum*, *Ehrlichia chaffeensis*, *Ehrlichia ewingii*, *Ehrlichia muris*, *Borrelia burgdorferi*, *Coxiella burnetii*, *Trypanosoma brucei*, *Leishmania major*, *Babesia microti*, *Plasmodium falciparum*, and *Plasmodium vivax*. In one non-limiting example, the probe set includes or consists of each of the probes of SEQ ID NOs: 2648-3207.

(46) In further embodiments, one or more of the disclosed probe sets are combined. Thus, some embodiments, the probe set includes probes for detecting at least one RNA virus, at least one DNA virus, at least one bacterial pathogen, at least one protozoan pathogen, or combinations of two or more thereof. In one example, a probe set includes probes capable of detecting nucleic acid molecules from one or more DNA viruses and one or more bacterial and/or protozoan pathogens. In one non-limiting example, a probe set includes probes including or consisting of each of the probes of SEQ ID NOs: 1770-3207. In another example, a probe set includes probes capable of detecting nucleic acid molecules from one or more RNA viruses, one or more DNA viruses, and one or more bacterial and/or protozoan pathogens. In a non-limiting example, the probe set includes probes including or consisting of each of the probes of SEQ ID NOs: 1-1300, SEQ ID NOs: 1391-1570, SEQ ID NOs: 1691-1769, and SEQ ID NOs: 1770-3207.

(47) In additional embodiments, a disclosed probe set further includes one or more control probes, such as one or more positive and/or negative control probes. For testing for validity of the run, intra-array reproducibility control and normalization. positive control probes may include one or more of: 1) one or more reference probes for intensity normalization, 2) one or more internal standards of known concentrations, and 3) one or more probes that are homologous to an internal control included in the hybridization mix. In some embodiments, positive control probes include one or more (such as 1, 10, 25, 50, 96, or more) ERCC probes (External RNA Controls Consortium) and one or more (such as 1, 10, 25, 50, 96, 250, 500, 900, or more) biological replicates targeting human genome sequences (for example, to define possible host contaminant).

(48) Negative control probes may include one or more probes for a virus that is known not to be present in human or mammalian subjects. In some non-limiting examples, negative control probes are specific for a plant virus. In other examples, negative control probes can be a structural negative probe, such as a sequence that forms a hairpin and does not hybridize with nucleic acids from any species.

(49) In some examples, the probe set includes at least 10, at least 20, at least 30, at least 40, at least 50, at least 60, at least 70, at least 80, at least 90, at least 100, at least 110, or at least 120 control probes. In some examples, the control probes are for one or more one or more different negative control viruses (such as 1, 2, 3, 4, 5, or more negative control viruses). In some examples, the probe set includes at least 10, at least 20, at least 30, at least 40, at least 50, at least 60, at least 70, at least 80, at least 90, at least 100, at least 110, at least 120, or more negative control probes. In some examples, the negative control probes are probes for one or more of White clover cryptic virus 1 (e.g. SEQ ID NOs: 1571-1580), Broad bean wilt virus 1 (e.g., SEQ ID NOs: 1582-1620), Lettuce necrotic yellows virus (e.g., SEQ ID NOs: 1621-1690), *Aedes albopictus* densovirus 2 (e.g., SEQ ID NOs: 3520-3557), Maize streak virus (e.g., SEQ ID NOs: 3558-3598), and/or Tomato pseudo-curly top virus (e.g., SEQ ID NOs: 3599-3628). In additional examples, the probe set includes at least 10, at least 20, at least 30, at least 40, at least 50, at least 60, at least 70, at least 80, at least 90, at least 100, at least 110, at least 120, or more positive control probes. In some examples, the positive control probes are probes for one or more housekeeping genes, such as one of more of ACTB (e.g., SEQ ID NOs: 3208-3301), ARL1 (e.g., SEQ ID NOs: 3302-3385), and/or CCDN1 (e.g., SEQ ID NOs: 3386-3519).

(50) In some embodiments, the disclosed probes are between 30 and 80 nucleotides in length (for example 30-50, 40-60, 50-70, or 60-80 nucleotides in length). In some examples, the probes are 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, or 80 nucleotides in length and are capable of hybridizing to the disclosed pathogen (e.g., viral, bacterial, or protozoan) nucleic acid molecules. In some examples, the probes are 60 nucleotides in length. In some examples, each of the probes in the probe set has a T_m between about 72-89° C., such as about 74-88° C., about 75-85° C., or about 76-82° C. In one specific example, each of the probes in the probe set has a T_m between 74.4 and 87.8° C. T_m ranges for exemplary RNA virus probes are shown in Table 1.

(51) In other embodiments the disclosed probe sets, or a subset thereof, are linked to a solid support. In some examples, the disclosed probe sets, or a subset thereof, are included on a microarray. In other examples, the solid support is a bead or plurality of beads, a microplate, column, or microfluidic device.

(52) In some embodiments, the microarray is a solid support or substrate including the probe set (or subset thereof) covalently linked to the support or substrate. Within an array, each arrayed probe is addressable, in that its location can be reliably and consistently determined within at least two dimensions of the array. Addressable arrays usually are computer readable, in that a computer can be programmed to correlate a particular address on the array with information about the sample at that position (such as hybridization or binding data, including for instance signal intensity).

(53) The microarray can include any of the probe sets described above, individually, or in combination. In some embodiments, the microarray includes nucleic acid probes that are at least 80%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical to the nucleic acid sequences of SEQ ID NOs: 1-1300, 1391-1570, and 1691-1769. In another embodiment, the microarray includes or consists of nucleic acid probes that are at least 80%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical to the nucleic acid sequences of SEQ ID NOs: 1-1300 and 1391-1769. In one non-limiting example, the microarray includes or consists of each of the probes of SEQ ID NOs: 1-1769. In another non-limiting example, the microarray includes or consists of each of the probes of SEQ ID NOs: 1-1300, 1391-1570, and 1691-1769. In a further non-limiting example, the microarray includes or consists of each of the probes of SEQ ID NOs: 1-1300 and 1391-1769.

(54) In another embodiment, the microarray includes nucleic acid probes that are at least 80%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical to the nucleic acid sequences of SEQ ID NOs: 1770-2647. In one non-limiting example, the microarray includes or consists of each of the probes of SEQ ID NOs: 1770-2647. In another non-limiting example, the microarray includes or consists of each of the probes of SEQ ID NOs: 1170-2647 and 3250-3628.

(55) In another embodiment, the microarray includes nucleic acid probes that are at least 80%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical to the nucleic acid sequences of SEQ ID NOs: 2648-3207. In one non-limiting example, the microarray includes or consists of each of the probes of SEQ ID NOs: 2648-3207. In another non-limiting example, the microarray includes or consists of each of the probes of SEQ ID NOs: 2648-3519.

(56) In a further embodiment, the microarray includes nucleic acid probes that are at least 80%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% identical to the nucleic acid sequences of SEQ ID NOs: 1770-3207. In one non-limiting example, the microarray includes or consists of each of the probes of SEQ ID NOs: 1770-3207. In a further non-limiting example, the microarray includes or consists of each of the probes of SEQ ID NOs: 1770-3628.

(57) In other embodiments, the microarray includes a subset of the probes of SEQ ID NOs: 1-1300, 1391-1570, and 1691-1769, such as at least 10%, at least 20%, at least 30%, at least 40%, at least 50%, at least 60%, at least 70%, at least 80%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or at least 99.9% of the probes of SEQ ID NOs: 1-1300, 1391-1570, and 1691-1769. In some examples, the microarray includes at least one probe for each of CHIKV, DEN1, DEN2, DEN3, DEN4, HAV, HCV type 1, HCV type 2, HCV type 3, HEV, HIV type 1, HIV type 2, HTLV type 1, HTLV type 2, WNV, and ZKV, such as at least 1, at least 2, at least 5, at least 10, at least 20, at least 30, or more probes for each virus. In some examples, the microarray includes at least 40 probes (such as at least 50, at least 60, at least 70, at least 80, at least 90, at least 100, or at least 110 probes) for one or more of CHIKV, DEN1, DEN2, DEN3, DEN4, HAV, HCV type 1, HCV type 2, HCV type 3, HEV, HIV type 1, HIV type 2, HTLV type 1, HTLV type 2, WNV, and ZKV.

(58) In other embodiments, the microarray includes a subset of the probes of SEQ ID NOs: 1770-2647, such as at least 10%, at least 20%, at least 30%, at least 40%, at least 50%, at least 60%, at least 70%, at least 80%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or at least 99.9% of the probes of SEQ ID NOs: 1770-2647. In some examples, the microarray includes at least one probe for each of CMV, EBV subtype B95-8, EBV subtype AG876, human herpes virus 8, Hepatitis B virus subtype adw, Hepatitis B virus subtype ayw, Hepatitis B virus subtype adr, Hepatitis B virus subtype ayr, human parvovirus B19, HPV type 6, HPV type 11, HPV type 16, and HPV type 18, such as at least 1, at least 2, at least 5, at least 10, at least 20, at least 30, or more probes for each virus. In some examples, the microarray includes at least 20 probes (such as at least 40, at least 50, at least 60, at least 70, at least 80, at least 90, at least 100, or at least 110 probes) for one or more of CMV, EBV subtype B95-8, EBV subtype AG876, human herpes virus 8, Hepatitis B virus subtype adw, Hepatitis B virus subtype ayw, Hepatitis B virus subtype adr, Hepatitis B virus subtype ayr, human parvovirus B19, HPV type 6, HPV type 11, HPV type 16, and HPV type 18.

(59) In other embodiments, the microarray includes a subset of the probes of SEQ ID NOs: 2648-3207, such as at least 10%, at least 20%, at least 30%, at least 40%, at least 50%, at least 60%, at least 70%, at least 80%, at least 90%, at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or at least 99.9% of the probes of SEQ ID NOs: 2648-3207. In some examples, the microarray includes at least one probe for each of *Treponema pallidum*, *Ehrlichia chaffeensis*, *Ehrlichia ewingii*, *Ehrlichia muris*, *Borrelia burgdorferi*, *Coxiella burnetii*, *Trypanosoma brucei*,

Trypanosoma cruzi, *Leishmania major*, *Babesia microti*, *Plasmodium falciparum*, and *Plasmodium vivax*, such as at least 1, at least 2, at least 5, at least 10, at least 20, at least 30, or more probes for each virus. In some examples, the microarray includes at least 10 probes (such as at least 30, at least 40, at least 50, at least 60, at least 70, at least 80, at least 90, at least 100, or at least 110 probes) for one or more of *Treponema pallidum*, *Ehrlichia chaffeensis*, *Ehrlichia ewingii*, *Ehrlichia muris*, *Borrelia burgdorferi*, *Coxiella burnetii*, *Trypanosoma brucei*, *Leishmania major*, *Babesia microti*, *Plasmodium falciparum*, and *Plasmodium vivax*.

(60) In additional embodiments, the microarray includes one or more control probes, such as one or more positive and/or negative control probes. In some examples, the microarray includes at least 10, at least 20, at least 30, at least 40, at least 50, at least 60, at least 70, at least 80, at least 90, at least 100, at least 110, or at least 120 negative control probes. In some examples, the microarray includes one or more negative control probes selected from SEQ ID NOs: 1571-1580, SEQ ID NOs: 1582-1620, SEQ ID NOs: 1621-1690, and SEQ ID NOs: 3520-3628. In additional examples, the microarray includes at least 10, at least 20, at least 30, at least 40, at least 50, at least 60, at least 70, at least 80, at least 90, at least 100, at least 110, or at least 120 positive control probes. In some examples, the microarray includes one or more positive control probes selected from SEQ ID NOs: 3208-3519.

(61) The solid support or substrate of the array can be formed from an organic polymer. Suitable materials for the solid support include, but are not limited to: polypropylene, polyethylene, polybutylene, polyisobutylene, polybutadiene, polyisoprene, polyvinylpyrrolidine, polytetrafluoroethylene, polyvinylidene difluoride, polyfluoroethylene-propylene, polyethylenevinyl alcohol, polymethylpentene, polychlorotrifluoroethylene, polysulfones, hydroxylated biaxially oriented polypropylene, aminated biaxially oriented polypropylene, thiolated biaxially oriented polypropylene, ethyleneacrylic acid, thylene methacrylic acid, and blends of copolymers thereof).

(62) A wide variety of array formats can be employed in accordance with the present disclosure. One example includes a two-dimensional pattern of discrete cells (such as 4096 squares in a 64 by 64 array). Other array formats including, but not limited to slot (rectangular) and circular arrays are equally suitable for use. In some examples, the array is a multi-well plate. In one example, the array is formed on a polymer medium, which is a thread, membrane or film. An example of an organic polymer medium is a polypropylene sheet having a thickness on the order of about 1 mil. (0.001 inch) to about 20 mil., although the thickness of the film is not critical and can be varied over a fairly broad range. The array can include biaxially oriented polypropylene (BOPP) films, which in addition to their durability, exhibit low background fluorescence.

(63) The array formats of the present disclosure can be included in a variety of different types of formats. A “format” includes any format to which the solid support can be affixed, such as microtiter plates (e.g., multi-well plates), test tubes, inorganic sheets, dipsticks, and the like. For example, membranes can be affixed to glass slides. The particular format is, in and of itself, unimportant. All that is necessary is that the solid support can be affixed thereto without affecting the functional behavior of the solid support or any biopolymer absorbed thereon, and that the format (such as the slide) is stable to any materials into which the device is introduced (such as clinical samples and hybridization solutions).

(64) The arrays of the present disclosure can be prepared by a variety of approaches. In one example, oligonucleotides (e.g., probes) are synthesized separately and then attached to a solid support (see U.S. Pat. No. 6,013,789). In another example, probes are synthesized directly onto the support to provide the desired array (see U.S. Pat. No. 5,554,501). Suitable methods for covalently coupling oligonucleotides to a solid support and for directly synthesizing oligonucleotides on the support are known; a summary of suitable methods can be found in Matson et al., Anal. Biochem. 217:306-10, 1994. In one example, the oligonucleotides are synthesized onto the support using conventional chemical techniques for preparing oligonucleotides on solid supports (such as PCT applications WO 85/01051 and WO 89/10977, or U.S. Pat. No. 5,554,501).

(65) The oligonucleotides can be bound to the support or substrate by either the 3' end of the oligonucleotide or by the 5' end of the oligonucleotide. In one example, the oligonucleotides are bound to the solid support by the 3' end. In general, the internal complementarity of an oligonucleotide probe in the region of the 3' end and the 5' end determines binding to the support.

III. Methods of Detecting Viral Nucleic Acids

(66) Disclosed herein are methods of detecting one or more pathogen nucleic acids (such as one or more viral, bacterial, and/or protozoan nucleic acids) in a sample from a subject. In some embodiments, the methods include preparing or isolating nucleic acids (such as DNA, RNA, or cDNA) from a sample, labeling the nucleic acids, and contacting the probe set, or a microarray including the probe set, with the labeled nucleic acids under conditions sufficient to allow pathogen nucleic acids present in the sample to hybridize with one or more of the probes. The presence and/or identity of pathogen nucleic acids in the sample is determined by detecting hybridization. In one example, hybridization is detected by measuring presence of labeled nucleic acid at an addressable location in an array.

(67) In particular embodiments, the methods include detecting one or more nucleic acids from RNA viruses in a sample, including one or more of CHIKV, DEN1, DEN2, DEN3, DEN4, HAV, HCV type 1, HCV type 2, HCV type 3, HEV, HIV type 1, HIV type 2, HTLV type I, HTLV type II, WNV, and ZKV. In other embodiments, the methods include detecting one or more nucleic acids from DNA viruses, including one or more of CMV, EBV subtype B95-8, EBV subtype AG876, human herpes virus 8, Hepatitis B virus subtype adw, Hepatitis B virus subtype ayw, Hepatitis B virus subtype adr, Hepatitis B virus subtype ayr, human parvovirus B19, HPV type 6, HPV type 11, HPV type 16, and HPV type 18. In still other embodiments, the methods include detecting one or more nucleic acids from bacteria and/or protozoans, including one or more of *Treponema pallidum*, *Ehrlichia chaffeensis*, *Ehrlichia ewingii*, *Ehrlichia muris*, *Borrelia burgdorferi*, *Coxiella burnetii*, *Trypanosoma brucei*, *Trypanosoma cruzi*, *Leishmania major*, *Babesia microti*, *Plasmodium falciparum*, and *Plasmodium vivax*. In further embodiments, the methods include detecting nucleic acids from at least one RNA virus, DNA virus, bacteria, and protozoan, such as at least one of the RNA viruses, DNA viruses, bacteria, and protozoans disclosed herein, or any combination thereof.

(68) Exemplary samples include peripheral blood, serum, plasma, cerebrospinal fluid, urine, saliva, feces, mucus, nasal wash, tissue biopsy, fine needle aspirate, surgical specimen, placenta, autopsy material, semen, vaginal fluid or tissue, and environmental samples. In particular examples, the sample is a blood sample, such as plasma. In non-limiting examples, the sample is blood or plasma from a blood donor. Thus, in some examples, the methods disclosed herein are used to screen donated blood for one or more pathogens potentially present and/or transmitted through blood transfusions.

(69) In some embodiments, the methods include isolating nucleic acids (such as RNA, cDNA, or a combination thereof) from the sample and contacting the probe set or microarray with the isolated nucleic acids. Methods of isolating RNA (e.g., viral RNA) from a sample are known and include commercially available kits, such as QIAGEN® RNeasy® mini-columns, MASTERPURE® Complete DNA and RNA Purification Kit (EPICENTRE® Madison, Wis.), Paraffin Block RNA Isolation Kit (Ambion, Inc.), and RNA Stat-60 (Tel-Test). cDNA is then prepared from the isolated RNA, and optionally labeled. In some examples, the methods include amplifying RNA prior to cDNA preparation and labeling, for example, using Quick Amp WT labeling kit (Agilent). Other methods of amplifying RNA include commercially available kits such as Ovation® RNA Amplification kit (Nugen), Arcturus™ RiboAmp™ HS kit (ThermoFisher), and Complete Whole Transcriptome Amplification Kit (WTA2, Sigma-Aldrich).

(70) In some embodiments, the methods do not include isolating and/or amplifying RNA from a sample prior to labeling. In some examples, the methods include generating amplified cDNA from a sample, followed by labeling the cDNA (for example with a fluorescent label, such as CyTM3 dye). In one non-limiting example, amplified cDNA is prepared from the sample using single-

primer isothermal amplification (for example, Ribo-SPIA® amplification, NuGen) prior to labeling. Methods of labeling cDNA are known and include commercially available kits, such as Genomic DNA Enzymatic Labeling Kit (Agilent). In some examples, the methods generate amplified and labeled cDNA from about 250 pg of target viral RNA (such as about 500 pg, about 750 pg, about 1 ng, about 2 ng, or more of target viral RNA).

(71) In other examples, the methods include isolating DNA from the sample and contacting the probe set or microarray with the isolated DNA. Methods of isolating DNA (such as viral DNA, bacterial DNA, or protozoan DNA) from a sample are known and include commercially available kits. In some examples, the methods include isolating viral DNA from a sample using a viral nucleic acid isolation kit. In one non-limiting example, the viral DNA is isolated using Dynabeads™ SILANE viral NA kit (Invitrogen). In other examples, bacterial or protozoan DNA is isolated from a sample using a DNA isolation kit. In one non-limiting example, bacterial or protozoan DNA is isolated using QIAamp® DNA Blood Mini kit (Qiagen). One of ordinary skill in the art can select appropriate methods or kits to isolate pathogen DNA from samples, for example, blood or plasma samples.

(72) The sample (such as nucleic acids isolated and/or amplified from a sample) can be labeled with any suitable label. Generally, the label will be selected based on the intended use of the sample or the desired readout. In some examples, the sample or nucleic acids from the sample is labelled with a fluorescent or chemiluminescent compound. In other examples, the label is an enzyme, a fluorophore, or a radioactive isotope. In one specific non-limiting example, the label is Cy™3 or Cy™5.

(73) Fluorophores suitable for use with the methods disclosed herein, include, but are not limited to, 6-carboxyfluorescein (FAM), tetrachlorofluorescein (TET), tetramethylrhodamine (TMR), hexachlorofluorescein (HEX), JOE, ROX, CAL Fluor™ dye, Pulsar™ dye, Quasar™ dye, Texas Red™ dye, Cy™3 dye and Cy™5 dye. Other examples of fluorophores that can be used with the methods provided herein include 4-acetamido-4'-isothiocyanatostilbene-2,2'-disulfonic acid, acridine and derivatives such as acridine and acridine isothiocyanate, 5-(2'-aminoethyl)aminonaphthalene-1-sulfonic acid (EDANS), 4-amino-N-[3-vinylsulfonyl]phenyl]-naphthalimide-3,5-disulfonate (Lucifer Yellow VS), N-(4-anilino-1-naphthyl)-maleimide, anthranilamide, Brilliant Yellow, coumarin and derivatives such as coumarin, 7-amino-4-methylcoumarin (AMC, Coumarin 120), 7-amino-4-trifluoromethylcoumarin (Coumarin 151); cyanosine; 4',6-diaminidino-2-phenylindole (DAPI); 5', 5''-dibromopyrogallol-sulfonephthalein (Bromopyrogallol Red); 7-diethylamino-3-(4'-isothiocyanatophenyl)-4-methylcoumarin; diethylenetriamine pentaacetate; 4,4'-diisothiocyanatodihydro-stilbene-2,2'-disulfonic acid; 4,4'-diisothiocyanatostilbene-2,2'-disulfonic acid; 5-[dimethyl-amino]naphthalene-1-sulfonyl chloride (DNS, dansyl chloride); 4-(4'-dimethyl-aminophenylazo)benzoic acid (DABCYL); 4-dimethylaminophenylazophenyl-4'-isothiocyanate (DABITC); eosin and derivatives such as eosin and eosin isothiocyanate; erythrosin and derivatives such as erythrosin B and erythrosin isothiocyanate; ethidium; fluorescein and derivatives such as 5-carboxyfluorescein (FAM), 5-(4,6-dichlorotriazin-2-yl)aminofluorescein (DTAF), 2'7'-dimethoxy-4'5'-dichloro-6-carboxyfluorescein (JOE), fluorescein, fluorescein isothiocyanate (FITC), and QFITC (XRITC); fluorescamine; IR144; IR1446; Malachite Green isothiocyanate; 4-methylumbelliferone; ortho cresolphthalein; nitrotyrosine; pararosaniline; Phenol Red; B-phycoerythrin; o-phthaldialdehyde; pyrene and derivatives such as pyrene, pyrene butyrate and succinimidyl 1-pyrene butyrate; Reactive Red 4 (Cibacron Brilliant Red 3B-A); rhodamine and derivatives such as 6-carboxy-X-rhodamine (ROX), 6-carboxyrhodamine (R6G), lissamine rhodamine B sulfonyl chloride, rhodamine (Rhod), rhodamine B, rhodamine 123, rhodamine X isothiocyanate, sulforhodamine B, sulforhodamine 101 and sulfonyl chloride derivative of sulforhodamine 101 (Texas Red™ dye); N,N,N',N'-tetramethyl-6-carboxyrhodamine (TAMRA); tetramethyl rhodamine; tetramethyl rhodamine isothiocyanate (TRITC); riboflavin; rosolic acid and terbium chelate derivatives.

(74) Other fluorophores that can be used include thiol-reactive europium chelates that emit at approximately 617 nm (Heyduk and Heyduk, *Analyt. Biochem.* 248:216-27, 1997; *J. Biol. Chem.* 274:3315-22, 1999). Other fluorophores that can be used include cyanine, merocyanine, styryl, and oxonyl compounds, such as those disclosed in U.S. Pat. Nos. 5,627,027; 5,486,616; 5,569,587; and 5,569,766, and in published PCT application no. US98/00475, each of which is incorporated herein by reference. Specific examples of fluorophores disclosed in one or more of these patent documents include CyTM3 and CyTM5, for instance, and substituted versions of these fluorophores. Additional fluorophores that can be used include GFP, LissamineTM, diethylaminocoumarin, fluorescein chlorotriazinyl, naphthofluorescein, 4,7-dichlororhodamine and xanthene (as described in U.S. Pat. No. 5,800,996 to Lee et al., herein incorporated by reference) and derivatives thereof. Other fluorophores are commercially available from known sources.

(75) The methods include contacting the sample (such as labeled nucleic acids from a sample) with a probe set disclosed herein (or subset thereof), or a microarray including the probe set (or subset thereof), under conditions sufficient to allow hybridization of pathogen nucleic acids present in the sample to one or more probes and detecting presence of pathogen nucleic acids hybridized to the probe set or microarray.

(76) Presence of one or more pathogen nucleic acids in the sample can be detected using any suitable means. For example, detection of hybridization can be accomplished by detecting nucleic acid molecules (such as RNA) using nucleic acid amplification methods (such as real-time RT-PCR) or array analysis. In a specific embodiment of the microarray technique, labeled cDNA prepared from a sample is applied to an array including a probe set disclosed herein. Labeled cDNA from the sample can hybridize specifically to one or more probes on the array. After washing to remove non-specifically bound probes, the chip is scanned by confocal laser microscopy or by another detection method, such as a CCD camera. Quantitation of hybridization of sample to each arrayed element allows for assessment of corresponding RNA abundance (e.g., if cDNA is analyzed). Microarray analysis can be performed by commercially available equipment, following manufacturer's protocols, such as are supplied with Affymetrix GeneChip® technology (Affymetrix, Santa Clara, CA), or Agilent's microarray technology (Agilent Technologies, Santa Clara, CA).

(77) In some examples, a sample is determined to contain nucleic acids from a particular pathogen by detecting hybridization between the sample (nucleic acid) and one or more probes of the pathogen-specific probe set. In some examples, a sample is determined to be positive for a pathogen when the log ratio between the signal intensity mean for the pathogen-specific probe set and the mean of a control group probe set is ≥ 1.5 . In other examples, a sample is determined to be negative for a pathogen when the log ratio between the signal intensity mean for the pathogen-specific probe set and the mean of a control group probe set is < 1 . In further examples, the sample is determined to be borderline for the pathogen when the log ratio between the signal intensity mean for the pathogen-specific probe set and the mean of a control group probe set is ≥ 1.0 to ≤ 1.5 . In some examples, a sample that is determined to be borderline for one or more pathogens is retested (for example, retested with the assay disclosed herein and/or tested using a virus-specific nucleic acid based test). In other examples, a sample that is determined to be borderline is discarded (e.g., not administered to a subject). In additional examples, a sample is determined to be positive for a particular pathogen when $\geq 50\%$ of the individual probes for the particular pathogen have a log ratio of > 1.5 . In some examples, a sample is determined to be positive for a particular pathogen when $\geq 50\%$ of the individual probes for the particular pathogen have a log ratio of > 1.5 and the log ratio between the signal intensity mean for the pathogen-specific probe set and the mean of a control group probe set is ≥ 1.5 .

EXAMPLES

(78) The following examples are provided to illustrate certain particular features and/or embodiments. These examples should not be construed to limit the disclosure to the particular

features or embodiments described.

Example 1

Materials and Methods

(79) Microarray-Based Platform Design

(80) Selection of transfusion-transmitted RNA viruses: Sequences of 16 RNA viruses of concern for transmission to blood recipients (released by AABB Transfusion-Transmitted Diseases Committee (Stramer et al., *Transfusion* 49:1S-29S, 2009)) were downloaded from GenBank at NCBI (available on the World Wide Web at ncbi.nlm.nih.gov/genbank).

(81) The complete genome for each RNA virus was uploaded in FASTA format using Agilent eArray software (available on the World Wide Web at earray.chem.agilent.com/earray/, Agilent Technologies Inc., Santa Clara, CA). Design settings were chosen to select 60-mer sense probes with 3' bias from each viral gene, according to the base composition methodology, which considers fusion temperature, GC % and cross-hybridization potential for probes. To get the best quality level probes for viral genome detection the “best probe” (BP) was selected. The probes were checked for vector and low complexity masking. Entire viral genome sequences were covered to the extent possible with all available Agilent-designed probes. The microarray was supplemented with additional predesigned GE (gene expression) array probes for 906 genes from the human genome (replicated 10 times), ERCC probes (replicated 45 times) and probes covering plant virus sequences (negative control). The selected probes and their characteristics are provided in Table 1.

(82) Oligonucleotide probe selection and methodology: Oligonucleotide probes were synthesized in situ from 3'-end base by base with Agilent SurePrint inkjet technology according to the manufacturer's protocol (Wolber et al., *Meth. Enzymol.* 410:28-57, 2006). The microarrays were manufactured with 60-mer oligonucleotides synthesized in 15,000 features on eight replicate arrays per slide.

(83) Sample collection and processing: Specimens positive for CHIKV, DENV1-4, HIV1-2, WNV strain NY99, and ZIKV were prepared, validated, and supplied by the FDA Center for Biologics Evaluation and Research (CBER) (Dong et al., *J. Appl. Microbiol.* 120:1119-1129, 2016).

(84) HCV genotypes 1a, 2a, and 3, and HEV RNA-positive plasma were purchased from Sera Care (Sera Care, Milford, MA). All positive specimens were diluted in negative plasma (Basematrix diluent, Sera Care) to create a range of concentrations. HAV RNA was obtained from Dr. Patrizia Farci, (National Institutes of Health, Bethesda, MD). HTLV types I and II NATtrol (Nucleic Acid Testing Control) were purchased from ZeptoMetrix (ZeptoMetrix, Buffalo, NY) (Table 2).

(85) TABLE-US-00001 TABLE 1 Selected viral probes SEQ BP ID Probe Product Virus Target ID

Start	NO:	Length	5' UTR	Hepatitis C	gi 22129792:1-341	243	1	60	genotype 1	5' UTR	Hepatitis C	gi 22129792:1-341	194	2	60	genotype 1	5' UTR	Hepatitis C	gi 22129792:1-341	173	4	60	genotype 1	5' UTR	Hepatitis C	gi 22129792:1-341	172	5	60	genotype 1	5' UTR	Hepatitis C	gi 22129792:1-341	171	6	60	genotype 1	5' UTR	Hepatitis C	gi 22129792:1-341	170	7	60	genotype 1	5' UTR	Hepatitis C	gi 22129792:1-341	169	8	60	genotype 1	5' UTR	Hepatitis C	gi 22129792:1-341	168	9	60	genotype 1	5' UTR	Hepatitis C	gi 22129792:1-341	167	10	60	genotype 1	core protein	Hepatitis C	gi 22129792:342-914	488	11	60	genotype 1	core protein	Hepatitis C	gi 22129792:342-914	487	12	60	genotype 1	core protein	Hepatitis C	gi 22129792:342-914	486	13	60	genotype 1	core protein	Hepatitis C	gi 22129792:342-914	485	14	60	genotype 1	core protein	Hepatitis C	gi 22129792:342-914	484	15	60	genotype 1	core protein	Hepatitis C	gi 22129792:342-914	483	16	60	genotype 1	core protein	Hepatitis C	gi 22129792:342-914	482	17	60	genotype 1	core protein	Hepatitis C	gi 22129792:342-914	481	18	60	genotype 1	core protein	Hepatitis C	gi 22129792:342-914	480	19	60	genotype 1	core protein	Hepatitis C	gi 22129792:342-914	479	20	60	genotype 1	E1 protein	Hepatitis C	gi 22129792:915-1494	342	21	60	genotype 1	E1 protein	Hepatitis C	gi 22129792:915-1494	341	22	60	genotype 1	E1 protein	Hepatitis C	gi 22129792:915-1494	340	23	60	genotype 1	E1 protein	Hepatitis C	gi 22129792:915-1494	339	24	60	genotype 1	E1 protein	Hepatitis C	gi 22129792:915-1494	504	25
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60 genotype 1 E1 protein Hepatitis C gi|22129792:915-1494 503 26 60 genotype 1 E1 protein
Hepatitis C gi|22129792:915-1494 502 27 60 genotype 1 E1 protein Hepatitis C gi|22129792:915-
1494 501 28 60 genotype 1 E1 protein Hepatitis C gi|22129792:915-1494 500 29 60 genotype 1 E1
protein Hepatitis C gi|22129792:915-1494 379 30 60 genotype 1 E2 protein Hepatitis C
gi|22129792:1491-2579 703 31 60 genotype 1 E2 protein Hepatitis C gi|22129792:1491-2579 702
32 60 genotype 1 E2 protein Hepatitis C gi|22129792:1491-2579 701 33 60 genotype 1 E2 protein
Hepatitis C gi|22129792:1491-2579 700 34 60 genotype 1 E2 protein Hepatitis C
gi|22129792:1491-2579 699 35 60 genotype 1 E2 protein Hepatitis C gi|22129792:1491-2579 697
36 60 genotype 1 E2 protein Hepatitis C gi|22129792:1491-2579 696 37 60 genotype 1 E2 protein
Hepatitis C gi|22129792:1491-2579 695 38 60 genotype 1 E2 protein Hepatitis C
gi|22129792:1491-2579 694 39 60 genotype 1 E2 protein Hepatitis C gi|22129792:1491-2579 693
40 60 genotype 1 p7 protein Hepatitis C gi|22129792:2580-2768 56 41 60 genotype 1 p7 protein
Hepatitis C gi|22129792:2580-2768 55 42 60 genotype 1 p7 protein Hepatitis C gi|22129792:2580-
2768 54 43 60 genotype 1 p7 protein Hepatitis C gi|22129792:2580-2768 53 44 60 genotype 1 p7
protein Hepatitis C gi|22129792:2580-2768 52 45 60 genotype 1 p7 protein Hepatitis C
gi|22129792:2580-2768 51 46 60 genotype 1 p7 protein Hepatitis C gi|22129792:2580-2768 50 47
60 genotype 1 p7 protein Hepatitis C gi|22129792:2580-2768 49 48 60 genotype 1 p7 protein
Hepatitis C gi|22129792:2580-2768 48 49 60 genotype 1 p7 protein Hepatitis C gi|22129792:2769-
3419 47 50 60 genotype 1 NS2 protein Hepatitis C gi|22129792:2580-2768 366 51 60 genotype 1
NS2 protein Hepatitis C gi|22129792:2769-3419 194 52 60 genotype 1 NS2 protein Hepatitis C
gi|22129792:2769-3419 193 53 60 genotype 1 NS2 protein Hepatitis C gi|22129792:2769-3419
192 54 60 genotype 1 NS2 protein Hepatitis C gi|22129792:2769-3419 191 55 60 genotype 1 NS2
protein Hepatitis C gi|22129792:2769-3419 190 56 60 genotype 1 NS2 protein Hepatitis C
gi|22129792:2769-3419 189 57 60 genotype 1 NS2 protein Hepatitis C gi|22129792:2769-3419
188 58 60 genotype 1 NS2 protein Hepatitis C gi|22129792:2769-3419 187 59 60 genotype 1 NS2
protein Hepatitis C gi|22129792:2769-3419 186 60 60 genotype 1 NS3 protease/ Hepatitis C
gi|22129792:3420-5312 1812 61 60 helicase genotype 1 NS3 protease/ Hepatitis C
gi|22129792:3420-5312 1620 62 60 helicase genotype 1 NS3 protease/ Hepatitis C
gi|22129792:3420-5312 849 63 60 helicase genotype 1 NS3 protease/ Hepatitis C
gi|22129792:3420-5312 1744 64 60 helicase genotype 1 NS3 protease/ Hepatitis C
gi|22129792:3420-5312 1497 65 60 helicase genotype 1 NS3 protease/ Hepatitis C
gi|22129792:3420-5312 1294 66 60 helicase genotype 1 NS3 protease/ Hepatitis C
gi|22129792:3420-5312 1234 67 60 helicase genotype 1 NS3 protease/ Hepatitis C
gi|22129792:5313-5474 1092 68 60 helicase genotype 1 NS3 protease/ Hepatitis C
gi|22129792:3420-5312 1020 69 60 helicase genotype 1 NS3 protease/ Hepatitis C
gi|22129792:3420-5312 765 70 60 helicase genotype 1 NS4A protein Hepatitis C
gi|22129792:3420-5312 103 71 60 genotype 1 NS4A protein Hepatitis C gi|22129792:5313-5474
102 72 60 genotype 1 NS4A protein Hepatitis C gi|22129792:5313-5474 101 73 60 genotype 1
NS4A protein Hepatitis C gi|22129792:5313-5474 100 74 60 genotype 1 NS4A protein Hepatitis C
gi|22129792:5313-5474 99 75 60 genotype 1 NS4A protein Hepatitis C gi|22129792:5313-5474 98
76 60 genotype 1 NS4A protein Hepatitis C gi|22129792:5313-5474 97 77 60 genotype 1 NS4A
protein Hepatitis C gi|22129792:5313-5474 96 78 60 genotype 1 NS4A protein Hepatitis C
gi|22129792:5313-5474 95 79 60 genotype 1 NS4A protein Hepatitis C gi|22129792:5313-5474 94
80 60 genotype 1 NS4B protein Hepatitis C gi|22129792:5475-6257 133 81 60 genotype 1 NS4B
protein Hepatitis C gi|22129792:5475-6257 723 82 60 genotype 1 NS4B protein Hepatitis C
gi|22129792:5475-6257 702 83 60 genotype 1 NS4B protein Hepatitis C gi|22129792:5475-6257
456 84 60 genotype 1 NS4B protein Hepatitis C gi|22129792:5475-6257 416 85 60 genotype 1
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gag HTLV 1 gb|AF033817.1|:450-1739 921 517 60 gag HTLV 1 gb|AF033817.1|:450-1739 917
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HTLV 1 gi|9626453:4829-6295 662 565 60 env HTLV 1 gi|9626453:4829-6295 621 566 60 env
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HTLV 2 gi|9626726:1-763 262 571 60 5' LTR HTLV 2 gi|9626726:1-763 261 572 60 5' LTR HTLV
2 gi|9626726:1-763 260 573 60 5' LTR HTLV 2 gi|9626726:1-763 258 574 60 5' LTR HTLV 2
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2 gi|9626726:6-119 52 581 60 gp1-tax protein HTLV 2 gi|9626726:6-119 51 582 60 gp1-tax protein
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membrane DEN 1 gi|9626685:437-934 436 914 60 glycoprotein precursor M membrane DEN 1
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membrane DEN 1 gi|9626685:437-934 430 920 60 glycoprotein precursor M envelope DEN 1
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pol. NS5 RNA- DEN 2 gi|158976983:7570-10269 2440 1094 60 dependent RNA pol. NS5 RNA-
DEN 2 gi|158976983:7570-10269 2379 1095 60 dependent RNA pol. NS5 RNA- DEN 2
gi|158976983:7570-10269 2152 1096 60 dependent RNA pol. NS5 RNA- DEN 2
gi|158976983:7570-10269 2092 1097 60 dependent RNA pol. NS5 RNA- DEN 2
gi|158976983:7570-10269 2032 1098 60 dependent RNA pol. NS5 RNA- DEN 2
gi|158976983:7570-10269 1969 1099 60 dependent RNA pol. NS5 RNA- DEN 2
gi|158976983:7570-10269 1894 1100 60 dependent RNA pol. NS5 anchored capsid DEN 3
gi|163644368:95-436 283 1101 60 protein C anchored capsid DEN 3 gi|163644368:95-436 282
1102 60 protein C anchored capsid DEN 3 gi|163644368:95-436 281 1103 60 protein C anchored
capsid DEN 3 gi|163644368:95-436 280 1104 60 protein C anchored capsid DEN 3
gi|163644368:95-436 279 1105 60 protein C anchored capsid DEN 3 gi|163644368:95-436 278
1106 60 protein C anchored capsid DEN 3 gi|163644368:95-436 277 1107 60 protein C anchored
capsid DEN 3 gi|163644368:95-436 276 1108 60 protein C anchored capsid DEN 3
gi|163644368:95-436 275 1109 60 protein C anchored capsid DEN 3 gi|163644368:95-436 274
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membrane DEN 3 gi|163644368:437-934 438 1112 60 glycoprotein precursor M membrane DEN 3
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gi|163644368:437-934 434 1116 60 glycoprotein precursor M membrane DEN 3
gi|163644368:437-934 432 1117 60 glycoprotein precursor M membrane DEN 3
gi|163644368:437-934 431 1118 60 glycoprotein precursor M membrane DEN 3
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1125 60 protein E envelope DEN 3 gi|163644368:935-2413 994 1126 60 protein E envelope DEN
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1128 60 protein E envelope DEN 3 gi|163644368:935-2413 805 1129 60 protein E envelope DEN
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997 1131 60 protein NS1 nonstructural DEN 3 gi|163644368:2414-3469 879 1132 60 protein NS1
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gi|163644368:2414-3469 380 1138 60 protein NS1 nonstructural DEN 3 gi|163644368:2414-3469
320 1139 60 protein NS1 nonstructural DEN 3 gi|163644368:2414-3469 256 1140 60 protein NS1
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protein NS2B nonstructural DEN 3 gi|163644368:4124-4513 312 1160 60 protein NS2B
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capsid DEN 4 gi|12084822:102-440 275 1204 60 protein C anchored capsid DEN 4
gi|12084822:102-440 274 1205 60 protein C anchored capsid DEN 4 gi|12084822:102-440 273
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gi|12084822:102-440 270 1209 60 protein C anchored capsid DEN 4 gi|12084822:102-440 269
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membrane DEN 4 gi|12084822:441-938 73 1218 60 glycoprotein precursor M membrane DEN 4
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DEN 4 gi|12084822:2424-3479 823 1232 60 protein NS1 non-structural DEN 4 gi|12084822:2424-
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NS1 non-structural DEN 4 gi|12084822:2424-3479 570 1235 60 protein NS1 non-structural DEN 4
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NS2B non-structural DEN 4 gi|12084822:4134-4523 329 1253 60 protein NS2B non-structural
DEN 4 gi|12084822:4134-4523 328 1254 60 protein NS2B non-structural DEN 4
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NS2B non-structural DEN 4 gi|12084822:4134-4523 324 1258 60 protein NS2B non-structural
DEN 4 gi|12084822:4134-4523 323 1259 60 protein NS2B non-structural DEN 4
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NS3 non-structural DEN 4 gi|12084822:4524-6377 1591 1263 60 protein NS3 non-structural DEN
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NS4A non-structural DEN 4 gi|12084822:6378-6758 317 1276 60 protein NS4A non-structural
DEN 4 gi|12084822:6378-6758 316 1277 60 protein NS4A non-structural DEN 4
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NS4A non-structural DEN 4 gi|12084822:6828-7562 657 1281 60 protein NS4B non-structural
DEN 4 gi|12084822:6828-7562 536 1282 60 protein NS4B non-structural DEN 4
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10262 2450 1293 60 protein NS5 non-structural DEN 4 gi|12084822:7563-10262 2379 1294 60
protein NS5 non-structural DEN 4 gi|12084822:7563-10262 2153 1295 60 protein NS5 non-
structural DEN 4 gi|12084822:7563-10262 2093 1296 60 protein NS5 non-structural DEN 4
gi|12084822:7563-10262 1925 1297 60 protein NS5 non-structural DEN 4 gi|12084822:7563-
10262 1865 1298 60 protein NS5 non-structural DEN 4 gi|12084822:7563-10262 1805 1299 60
protein NS5 non-structural DEN 4 gi|12084822:7563-10262 1702 1300 60 protein NS5 polyprotein
GB virus C/ gi|9628705:459-9080 8205 1301 60 precursor Hepatitis G polyprotein GB virus C/
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gi|9628705:459-9080 7305 1303 60 precursor Hepatitis G polyprotein GB virus C/
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gi|9628705:459-9080 6638 1306 60 precursor Hepatitis G polyprotein GB virus C/
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gi|9628705:459-1538 545 1311 60 protein Hepatitis G putative E1 GB virus C/ gi|9628705:459-
1538 1021 1312 60 protein Hepatitis G putative E1 GB virus C/ gi|9628705:459-1538 964 1313 60
protein Hepatitis G putative E1 GB virus C/ gi|9628705:459-1538 924 1314 60 protein Hepatitis G
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virus C/ gi|9628705:459-1538 783 1316 60 protein Hepatitis G putative E1 GB virus C/
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1538 644 1318 60 protein Hepatitis G putative E1 GB virus C/ gi|9628705:459-1538 601 1319 60
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putative E2 GB virus C/ gi|9628705:1539-2474 877 1321 60 protein Hepatitis G putative E2 GB
virus C/ gi|9628705:1539-2474 847 1322 60 protein Hepatitis G putative E2 GB virus C/

gi|9628705:1539-2474 817 1323 60 protein Hepatitis G putative E2 GB virus C/ gi|9628705:1539-2474 671 1324 60 protein Hepatitis G putative E2 GB virus C/ gi|9628705:1539-2474 641 1325 60 protein Hepatitis G putative E2 GB virus C/ gi|9628705:1539-2474 611 1326 60 protein Hepatitis G putative E2 GB virus C/ gi|9628705:1539-2474 573 1327 60 protein Hepatitis G putative E2 GB virus C/ gi|9628705:1539-2474 543 1328 60 protein Hepatitis G putative E2 GB virus C/ gi|9628705:1539-2474 502 1329 60 protein Hepatitis G putative E2 GB virus C/ gi|9628705:1539-2474 310 1330 60 protein Hepatitis G putative protein GB virus C/ gi|9628705:2475-3233 700 1331 60 p7-NS2 Hepatitis G putative protein GB virus C/ gi|9628705:2475-3233 594 1332 60 p7-NS2 Hepatitis G putative protein GB virus C/ gi|9628705:2475-3233 564 1333 60 p7-NS2 Hepatitis G putative protein GB virus C/ gi|9628705:2475-3233 532 1334 60 p7-NS2 Hepatitis G putative protein GB virus C/ gi|9628705:2475-3233 502 1335 60 p7-NS2 Hepatitis G putative protein GB virus C/ gi|9628705:2475-3233 472 1336 60 p7-NS2 Hepatitis G putative protein GB virus C/ gi|9628705:2475-3233 350 1337 60 p7-NS2 Hepatitis G putative protein GB virus C/ gi|9628705:2475-3233 320 1338 60 p7-NS2 Hepatitis G putative protein GB virus C/ gi|9628705:2475-3233 286 1339 60 p7-NS2 Hepatitis G putative protein GB virus C/ gi|9628705:2475-3233 194 1340 60 p7-NS2 Hepatitis G NS3 GB virus C/ gi|9628705:3234-5111 1168 1341 60 proteinase/ Hepatitis G ATPase/helicase NS3 GB virus C/ gi|9628705:3234-5111 1162 1342 60 proteinase/ Hepatitis G ATPase/helicase NS3 GB virus C/ gi|9628705:3234-5111 1156 1343 60 proteinase/ Hepatitis G ATPase/helicase NS3 GB virus C/ gi|9628705:3234-5111 1155 1344 60 proteinase/ Hepatitis G ATPase/helicase NS3 GB virus C/ gi|9628705:3234-5111 1154 1345 60 proteinase/ Hepatitis G ATPase/helicase NS3 GB virus C/ gi|9628705:3234-5111 1153 1346 60 proteinase/ Hepatitis G ATPase/helicase NS3 GB virus C/ gi|9628705:3234-5111 1152 1347 60 proteinase/ Hepatitis G ATPase/helicase NS3 GB virus C/ gi|9628705:3234-5111 1146 1348 60 proteinase/ Hepatitis G ATPase/helicase NS3 GB virus C/ gi|9628705:3234-5111 1145 1349 60 proteinase/ Hepatitis G ATPase/helicase NS3 GB virus C/ gi|9628705:3234-5111 1144 1350 60 proteinase/ Hepatitis G ATPase/helicase putative NS4A GB virus C/ gi|9628705:5112-5309 45 1351 60 protein Hepatitis G putative NS4A GB virus C/ gi|9628705:5112-5309 44 1352 60 protein Hepatitis G putative NS4A GB virus C/ gi|9628705:5112-5309 43 1353 60 protein Hepatitis G putative NS4A GB virus C/ gi|9628705:5112-5309 42 1354 60 protein Hepatitis G putative NS4A GB virus C/ gi|9628705:5112-5309 16 1355 60 protein Hepatitis G putative NS4A GB virus C/ gi|9628705:5112-5309 13 1356 60 protein Hepatitis G putative NS4A GB virus C/ gi|9628705:5112-5309 12 1357 60 protein Hepatitis G putative NS4A GB virus C/ gi|9628705:5112-5309 11 1358 60 protein Hepatitis G putative NS4A GB virus C/ gi|9628705:5112-5309 6 1359 60 protein Hepatitis G putative NS4A GB virus C/ gi|9628705:5112-5309 5 1360 60 protein Hepatitis G putative NS4B GB virus C/ gi|9628705:5310-6152 700 1361 60 protein Hepatitis G putative NS4B GB virus C/ gi|9628705:5310-6152 699 1362 60 protein Hepatitis G putative NS4B GB virus C/ gi|9628705:5310-6152 698 1363 60 protein Hepatitis G putative NS4B GB virus C/ gi|9628705:5310-6152 696 1364 60 protein Hepatitis G putative NS4B GB virus C/ gi|9628705:5310-6152 695 1365 60 protein Hepatitis G putative NS4B GB virus C/ gi|9628705:5310-6152 694 1366 60 protein Hepatitis G putative NS4B GB virus C/ gi|9628705:5310-6152 693 1367 60 protein Hepatitis G putative NS4B GB virus C/ gi|9628705:5310-6152 692 1368 60 protein Hepatitis G putative NS4B GB virus C/ gi|9628705:5310-6152 691 1369 60 protein Hepatitis G putative NS4B GB virus C/ gi|9628705:5310-6152 690 1370 60 protein Hepatitis G putative NS5A GB virus C/ gi|9628705:6153-7388 1174 1371 60 protein Hepatitis G putative NS5A GB virus C/ gi|9628705:6153-7388 1173 1372 60 protein Hepatitis G putative NS5A GB virus C/ gi|9628705:6153-7388 1155 1373 60 protein Hepatitis G putative NS5A GB virus C/ gi|9628705:6153-7388 1154 1374 60 protein Hepatitis G putative NS5A GB virus C/ gi|9628705:6153-7388 1153 1375 60 protein Hepatitis G putative NS5A GB virus C/

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gi|9628705:6153-7388 941 1377 60 protein Hepatitis G putative NS5A GB virus C/
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virus C/ gi|9628705:7389-9077 1274 1382 60 RNA- Hepatitis G dependent RNA pol. putative
NS5B GB virus C/ gi|9628705:7389-9077 856 1383 60 RNA- Hepatitis G dependent RNA pol.
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pol. putative NS5B GB virus C/ gi|9628705:7389-9077 374 1385 60 RNA- Hepatitis G dependent
RNA pol. putative NS5B GB virus C/ gi|9628705:7389-9077 373 1386 60 RNA- Hepatitis G
dependent RNA GB virus C/ pol. Hepatitis G putative NS5B GB virus C/ gi|9628705:7389-9077
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9077 371 1388 60 RNA- Hepatitis G dependent RNA pol. putative NS5B GB virus C/
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C/ gi|9628705:7389-9077 73 1390 60 RNA- Hepatitis G dependent RNA pol. 1A VP4b Hepatitis
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1392 60 mature peptide A virus 1A VP4b Hepatitis gi|9626732:1-805 568 1393 60 mature peptide
A virus 1A VP4b Hepatitis gi|9626732:1-805 513 1394 60 mature peptide A virus 1A VP4b
Hepatitis gi|9626732:1-805 466 1395 60 mature peptide A virus 1A VP4b Hepatitis gi|9626732:1-
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Hepatitis gi|9626732:1-805 286 1399 60 mature peptide A virus 1A VP4b Hepatitis gi|9626732:1-
805 159 1400 60 mature peptide A virus 1B VP2 Hepatitis gi|9626732:804-1469 606 1401 60
mature peptide A virus 1B VP2 Hepatitis gi|9626732:804-1469 546 1402 60 mature peptide A virus
1B VP2 Hepatitis gi|9626732:804-1469 486 1403 60 mature peptide A virus 1B VP2 Hepatitis
gi|9626732:804-1469 397 1404 60 mature peptide A virus 1B VP2 Hepatitis gi|9626732:804-1469
329 1405 60 mature peptide A virus 1B VP2 Hepatitis gi|9626732:804-1469 269 1406 60 mature
peptide A virus 1B VP2 Hepatitis gi|9626732:804-1469 203 1407 60 mature peptide A virus 1B
VP2 Hepatitis gi|9626732:804-1469 143 1408 60 mature peptide A virus 1B VP2 Hepatitis
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peptide A virus 1C VP3 Hepatitis gi|9626732:1470-2207 619 1412 60 mature peptide A virus 1C
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gi|9626732:1470-2207 499 1414 60 mature peptide A virus 1C VP3 Hepatitis gi|9626732:1470-
2207 439 1415 60 mature peptide A virus 1C VP3 Hepatitis gi|9626732:1470-2207 379 1416 60
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virus 1C VP3 Hepatitis gi|9626732:1470-2207 246 1418 60 mature peptide A virus 1C VP3
Hepatitis gi|9626732:1470-2207 185 1419 60 mature peptide A virus 1C VP3 Hepatitis
gi|9626732:1470-2207 125 1420 60 mature peptide A virus 1D VP1 Hepatitis gi|9626732:2208-
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Hepatitis gi|9626732:2208-3107 534 1425 60 mature peptide A virus 1D VP1 Hepatitis
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virus 1D VP1 Hepatitis gi|9626732:2208-3107 189 1430 60 mature peptide A virus 2A mature
Hepatitis gi|9626732:3108-3674 494 1431 60 peptide A virus 2A mature Hepatitis
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[illegible]

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gi|9626440:4-5085 2548 1508 60 polyprotein E virus ORF 1- Hepatitis gi|9626440:4-5085 2130
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1537 60 helicase 1 E virus Viral Hepatitis gi|9626440:2914-3562 546 1538 60 helicase 1 E virus
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Hepatitis gi|9626440:4191-4684 207 1545 60 RNA pol. E virus RNA dependent Hepatitis
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RNA pol. E virus RNA dependent Hepatitis gi|9626440:4191-4684 202 1549 60 RNA pol. E virus
RNA dependent Hepatitis gi|9626440:4191-4684 201 1550 60 RNA pol. E virus ORF 3- Hepatitis
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phosphoprotein LNYV gi|83659771:1631-2712 431 1638 60 phosphoprotein LNYV
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230 1769 60 End X-Hyb Product Distance Tm Pot % G % C % A % T % GC PolyX 5 UTR 99
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5 dependent RNA polymerase 5 UTR 80 86.27 0 36.67 18.33 20 25 55 3 5 UTR 150 86.34 0 26.67
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78.85 0 20 18.33 43.33 18.33 38.33 4 gag 545 79.58 0 21.67 20 31.67 26.67 41.67 4 gag 605 78.28
0 21.67 16.67 43.33 18.33 38.33 6 gag 667 80.04 0 21.67 23.33 36.67 18.33 45 3 gag 754 75.62 0

18.33 16.67 40 25 35 gag 814 80.67 0 26.67 18.33 38.33 16.67 45 4 gag 903 81.87 0 28.33 16.67 36.67 18.33 45 4 Vif 114 78.8 0 21.67 20 40 18.33 41.67 6 Vif 154 77.98 0 13.33 25 43.33 18.33 38.33 6 Vif 194 78.39 0 23.33 18.33 33.33 25 41.67 3 Vif 234 77.97 0 21.67 18.33 31.67 28.33 40 3 Vif 276 75.46 0 13.33 21.67 28.33 36.67 35 5 Vif 316 78.46 0 20 20 45 15 40 5 Vif 356 79.26 0 33.33 10 38.33 18.33 43.33 5 Vif 396 78.73 0 28.33 13.33 33.33 25 41.67 4 Vif 436 77.87 0 20 20 38.33 21.67 40 4 Vif 476 77.27 0 21.67 18.33 35 25 40 4 vpr 78 81.1 0 30 15 35 20 45 3 vpr 98 79.56 0 21.67 20 26.67 31.67 41.67 4 vpr 118 78.61 0 18.33 20 30 31.67 38.33 4 vpr 138 79.69 0 21.67 18.33 31.67 28.33 40 3 vpr 158 78.79 0 26.67 13.33 33.33 26.67 40 4 vpr 184 81.7 0 28.33 16.67 26.67 28.33 45 4 vpr 204 78.58 0 20 20 25 35 40 4 vpr 224 79.94 0 28.33 15 26.67 30 43.33 4 vpr 244 77.34 0 26.67 13.33 31.67 28.33 40 4 vpr 264 79.95 0 28.33 16.67 38.33 16.67 45 4 tat 91 80.29 0 18.33 26.67 30 25 45 3 tat 218 77.17 0 28.33 6.67 33.33 31.67 35 3 tat 285 76.51 0 15 20 48.33 16.67 35 4 tat 345 79.94 0 26.67 15 38.33 20 41.67 3 tat 405 80.08 0 21.67 21.67 26.67 30 43.33 4 tat 572 82.24 0 28.33 16.67 28.33 26.67 45 3 tat 692 81.72 0 35 10 30 25 45 6 tat 824 79.96 0 30 15 35 20 45 3 tat 884 77.23 0 25 13.33 35 26.67 38.33 3 tat 944 77.7 0 16.67 18.33 45 20 35 4 rev 60 78.01 0 25 16.67 35 23.33 41.67 3 rev 134 80.4 0 23.33 21.67 23.33 31.67 45 2 rev 231 80.23 0 31.67 13.33 38.33 16.67 45 2 rev 320 80.29 0 18.33 26.67 30 25 45 3 rev 447 77.17 0 28.33 6.67 33.33 31.67 35 3 rev 514 76.51 0 15 20 48.33 16.67 35 4 rev 574 79.94 0 26.67 15 38.33 20 41.67 3 rev 634 80.08 0 21.67 21.67 26.67 30 43.33 4 rev 801 82.24 0 28.33 16.67 28.33 26.67 45 3 rev 921 81.72 0 35 10 30 25 45 6 vpu 114 78.38 0 30 11.67 41.67 16.67 41.67 3 vpu 61 85.51 0 40 13.33 18.33 28.33 53.33 5 vpu 62 85.89 0 40 15 16.67 28.33 55 5 vpu 64 86.58 0 40 15 18.33 26.67 55 5 vpu 66 86.64 0 40 15 18.33 26.67 55 5 vpu 67 86.64 0 40 15 20 25 55 5 vpu 68 85.92 0 40 15 18.33 26.67 55 5 vpu 69 85.79 0 38.33 15 20 26.67 53.33 5 vpu 70 85.81 0 38.33 15 21.67 25 53.33 5 vpu 71 85.37 0 38.33 15 21.67 25 53.33 5 asp 93 82.24 0 28.33 16.67 28.33 26.67 45 3 asp 210 81.72 0 35 10 30 25 45 6 asp 256 80.7 0 26.67 16.67 41.67 15 43.33 5 asp 359 80.63 0 30 15 35 20 45 2 asp 389 76.87 0 21.67 13.33 38.33 26.67 35 3 asp 419 79.95 0 18.33 23.33 30 28.33 41.67 4 asp 449 80.68 0 21.67 20 38.33 20 41.67 4 asp 479 78.34 0 16.67 20 46.67 16.67 36.67 4 asp 509 79.23 0 16.67 23.33 43.33 16.67 40 4 asp 539 77.8 0 25 13.33 36.67 25 38.33 3 nef 179 80.47 0 25 20 38.33 16.67 45 3 nef 199 79.2 0 28.33 15 41.67 15 43.33 3 nef 220 79.81 0 23.33 20 31.67 25 43.33 3 nef 300 79.39 0 16.67 28.33 28.33 26.67 45 3 nef 320 80.05 0 15 28.33 33.33 23.33 43.33 3 nef 348 80.65 0 26.67 16.67 40 16.67 43.33 6 nef 368 79.99 0 23.33 18.33 35 23.33 41.67 6 nef 388 80.05 0 26.67 15 35 23.33 41.67 6 nef 408 78.88 0 16.67 25 28.33 30 41.67 5 nef 517 81.69 0 21.67 23.33 38.33 16.67 45 5 5' LTR 155 79.1 0 18.33 25 25 31.67 43.33 3 5' LTR 343 79.71 0 20 23.33 21.67 35 43.33 3 5' LTR 472 79.15 0 25 18.33 41.67 15 43.33 3 5' LTR 502 77.55 0 21.67 16.67 45 16.67 38.33 3 5' LTR 532 79.96 0 28.33 13.33 43.33 15 41.67 4 5' LTR 647 81.97 0 28.33 16.67 33.33 21.67 45 4 5' LTR 715 80.36 0 26.67 18.33 36.67 18.33 45 2 5' LTR 788 78.77 0 23.33 18.33 33.33 25 41.67 3 5' LTR 818 75.67 0 20 16.67 43.33 20 36.67 5 5' LTR 848 75.56 0 26.67 11.67 41.67 20 38.33 5 gag 430 82.46 0 30 15 38.33 16.67 45 3 polyprotein gag 577 80.03 0 25 20 36.67 18.33 45 4 polyprotein gag 637 81.88 0 16.67 28.33 41.67 13.33 45 6 polyprotein gag 697 81.02 0 21.67 21.67 40 16.67 43.33 4 polyprotein gag 757 79.55 0 21.67 20 41.67 16.67 41.67 4 polyprotein gag 817 80.48 0 23.33 21.67 36.67 18.33 45 4 polyprotein gag 961 80.93 0 25 20 38.33 16.67 45 3 polyprotein gag 1021 79.57 0 20 20 31.67 28.33 40 3 polyprotein gag 1094 81.5 0 31.67 13.33 33.33 21.67 45 4 polyprotein gag 1219 80.48 0 20 25 45 10 45 4 polyprotein gag-pol 60 77.83 0 20 21.67 35 23.33 41.67 4 gag-pol 120 79.05 0 18.33 23.33 35 23.33 41.67 5 gag-pol 180 79.42 0 23.33 20 36.67 20 43.33 5 gag-pol 265 80.72 0 23.33 21.67 33.33 21.67 45 4 gag-pol 380 78.61 0 28.33 15 43.33 13.33 43.33 4 gag-pol 480 80.85 0 28.33 16.67 33.33 21.67 45 3 gag-pol 557 76.38 0 11.67 23.33 50 15 35 3 gag-pol 617 79.46 0 30 11.67 33.33 25 41.67 4 gag-pol 703 76.51 0 23.33 15 45 16.67 38.33 3 gag-pol 764 81.32 0 26.67 18.33 31.67 23.33 45 3 gp2-vif 209 79 0 18.33 25 40 16.67 43.33 4 protein gp2-vif 261 80.96 0 13.33 30 31.67 25 43.33 4 protein gp2-vif 311 80.06 0 28.33 13.33 30 28.33 41.67 4 protein gp2-vif 341 78.94 0 18.33 21.67 28.33 31.67 40 3 protein gp2-vif 372 80.21 0 18.33 25 35 21.67 43.33 4

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23.33 43.33 6 Protein gp3-vpx 122 81.16 0 28.33 15 33.33 23.33 43.33 6 Protein gp3-vpx 123
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Protein gp3-vpx 126 81.53 0 28.33 16.67 30 25 45 6 Protein gp3-vpx 127 81.6 0 28.33 16.67 30 25
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16.67 28.33 28.33 43.33 5 Protein gp3-vpx 130 80.29 0 25 16.67 28.33 30 41.67 5 Protein gp4-vpr
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80.43 0 16.67 28.33 28.33 26.67 45 3 protein gp7-env 503 79.14 0 21.67 20 23.33 35 41.67 6
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protein gp7-env 1307 81.02 0 33.33 11.67 30 25 45 5 protein gp8-nef 66 79.96 0 28.33 13.33 43.33
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5 rex 2151 78.28 0 16.67 21.67 40 21.67 38.33 3 rex 2516 81.15 0 16.67 28.33 20 35 45 2 tax 109
81.24 0 18.33 26.67 35 20 45 5 tax 169 76.49 0 6.67 30 26.67 36.67 36.67 4 tax 262 77.12 0 6.67
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2091 82.63 0 18.33 26.67 36.67 18.33 45 5 tax 2151 78.28 0 16.67 21.67 40 21.67 38.33 3 tax
2516 81.15 0 16.67 28.33 20 35 45 2 env 276 75.75 0 13.33 23.33 33.33 30 36.67 4 env 316 79.9 0
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0 13.33 30 23.33 33.33 43.33 4 env 1211 80.16 0 20 23.33 31.67 25 43.33 5 5' LTR 502 80.91 0
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2089 80.18 0 15 28.33 28.33 28.33 43.33 3 protein gp4-rex 26 kD 2251 80.81 0 18.33 26.67 35 20
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81.95 0 21.67 20 18.33 40 41.67 4 protein NS2A nonstructural 401 81.59 0 26.67 16.67 21.67 35
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137 80.84 0 30 15 38.33 16.67 45 3 protein NS3 nonstructural 267 81.55 0 26.67 18.33 28.33 26.67
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36.67 43.33 3 protein NS4A nonstructural 216 79.62 0 21.67 21.67 20 36.67 43.33 3 protein NS4A
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80.14 0 20 23.33 35 21.67 43.33 3 protein C membrane 85 82.64 0 25 20 25 30 45 3 glycoprotein
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80.1 0 20 21.67 33.33 25 41.67 3 protein NS1 nonstructural 235 79.64 0 21.67 21.67 33.33 23.33
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0 26.67 13.33 33.33 26.67 40 6 protein NS2A nonstructural 198 78.54 0 25 13.33 26.67 35 38.33 4
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81.83 0 26.67 16.67 31.67 25 43.33 4 protein NS2B nonstructural 150 81.44 0 28.33 16.67 30 25
45 4 protein NS2B nonstructural 151 81.25 0 26.67 16.67 30 26.67 43.33 4 protein NS2B
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43.33 4 protein NS4A nonstructural 176 79.45 0 30 11.67 26.67 31.67 41.67 5 protein NS4A
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11.67 25 33.33 41.67 6 protein NS4A nonstructural 179 80.32 0 30 11.67 25 33.33 41.67 6 protein
NS4A nonstructural 180 80.02 0 31.67 11.67 25 31.67 43.33 6 protein NS4A nonstructural 181
80.49 0 31.67 10 25 33.33 41.67 6 protein NS4A nonstructural 182 80.65 0 33.33 10 25 31.67
43.33 6 protein NS4A nonstructural 252 81.84 0 25 20 26.67 28.33 45 3 protein NS4B

nonstructural 256 80.6 0 26.67 18.33 28.33 26.67 45 6 protein NS4B nonstructural 257 80.6 0
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6 protein NS4B nonstructural 259 81.56 0 28.33 16.67 30 25 45 6 protein NS4B nonstructural 260
82.07 0 28.33 16.67 28.33 26.67 45 6 protein NS4B nonstructural 261 81.53 0 28.33 16.67 28.33
26.67 45 6 protein NS4B nonstructural 262 81.63 0 28.33 16.67 28.33 26.67 45 6 protein NS4B
nonstructural 263 81.53 0 28.33 16.67 28.33 26.67 45 6 protein NS4B nonstructural 264 80.96 0
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protein NS5 nonstructural 120 78.8 0 28.33 13.33 33.33 25 41.67 4 protein NS5 nonstructural 221
81.04 0 31.67 13.33 31.67 23.33 45 3 protein NS5 nonstructural 600 81.29 0 21.67 23.33 28.33
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18.33 33.33 21.67 45 4 protein NS5 gp1- 103 81.63 0 20 25 26.67 28.33 45 3 nonstructural
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polyprotein gp1- 809 81.32 0 18.33 26.67 26.67 28.33 45 3 nonstructural polyprotein gp1- 958
80.67 0 26.67 18.33 36.67 18.33 45 3 nonstructural polyprotein gp2-structural 307 80.44 0 25 20
28.33 26.67 45 3 polyprotein gp2-structural 372 82.09 0 25 20 33.33 21.67 45 3 polyprotein gp2-
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82.38 0 21.67 23.33 33.33 21.67 45 5 polyprotein gp3-truncated 844 82.49 0 21.67 23.33 33.33
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28.33 31.67 40 4 protein C anchored capsid 238 80.84 0 25 18.33 26.67 30 43.33 4 protein C
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62 80.15 0 18.33 25 25 31.67 43.33 5 glycoprotein precursor M membrane 63 79.96 0 16.67 25
26.67 31.67 41.67 5 glycoprotein precursor M membrane 64 79.82 0 15 25 28.33 31.67 40 5
glycoprotein precursor M membrane 65 79.89 0 16.67 25 28.33 30 41.67 5 glycoprotein precursor
M membrane 66 80.09 0 18.33 25 26.67 30 43.33 5 glycoprotein precursor M membrane 67 79.82
0 18.33 23.33 28.33 30 41.67 5 glycoprotein precursor M membrane 68 79.49 0 18.33 21.67 28.33
31.67 40 5 glycoprotein precursor M membrane 69 79.49 0 18.33 21.67 30 30 40 5 glycoprotein
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41.67 23.33 35 5 protein E envelope 752 81.12 0 26.67 18.33 40 15 45 6 protein E envelope 812
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81.26 0 20 25 35 20 45 3 protein NS1 nonstructural 220 80.04 0 28.33 15 26.67 30 43.33 3 protein
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NS1 nonstructural 461 79.88 0 30 15 35 20 45 4 protein NS1 nonstructural 561 81.72 0 21.67
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NS1 nonstructural 698 81.74 0 16.67 28.33 40 15 45 4 protein NS1 nonstructural 63 78.19 0 18.33
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NS2B nonstructural 62 81.59 0 21.67 21.67 31.67 25 43.33 5 protein NS2B nonstructural 63 80.89
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81.64 0 20 25 38.33 16.67 45 3 protein NS4A nonstructural 379 78.87 0 20 21.67 36.67 21.67
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82.27 0 26.67 18.33 40 15 45 6 protein NS5 nonstructural 841 81.36 0 21.67 23.33 35 20 45 4
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80.14 0 11.67 30 28.33 30 41.67 4 glycoprotein precursor M membrane 80 80.21 0 16.67 26.67
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81.44 0 30 15 38.33 16.67 45 6 glycoprotein precursor M membrane 270 81.21 0 28.33 16.67
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envelope 64 81.37 0 31.67 13.33 21.67 33.33 45 3 protein E envelope 124 78.25 0 18.33 21.67 35
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695 80.24 0 15 30 31.67 23.33 45 5 protein NS1 nonstructural 105 79.5 0 15 25 38.33 21.67 40 6
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81.33 0 23.33 21.67 26.67 28.33 45 3 protein NS2A nonstructural 349 79.47 0 16.67 28.33 30 25
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81.01 0 21.67 23.33 40 15 45 4 protein NS3 nonstructural 466 77.78 0 21.67 15 45 18.33 36.67 6
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76.49 0 18.33 18.33 25 38.33 36.67 6 protein NS4A nonstructural 72 76.83 0 16.67 18.33 25 40 35
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76.29 0 16.67 18.33 26.67 38.33 35 6 protein NS4A nonstructural 75 76.29 0 16.67 18.33 26.67
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envelope 343 81.88 0 30 15 36.67 18.33 45 4 protein E envelope 403 80.4 0 30 15 28.33 26.67 45 4
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E nonstructural 60 78.87 0 25 16.67 38.33 20 41.67 3 protein NS1 nonstructural 178 81.3 0 28.33
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protein putative NS4A 193 85.09 0 31.67 23.33 20 25 55 6 protein putative NS4A 194 85.03 0
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NS4B 145 80.68 0 21.67 23.33 25 30 45 3 protein putative NS4B 146 80.77 0 21.67 23.33 25 30 45 3 protein putative NS4B 148 81.51 0 21.67 23.33 26.67 28.33 45 3 protein putative NS4B 149 80.72 0 21.67 23.33 26.67 28.33 45 3 protein putative NS4B 150 80.02 0 20 23.33 26.67 30 43.33 3 protein putative NS4B 151 80.1 0 20 23.33 26.67 30 43.33 3 protein putative NS4B 152 80.8 0 20 25 25 30 45 3 protein putative NS4B 153 80.8 0 20 25 25 30 45 3 protein putative NS4B 154 80.39 0 20 23.33 25 31.67 43.33 3 protein putative NS5A 63 81.64 0 30 15 28.33 26.67 45 3 protein putative NS5A 64 81.57 0 28.33 16.67 28.33 26.67 45 3 protein putative NS5A 82 82.61 0 33.33 11.67 23.33 31.67 45 3 protein putative NS5A 83 82.19 0 33.33 11.67 21.67 33.33 45 3 protein putative NS5A 84 82.12 0 31.67 13.33 21.67 33.33 45 3 protein putative NS5A 293 81.93 0 20 25 26.67 28.33 45 4 protein putative NS5A 296 81.15 0 21.67 23.33 26.67 28.33 45 4 protein putative NS5A 297 80.48 0 21.67 23.33 26.67 28.33 45 4 protein putative NS5A 298 81.04 0 23.33 21.67 26.67 28.33 45 4 protein putative NS5A 301 80.92 0 25 20 26.67 28.33 45 4 protein putative NS5B 415 81.82 0 20 25 28.33 26.67 45 3 RNA- dependent RNA pol. putative NS5B 416 81.85 0 20 25 28.33 26.67 45 3 RNA- dependent RNA pol. putative NS5B 834 81.13 0 26.67 18.33 28.33 26.67 45 5 RNA- dependent RNA pol. putative NS5B 1315 81.07 0 23.33 21.67 26.67 28.33 45 5 RNA- dependent RNA pol. putative NS5B 1316 80.37 0 23.33 21.67 25 30 45 5 RNA- dependent RNA pol. putative NS5B 1317 80.28 0 21.67 21.67 26.67 30 43.33 5 RNA- dependent RNA pol. putative NS5B 1318 80.32 0 23.33 21.67 25 30 45 5 RNA- dependent RNA pol. putative NS5B 1319 80.76 0 23.33 21.67 25 30 45 5 RNA- dependent RNA pol. putative NS5B 1616 80.79 0 25 20 23.33 31.67 45 3 RNA- dependent RNA pol. putative NS5B 1617 80.23 0 25 20 23.33 31.67 45 3 RNA- dependent RNA pol. 1A VP4b 68 81.03 0 23.33 21.67 23.33 31.67 45 4 mature peptide 1A VP4b 115 75.84 0 18.33 16.67 30 35 35 5 mature peptide 1A VP4b 238 80.5 0 21.67 21.67 23.33 33.33 43.33 3 mature peptide 1A VP4b 293 81.47 0 26.67 18.33 20 35 45 3 mature peptide 1A VP4b 340 80.79 0 21.67 21.67 31.67 25 43.33 5 mature peptide 1A VP4b 380 80.48 0 33.33 11.67 26.67 28.33 45 3 mature peptide 1A VP4b 428 80.46 0 25 20 28.33 26.67 45 4 mature peptide 1A VP4b 468 80.54 0 21.67 23.33 20 35 45 4 mature peptide 1A VP4b 520 81.35 0 30 15 28.33 26.67 45 4 mature peptide 1A VP4b 647 77.38 0 16.67 21.67 21.67 40 38.33 5 mature peptide 1B VP2 61 76.15 0 16.67 18.33 26.67 38.33 35 4 mature peptide 1B VP2 121 75.88 0 20 15 33.33 31.67 35 3 mature peptide 1B VP2 181 77.16 0 15 20 33.33 31.67 35 3 mature peptide 1B VP2 270 76.51 0 18.33 16.67 25 40 35 3 mature peptide 1B VP2 338 80.84 0 21.67 23.33 25 30 45 6 mature peptide 1B VP2 398 79.36 0 25 13.33 30 31.67 38.33 3 mature peptide 1B VP2 464 77.13 0 20 15 28.33 36.67 35 4 mature peptide 1B VP2 524 77.9 0 20 16.67 33.33 30 36.67 5 mature peptide 1B VP2 585 81.32 0 21.67 23.33 23.33 31.67 45 3 mature peptide 1B VP2 645 78.29 0 21.67 16.67 23.33 38.33 38.33 4 mature peptide 1C VP3 60 77.07 0 15 20 26.67 38.33 35 4 mature peptide 1C VP3 120 77.83 0 15 23.33 23.33 38.33 38.33 3 mature peptide 1C VP3 180 80.34 0 25 16.67 31.67 26.67 41.67 3 mature peptide 1C VP3 240 79.49 0 20 21.67 23.33 35 41.67 3 mature peptide 1C VP3 300 79.56 0 21.67 20 33.33 25 41.67 3 mature peptide 1C VP3 360 77.2 0 23.33 13.33 21.67 41.67 36.67 4 mature peptide 1C VP3 423 77.17 0 20 15 21.67 43.33 35 5 mature peptide 1C VP3 493 76.77 0 11.67 23.33 35 30 35 6 mature peptide 1C VP3 554 76.13 0 15 20 28.33 36.67 35 3 mature peptide 1C VP3 614 78.67 0 21.67 18.33 25 35 40 4 mature peptide 1D VP1 60 78.57 0 31.67 11.67 33.33 23.33 43.33 4 mature peptide 1D VP1 120 79.89 0 18.33 21.67 35 25 40 3 mature peptide 1D VP1 247 76.59 0 20 15 31.67 33.33 35 3 mature peptide 1D VP1 307 77.15 0 23.33 13.33 21.67 41.67 36.67 3 mature peptide 1D VP1 367 79.08 0 21.67 18.33 36.67 23.33 40 4 mature peptide 1D VP1 467 80.54 0 26.67 18.33 23.33 31.67 45 2 mature peptide 1D VP1 527 78.63 0 23.33 16.67 25 35 40 5 mature peptide 1D VP1 587 78.77 0 10 30 25 35 40 3 mature peptide 1D VP1 652 76.68 0 15 20 26.67 38.33 35 4 mature peptide 1D VP1 712 77.66 0 16.67 21.67 36.67 25 38.33 3 mature peptide 2A mature 74 75.99 0 25 10 35 30 35 4 peptide 2A mature 116 77.96 0 18.33 16.67 30 35 35 4 peptide 2A mature 156 77.7 0 20 16.67 30 33.33 36.67 4 peptide 2A mature 196 77.49 0 21.67 13.33 36.67 28.33 35 3 peptide 2A mature 287 77.59 0 33.33 6.67 35 25 40 3 peptide 2A mature 389 80.03 0 30 13.33 20 36.67 43.33 4 peptide 2A mature 432

76.63 0 21.67 13.33 30 35 35 6 peptide 2A mature 472 79.08 0 13.33 25 30 31.67 38.33 6 peptide
2A mature 512 78.94 0 18.33 21.67 40 20 40 5 peptide 2A mature 552 77.38 0 25 13.33 43.33
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79.2 0 25 15 35 25 40 4 peptide 2B mature 67 78.5 0 23.33 15 35 26.67 38.33 4 peptide 2B mature
68 78.43 0 21.67 15 35 28.33 36.67 4 peptide 2B mature 69 78.43 0 21.67 15 33.33 30 36.67 4
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77.74 0 21.67 18.33 36.67 23.33 40 3 mature peptide 3B (VPg) 61 78.06 0 20 18.33 36.67 25 38.33
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1 Viral 104 86.14 0 23.33 31.67 18.33 26.67 55 3 helicase 1 Viral 106 86.75 0 21.67 33.33 20 25 55
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81.05 0 20 25 21.67 33.33 45 3 protease cofactor putative 536 79.88 0 20 21.67 28.33 30 41.67 3
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79.32 0 31.67 10 30 28.33 41.67 3 protein nucleocapsid 226 82.08 0 18.33 26.67 31.67 23.33 45 5
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23.33 21.67 31.67 23.33 45 3 phosphoprotein 524 80.06 0 25 20 28.33 26.67 45 3 phosphoprotein
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25 13.33 18.33 43.33 38.33 5 5' trailer RNA 62 77 0 23.33 13.33 18.33 45 36.67 5 5' trailer RNA
63 76.39 0 23.33 13.33 18.33 45 36.67 5 5' trailer RNA 64 76.82 0 23.33 11.67 20 45 35 5 5' trailer

RNA 65 76.05 0 23.33 11.67 20 45 35 5' trailer RNA 73 75.88 0 23.33 11.67 21.67 43.33 35 5' trailer RNA 74 76.29 0 23.33 11.67 23.33 41.67 35 5' trailer RNA 75 76.29 0 23.33 11.67 25 40 35 5' trailer RNA 76 76.39 0 25 11.67 25 38.33 36.67 5 NS5 protein 498 80.74 0 23.33 21.67 33.33 21.67 45 4 NS5 protein 559 81.87 0 25 20 35 20 45 5 NS5 protein 632 80.62 0 30 15 28.33 26.67 45 5 NS5 protein 872 82.48 0 20 25 28.33 26.67 45 4 NS5 protein 1080 81.39 0 28.33 16.67 40 15 45 4 NS5 protein 1140 81.31 0 26.67 16.67 23.33 33.33 43.33 3 NS5 protein 1284 81.05 0 31.67 13.33 28.33 26.67 45 2 NS5 protein 1350 81.53 0 23.33 21.67 31.67 23.33 45 5 NS5 protein 1411 78.37 0 23.33 15 43.33 18.33 38.33 5 NS5 protein 1506 81.82 0 25 20 36.67 18.33 45 3 NS5 protein 336 81.2 0 26.67 18.33 26.67 28.33 45 3 NS5 protein 399 81.27 0 25 20 33.33 21.67 45 4 NS5 protein 508 80.62 0 30 15 28.33 26.67 45 5 NS5 protein 748 82.48 0 20 25 28.33 26.67 45 4 NS5 protein 956 81.39 0 28.33 16.67 40 15 45 4 NS5 protein 1016 81.31 0 26.67 16.67 23.33 33.33 43.33 3 NS5 protein 1160 81.05 0 31.67 13.33 28.33 26.67 45 2 NS5 protein 1226 81.53 0 23.33 21.67 31.67 23.33 45 5 NS5 protein 1287 78.37 0 23.33 15 43.33 18.33 38.33 5 NS5 protein 1381 81.05 0 26.67 18.33 36.67 18.33 45 3 NS5 protein 101 81.2 0 26.67 18.33 26.67 28.33 45 3 NS5 protein 164 81.27 0 25 20 33.33 21.67 45 4 NS5 protein 273 80.62 0 30 15 28.33 26.67 45 5 NS5 protein 513 82.48 0 20 25 28.33 26.67 45 4 NS5 protein 721 81.39 0 28.33 16.67 40 15 45 4 NS5 protein 781 81.31 0 26.67 16.67 23.33 33.33 43.33 3 NS5 protein 925 81.05 0 31.67 13.33 28.33 26.67 45 2 NS5 protein 991 81.53 0 23.33 21.67 31.67 23.33 45 5 NS5 protein 1052 78.37 0 23.33 15 43.33 18.33 38.33 5 NS5 protein 1146 81.05 0 26.67 18.33 36.67 18.33 45 3 NS5 protein 375 80.74 0 23.33 21.67 33.33 21.67 45 4 NS5 protein 510 80.62 0 28.33 16.67 30 25 45 5 NS5 protein 749 82.48 0 20 25 28.33 26.67 45 4 NS5 protein 957 81.39 0 28.33 16.67 40 15 45 4 NS5 protein 1017 81.31 0 26.67 16.67 23.33 33.33 43.33 3 NS5 protein 1161 81.05 0 31.67 13.33 28.33 26.67 45 2 NS5 protein 1227 81.53 0 23.33 21.67 31.67 23.33 45 5 NS5 protein 1288 78.37 0 23.33 15 43.33 18.33 38.33 5 NS5 protein 1353 82.74 0 21.67 23.33 38.33 16.67 45 5 NS5 protein 1459 80.12 0 30 15 31.67 23.33 45 2 5UTR 60 78.39 0 25 15 30 30 40 4 5UTR 69 78.43 0 26.67 11.67 35 26.67 38.33 4 capsid 144 77.34 0 21.67 13.33 41.67 23.33 35 3 capsid 160 79.12 0 28.33 11.67 41.67 18.33 40 6 propeptide 174 80.99 0 23.33 21.67 28.33 26.67 45 3 propeptide 183 78.98 0 23.33 16.67 31.67 28.33 40 2 membrane 64 81.94 0 20 25 31.67 23.33 45 5 protein membrane 78 80.99 0 25 20 26.67 28.33 45 5 protein envelope 74 80.57 0 25 20 16.67 38.33 45 5 protein envelope 194 81.61 0 26.67 15 26.67 31.67 41.67 5 protein envelope 447 80.6 0 21.67 23.33 33.33 21.67 45 4 protein envelope 960 79.05 0 23.33 18.33 23.33 35 41.67 4 protein envelope 1198 82.15 0 26.67 18.33 28.33 26.67 45 5 protein NS1 protein 64 79.96 0 21.67 21.67 31.67 25 43.33 2 NS1 protein 439 80.43 0 26.67 16.67 38.33 18.33 43.33 4 NS1 protein 608 77.05 0 26.67 13.33 26.67 33.33 40 3 NS1 protein 866 81.68 0 28.33 16.67 31.67 23.33 45 5 NS2A protein 148 80.85 0 25 20 28.33 26.67 45 4 NS2A protein 152 80.03 0 25 18.33 28.33 28.33 43.33 4 NS2A protein 451 80.55 0 23.33 21.67 25 30 45 3 NS2B protein 287 81.46 0 28.33 16.67 40 15 45 5 NS2B protein 292 80.68 0 28.33 15 40 16.67 43.33 5 NS2B protein 297 81.32 0 30 15 36.67 18.33 45 5 NS3 protein 192 81.23 0 26.67 18.33 36.67 18.33 45 3 NS3 protein 395 80.79 0 15 30 28.33 26.67 45 3 NS3 protein 797 82.73 0 28.33 16.67 28.33 26.67 45 5 NS3 protein 1131 81.83 0 18.33 26.67 33.33 21.67 45 3 NS4A protein 246 81.88 0 33.33 11.67 23.33 31.67 45 5 NS4A protein 204 87.21 0 33.33 21.67 18.33 26.67 55 4 NS4A protein 383 86.35 0 28.33 26.67 28.33 16.67 55 3 NS4B protein 112 80.14 0 23.33 21.67 33.33 21.67 45 5 NS4B protein 819 80.74 0 23.33 21.67 28.33 26.67 45 3 NS4B protein 1082 80.52 0 26.67 16.67 33.33 23.33 43.33 3 NS4B protein 1506 80.98 0 31.67 13.33 38.33 16.67 45 3 NS5 protein 74 80.74 0 23.33 21.67 33.33 21.67 45 4 NS5 protein 716 81.31 0 26.67 16.67 23.33 33.33 43.33 3 NS5 protein 1172 79.51 0 30 15 35 20 45 3 3UTR 61 84.45 0 40 15 25 20 55 4 3UTR 70 84.24 0 33.33 21.67 23.33 21.67 55 4 WCCV - White clover cryptic virus; BBWV - Broad bean wilt virus; LNYV - Lettuce necrotic yellows virus
 (86) TABLE-US-00002 TABLE 2 Viral pathogens used in testing the Pathogen Chip Virus
 Type/Strain Source References CHIKV R91064 FDA/CBER Lot Release Panels* HAV SD11 Dr. Farci Lab HCV Genotype 1b Sera Care (Sera Care, Milford, MA) HCV Genotype 2a Sera Care

(Sera Care, Milford, MA) HCV Genotype 3 Sera Care (Sera Care, Milford, MA) HEV Genotype 3a WHO Standard HIV-1 Group M, FDA/CBER Lot Release Panels 1, 2 Subtype B HIV-2 Subtype B FDA/CBER Lot Release Panels 3 DENGUE Serotype 1, *Aedes albopictus* 4 2, 3 and 4 C6/36 cell culture HTLV-I ZeptoMetrix HTLV-II ZeptoMetrix WEST NY99 Cell culture 5 NILE ZIKA PRVABC62 FDA/CBER Lot Release Panels ZIKA FSS13025 FDA/CBER Lot Release Panels

*The FDA Center for Biologics Evaluation and Research (CBER), Division of Emerging and Transfusion Transmitted Diseases produces and makes available to blood donor screening test manufacturers panels which are sets of vialled human plasma containing virus particles that are carefully quantified for evaluating virus detection devices. Each set has several vials each one a different virus concentration and some with virus-free plasma. These panels are also used to test each new lot of a licensed blood donor screening device for release to the public, hence they are called Lot Release Panels. There are separate panels prepared for each type of virus. 1 Davis et al., *J Virol Methods*, 107: 37-44 (2003) 2 Lee et al., *J Virol Methods*, 137: 287-291 (2006) 3 Lee et al., *J Virol Methods*, 137: 287-291 (2005). 4 Dong et al., *J Appl Microbiol*, 120: 1119-1129 (2016). 5 Grinev et al., *J Virol Methods*, 154: 27-40 (2008).

(87) Nucleic acids from positive plasma and from NATtrol were extracted using the Dynabeads™ SILANE Viral NA Kit (ThermoFisher Scientific, Waltham, MA) according to the manufacturer's protocol.

(88) cDNA from random-primed, reverse-transcribed total RNA was performed with the Ovation® Pico WTA System (NuGEN, San Carlos, CA) using the manufacturer's recommended protocols and input amounts. For this study, the Agilent SureTag® Labeling Kit was used for generating Cy™3 dye labeled cDNA targets. Labeled cDNA was purified with SureTag® Kit spin columns and specific activities (degree of labeling) were calculated for use in hybridization reactions. A master mix containing 10× blocking agent and 2× GE hybridization buffer HI-RPM, was added to 3-5 µg of labeled cDNA, denatured, and hybridized to arrays under 8-chamber gasket slides at 65° C. with 20-rpm rotation for 24 hours in an Agilent hybridization oven. Arrays were processed using wash procedure A and scanned on an Agilent SureScan® G4900DA microarray scanner using 5-µm resolution.

(89) Microarray-based platform data analysis: After scanning, microarray images were analyzed using Agilent Feature Extraction software (Agilent Technologies, Inc., Santa Clara, CA) with default protocols and settings. Average pixel intensity and subtraction of local background for each feature was calculated. Images were manually examined to note any arrays affected by high background, scratches, or other technical artifacts. Probe sets associated with low signal intensity or bad quality features were considered unreliable and excluded from the analysis. Feature intensities for Cy™3 dye channels were imported into the Partek™ Genomics Suite™ software (Partek Inc., St. Louis, MO, USA).

(90) First, microarray analysis was performed by ranking the highest signal intensity probes by the mean of the set of probes defining each pathogen on the platform. Next, an experimental threshold was defined as a log ratio of signal intensity mean for the set of probes defining each pathogen and the mean of the Agilent control probes set. The threshold was applied to all the arrays tested to define the final parameters for test validation.

(91) RT-qPCR Validation

(92) Altona RT-qPCR: CHIKV, DENV 1-4 and ZIKV positive specimens were quantified using the Altona RealStar RT-qPCR kit (Altona Diagnostic GmbH, Hamburg, Germany) according to the manufacturer's instructions. The positive control and the internal control were provided by the manufacturer. Serial dilutions of CHIKV (ATCC VR-3246SD), DENV (ATCC VR-3231SD), and ZIKV (ATCC VR-1843DQ) quantitative genomic RNA (specification range: 1×10^{sup.5}-1×10^{sup.6} copies/4) obtained from ATCC (American Type Culture Collection Manassas, VA) were prepared to generate a standard curve for copy number quantification.

(93) Primer Design (Genesig) RT-qPCR: HAV (target/5' NCR), HCV (5'UTR), HEV (ORF2), HIV-

1 (target/POL), HIV-2 (target/POL), HTLVII (target/POL), and WNV (5'UTR) positive specimens were quantified using the Primer Design Genesig kit (Primerdesign Ltd, United Kingdom) according to the manufacturer's protocol (OneStep RT-qPCR protocol). Each kit contained a positive control template for the PCR set up and for copy number determination (generated serial dilutions for the standard curve).

(94) The RT-qPCR assays were performed on a ViiA7 Applied Biosystems real-time PCR system (Thermo Fisher Scientific Inc., Waltham, MA, USA). Each sample was tested in duplicate and the mean C.sub.q value was calculated.

Example 2

Microarray Design, Specificity, and Validation

(95) Microarray design: The pathogen chip design strategy was to cover all high priority blood-borne RNA viruses (retroviruses and both positive- and negative-strand RNA viruses) using multiple probes to independent targets sites in the genome of each species. In total, 1,769 unique viral oligonucleotides derived from 16 distinct viral genomes (Table 1) were included that allowed discrimination of pathogens at the level of species, subtypes and genotypes. The microarray was supplemented with an additional number of predesigned GE array probes for 906 genes from the human genome, 84 ERCC probes and 120 probes specific for plant viruses representing negative controls (Table 3).

(96) TABLE-US-00003 TABLE 3 Probe distribution on pathogen chip

Probe group	Number of
Number of type targets probes	1010
Purpose	14,716 RNA pathogens coverage and internal controls
Pathogen Specific	17 1,769 Probes intensity analysis (not replicated) of pathogen specific genes
Internal Control	902 902 Agilent requirement (replicated for probes normalization 10 times)
ERCC probes	84 84 Determination of (replicated intra-probe variance 45 times)
Negative Control	3 120 Determination of (not replicated) probes cross reactivity

(97) The design included multiple gene targets for each pathogen genome in order to select the best probes for the final platform design. The design strategy was to balance the number of probes for each pathogen with a final count of 90-110 probes each. Probes selected in the final design generated a more intense signal and produced higher percentage coverage of the specific genome across the different experiments (FIG. 1A).

(98) Microarray specificity: One of the challenges impacting the sensitivity of microarray based multi-pathogen nucleic acid detection in blood specimens is the relatively small concentration of target nucleic acids compared to a high background concentration of human DNA. A novel workflow was designed, combining two different applications (Agilent and Nugen), that had not been previously combined, to address this challenge. Typically, the Agilent amplification WT kit (Oligo dT) is used to amplify total RNA, with a minimum nucleic acid requirement of 25 nanograms, and produces a cRNA final product that is labelled with Cy3 fluorophore. The workflow was modified using a method that generates amplified cDNA from as little as 500 picograms of target viral RNA. One single-primer isothermal amplification using Nugen Ribo-SPIA technology was combined with the Agilent Genomic DNA Enzymatic Labeling Kit for generating Cy3 labeled cDNA. This kit was not previously developed for single color RNA probes and produces 300% the amplified product compared to the standard methodology (FIG. 1B, FIG. 2A). Nearly all samples were detected on the platform and all probes generated a strong signal specific for each positive plasma specimen analyzed. No specific signal was produced by negative control plasma (FIG. 2B). Random non-specific intensity signal was produced in only a few arrays. This indicated that the generation of cDNA instead of amplified RNA followed by Cy3 labelling and hybridization based on a DNA application was successful (FIG. 2C).

(99) Analysis strategies: Quality of signals generated by probes for each species was assessed according to two experimental criteria: 1) defining a threshold able to distinguish a true signal from its background; and 2) defining true positives only when 50% of probes generated a signal above the set threshold. These two levels of data analysis were needed to detect positive probes in the

presence of multi-pathogen testing at the same time and at different concentrations.

(100) The threshold was defined as the log ratio between the signal intensity mean for each pathogen specific probe set and the mean of the Agilent control group probe set. After comparing the results of the same set of probes across different arrays and selecting the probes showing an inter-array reproducibility, an experimental threshold value was defined as follows: Log Ratio <1 negative; Log Ratio ≥ 1.0 to ≤ 1.5 borderline; Log Ratio ≥ 1.5 positive.

(101) Data analysis at the individual probe level was also performed to assess if the tested samples were false positives. Only when at least 50% of specific probes had Log Ratio >1.5 was the test considered valid (FIG. 1C).

(102) For nearly all borderline results, only 20-25% of the specific probes showed mean intensity in the correct range, so the test was defined negative. For positive results (Log Ratio >1.5) more than 50% of the specific probes set were in the correct range. One example was an HCV 1a positive plasma samples test that was detected by 110 out of 110 probes at a concentration of 10.sup.5 copies/mL, 90 out of 110 probes at a concentration of 10.sup.4 copies/mL, and 70 out of 110 probes at a concentration of 10.sup.3 copies/mL. On average, at 10.sup.2 copies/mL more than 50% of the probes were generating a fluorescence signal above the set threshold.

(103) Data from more than 168 tested samples (one or multiple targets per array) showed consistent results. The mean of the probes specific for any positive plasma sample was always at least 10-fold higher than the mean of internal control probes (background), showing a wide probe population range of intensity. As shown in Table 4 and Table 5, the Log Ratio was above 1.5 for all the pathogens tested at a concentration of 10.sup.2 copies/mL and there were no cross reactions with other probes across the platform.

(104) Microarray sensitivity: HAV, CHIKV, DENV1-4, HCV Genotypes 1a, 2b, and 3, HIV-1,2 and WNV strain NY99 had 10.sup.2 copies/mL limits of detection. The lowest detectable level for HEV was 10.sup.4 copies/mL. The analytical sensitivity for each assay was determined using a concentration range based on the clinical requirement for pathogen detection. There were no false negatives or false positives when testing the positive plasma. In the presence of very low pathogen concentrations, the log ratio was at the borderline level so the results were qualified according to double level analysis (at least 50% of the probes generated a fluorescence signal above the set threshold). In the presence of negative plasma samples, the log ratio value was always negative (Table 4).

(105) A mix of different positive plasma samples at different concentrations was simultaneously tested in a single experiment. Four different combinations were generated. The multi-pathogens-mixes were composed of 8 (CHIKV, DEN3, DEN1, HAV, HCV1a, HEV, WNV and ZIKV), 4 (CHIKV, DEN1, ZKV, WNV), 4 (DENV3, HAV HCV1a HEV) and 3 (CHIKV, DEN1, ZIKV) different pathogens, respectively at a concentration range from 10.sup.5 to 10.sup.3 copies/mL. (Table 6).

(106) TABLE-US-00004 TABLE 4 Test results based on Log ratio

	CK	DEN1	DEN2	DEN3	DEN4
HAV	2.42	-1.15	-1.22	-0.04	-0.28
HCV1a	-0.50	-0.24	-0.35	DEN1	-0.12
HCV2a	1.60	-0.33	0.09	0.14	-0.08
CK	-0.20	-0.31	DEN2	-0.24	-0.82
2.42	1.84	-0.30	-0.25	-0.24	-0.17
-1.15	-0.28	DEN3	0.03	-0.19	-0.02
-1.22	1.62	-0.24	-0.28	-0.24	-0.34
-0.04	DEN4	0.07	-0.31	-0.64	0.18
-0.28	1.80	-0.64	-0.17	-0.28	HAV
-0.50	-0.06	-0.97	-1.24	-0.25	-0.34
-0.24	2.97	-0.21	-0.31	HCV1	0.46
-0.35	-0.53	-0.55	0.63	0.56	0.26
DEN1	2.91	2.80	HCV2	0.47	-0.60
1.60	-0.72	0.66	0.31	0.09	2.16
-0.33	2.85	HCV3	0.39	-0.73	-0.77
0.09	0.06	0.06	-0.07	2.63	2.53
0.14	HEV	0.41	-0.62	-0.63	0.23
-0.08	0.35	1.06	-0.14	-0.24	HIV1
-0.20	0.96	0.55	0.08	0.77	0.56
-0.31	0.01	0.05	-0.06	HIV2	-0.13
DEN2	-0.90	-0.94	0.02	-0.29	-0.07
-0.82	-0.09	-0.20	HTLVI	-0.57	-1.29
1.84	-0.45	-0.40	-0.21	0.19	0.08
-0.30	WNV	0.10	-0.50	-0.70	0.21
-0.25	-0.34	-0.33	-0.19	-0.29	ZKV
-0.24	-0.29	-0.64	-0.92	-0.49	-0.41
-0.17	-0.45	0.12	0.26	HCV3	HEV
-0.28	HIV-1	HIV-2	HTLVI	HTLVII	WNV
DEN3	ZKV	NC	CK	-0.56	-0.02
0.03	-0.99	-0.68	-0.42	-0.96	-0.97
-0.19	-0.20	-0.96	DEN1	-0.52	0.46
-0.34	-0.76	-0.55	-0.28	-0.76	-0.53
-0.33	0.00	-0.86	DEN2	-0.49	-0.11
-0.19	-1.13	-0.89	-0.52	-0.84	-1.00
-0.29	-0.28	-1.01	DEN3	-0.55	0.01
-0.92	-0.95	-0.81			

-0.56 -0.91 -0.22 0.05 -0.96 DEN4 -0.49 -0.20 -1.02 -0.80 -0.66 -1.05 -0.30 -0.36 -1.07
HAV -0.52 0.35 -0.78 -0.47 -0.35 -0.71 -1.22 -0.04 -0.69 HCV1 2.59 0.79 -0.30 0.94 0.44
-0.05 -0.43 0.60 -0.37 HCV2 1.84 0.76 -0.53 0.65 -0.10 -0.37 -0.05 0.50 -0.32 HCV3 2.32
0.34 -0.28 0.92 0.02 -0.51 -0.68 0.28 -0.46 HEV -0.45 1.89 -0.31 -0.17 0.10 -0.04 -0.51 0.54
-0.39 HIV1 -0.27 1.07 1.89 0.46 -0.22 0.06 0.13 0.92 0.23 HIV2 -0.41 0.23 0.91 1.68 -0.07
-0.44 -0.75 0.09 -0.66 HTLVI -0.38 -0.21 -1.08 -0.87 2.67 0.88 -1.47 -0.15 -0.99 HTLVII
-0.13 -0.09 -0.92 -0.59 0.38 3.30 -1.13 -0.10 -0.86 WNV -0.51 0.25 -0.62 -0.52 -0.40 -0.64
2.24 0.00 -0.82 ZKV 0.32 0.68 -0.96 -0.68 -0.52 -0.78 0.16 2.07 -0.83 CK, Chikungunya virus;
DEN, dengue; HAV, hepatitis A virus; HCV, hepatitis C virus; HEV, hepatitis E virus; HIV, human
immunodeficiency virus; HTLV, Human T-cell lymphotropic virus; WNV, West Nile Virus; ZKV,
Zika Virus; NC, negative control.

(107) TABLE-US-00005 TABLE 5 Multi-pathogen mix test results based on Log ratio MPM1
MPM2 MPM3 MPM4 CHIKV 3.42 3.37 0.24 2.25 DEN1 3.14 3.10 1.80 2.45 DEN2 1.11 1.10
-0.23 0.35 DEN3 2.72 1.19 3.00 0.51 DEN4 1.31 1.19 0.83 0.59 HAV 1.33 0.12 2.18 -1.13 HCV-
1a 2.53 0.61 2.72 -0.59 HCV-2a 2.16 0.66 2.44 -0.65 HCV-3 2.45 0.65 2.60 -0.72 HEV 1.64 0.67
1.21 -0.71 HIV-1 1.05 1.24 1.11 -0.36 HIV-2 0.13 0.24 0.17 -0.81 HTLV-I -0.17 0.06 -0.07
-1.37 HTLV-II -0.02 0.12 -0.09 -1.07 WNV 1.63 1.65 0.11 -0.03 ZKV 3.09 3.04 0.30 1.98
MPM1 = CHIKV, HAV, HCV-1a, HEV, DEN3, DEN1, ZKV, WNV MPM2 = CHIKV, DEN1, ZKV,
WNV MPM3 = HAV, HEV, DEN3, HCV-1a MPM4 = CHIKV, ZKV, DEN1

(108) TABLE-US-00006 TABLE 6 Pathogen Chip performance based plasma panel test results
Pathogen Copies/mL pos/total qPCR Validation Chikungunya 10{circumflex over ()}3 1/1 Y
Chikungunya 10{circumflex over ()}2 4/4 Y Dengue1 10{circumflex over ()}3 3/3 Y Dengue1
10{circumflex over ()}2 2/2 Y Dengue1 10{circumflex over ()}1 0/1 Y Dengue2 10{circumflex over ()}3 3/3 Y Dengue2
10{circumflex over ()}2 3/3 Y Dengue2 10{circumflex over ()}1 0/1 Y Dengue3 10{circumflex over ()}3 3/3 Y Dengue3
10{circumflex over ()}2 3/3 Y Dengue3 10{circumflex over ()}1 0/1 Y Dengue4 10{circumflex over ()}3 3/3 Y Dengue4
10{circumflex over ()}2 3/3 Y Dengue4 10{circumflex over ()}1 0/1 Y HAV 10{circumflex over ()}3 2/2 Y HAV
10{circumflex over ()}2 2/2 Y HCV-1a 10{circumflex over ()}3 3/3 Y HCV-1a 10{circumflex over ()}2 3/3 Y HCV-2a
10{circumflex over ()}2 2/2 Y HCV-3 10{circumflex over ()}2 2/2 Y HEV 10{circumflex over ()}4 3/3 Y HEV 10{circumflex over ()}3
0/2 Y HEV 10{circumflex over ()}2 0/2 NA HIV-1 10{circumflex over ()}3 2/2 y HIV-1 10{circumflex over ()}2 2/2 y
HIV-2 10{circumflex over ()}3 3/3 y HIV-2 10{circumflex over ()}2 3/3 y HTLV-I 10{circumflex over ()}3 2/2 y
HTLV-I 10{circumflex over ()}2 2/2 y HTLV-II 10{circumflex over ()}3 2/2 y HTLV-II 10{circumflex over ()}2 2/2 y
WNV (NY99) 10{circumflex over ()}5 1/1 y WNV (NY99) 10{circumflex over ()}4 1/1 y WNV (NY99)
10{circumflex over ()}3 3/3 y WNV (NY99) 10{circumflex over ()}2 4/4 y WNV (NY99) 10{circumflex over ()}1 0/2 NA
ZIKA PRVABC60 10{circumflex over ()}3 3/3 Y ZIKA PRVABC61 10{circumflex over ()}2 3/3 Y ZIKA
PRVABC62 10{circumflex over ()}1 0/2 Y ZIKV FSS13025 10{circumflex over ()}3 3/3 Y ZIKV
FSS13025 10{circumflex over ()}2 3/3 Y ZIKV FSS13025 10{circumflex over ()}1 0/2 Y MPM1
10{circumflex over ()}5-10{circumflex over ()}3 3/3 y MPM2 10{circumflex over ()}5-
10{circumflex over ()}3 3/3 y MPM3 10{circumflex over ()}5-10{circumflex over ()}3 3/3 y
MPM4 10{circumflex over ()}5-10{circumflex over ()}3 3/3 y NA = not applicable

(109) Among the 99 positive samples tested at a concentration ranking from 10.sup.5 to 10.sup.2
copies/mL, 92 out 92 samples were correctly detected. Only HEV testing resulted correct detection
in 3 out of 7 positive samples (42%) at a final concentration of 10.sup.4 copies/mL. No specific
signal was detected below this value. There were 21 positive samples that were not detected
because the concentration was below the limit of detection of the platform (<10.sup.2 copies/mL).
In all four mix combinations all pathogens were detected without interference among the targets.

(110) All of the samples tested (single or multiple pathogens at the same time) were performed at
least 3 times each, with at least a week interval between the experiments, in order to test the

reproducibility of the results. The consistency of positive results across the different arrays confirmed that the array design together with the double level analysis model performed well. (111) Validation of the limit of microarray data by RT-qPCR: Microarray-based pathogen chip results were confirmed by RT-qPCR of the RNA aliquots used for testing. All positive results were confirmed and the copy numbers for each pathogen were calculated to define the limit of the detection for each species on the array (Table 7).

(112) TABLE-US-00007 TABLE 7 Validation of Pathogen Chip detection results Pathogen Chip qPCR Ct Virus Virus Results Value Copy No. CHIKV POS 26.9 $3.1 \times 10^{sup.3}$ CHIKV POS 31.6 $1.4 \times 10^{sup.2}$ CHIKV POS 31.5 $1.6 \times 10^{sup.2}$ CHIKV POS 29.4 $6.3 \times 10^{sup.2}$ CHIKV POS 29.7 $5.8 \times 10^{sup.2}$ DENGUE-1 POS 31.1 $2.3 \times 10^{sup.3}$ DENGUE-1 POS 34.4 $3.0 \times 10^{sup.2}$ DENGUE-1 NEG 38.7 41 DENGUE-2 POS 29.1 $8.4 \times 10^{sup.3}$ DENGUE-2 POS 35.9 $1.2 \times 10^{sup.2}$ DENGUE-2 NEG 38.9 18 DENGUE-3 POS 29.3 $7.1 \times 10^{sup.3}$ DENGUE-3 POS 32.7 $8.4 \times 10^{sup.2}$ DENGUE-3 NEG 37.3 31 DENGUE-4 POS 30.3 $3.3 \times 10^{sup.3}$ DENGUE-4 POS 34.2 $2.6 \times 10^{sup.2}$ DENGUE-4 NEG 37.0 79 HAV POS 28.4 $3.2 \times 10^{sup.3}$ HAV POS 39.1 $2.8 \times 10^{sup.3}$ HAV POS 29.2 $8.2 \times 10^{sup.2}$ HAV POS 32.2 $1.2 \times 10^{sup.2}$ HCV-1a POS 26.2 $4.1 \times 10^{sup.3}$ HCV-1a POS 31.4 $1.6 \times 10^{sup.2}$ HCV-2a POS 27.4 $3.8 \times 10^{sup.3}$ HCV-2a POS 32.1 $1.8 \times 10^{sup.2}$ HCV-3 POS 33.1 $1.4 \times 10^{sup.2}$ HEV POS 25.4 $1.9 \times 10^{sup.4}$ HEV NEG 28.4 $1.8 \times 10^{sup.3}$ HIV-1 POS 27.3 $4.6 \times 10^{sup.3}$ HIV-1 POS 32.8 $1.6 \times 10^{sup.2}$ HIV-2 POS 27.4 $4.3 \times 10^{sup.3}$ HIV-2 POS 30.7 $1.8 \times 10^{sup.2}$ HTLV-I POS 28.7 $3.9 \times 10^{sup.3}$ HTLV-I POS 28.3 $2.9 \times 10^{sup.2}$ HTLV-II POS 25.574 $2.7 \times 10^{sup.3}$ HTLV-II POS 29.289 $2.4 \times 10^{sup.2}$ WNV (NY99) POS 21.5 $1.9 \times 10^{sup.5}$ WNV (NY99) POS 24.5 $1.5 \times 10^{sup.4}$ WNV (NY99) POS 27.6 $2.1 \times 10^{sup.3}$ WNV (NY99) POS 31.4 $1.0 \times 10^{sup.2}$ ZIKA PRVABC60 POS 26.2 $1.5 \times 10^{sup.3}$ ZIKA PRVABC60 POS 30.1 $1.2 \times 10^{sup.2}$ ZIKA PRVABC60 NEG 31.8 41 ZIKA FSS13025 POS 25.3 $2.4 \times 10^{sup.3}$ ZIKA FSS13025 POS 29.1 $1.3 \times 10^{sup.2}$ ZIKA FSS13025 NEG 33.0 17

Example 3

Microarray for Detection of DNA Viruses, Bacteria, and Protozoan Pathogens

(113) A microarray for DNA viruses, bacteria, and protozoan pathogens was developed. The design included multiple gene targets for each pathogen genome in order to select the best probes for the final platform design. The design strategy was to choose the probes with the best “scores” (homology, thermodynamics, secondary structure and sequence complexity) balancing the cross-hybridization with the host genome and with other pathogens' genomes. The second design strategy was to balance the number of probes for each pathogen with a final count of 50-110 probes each. Probes selected in the final design generated a more intense signal and produced higher percentage coverage of the specific genome across the different experiments.

(114) The final design was supplemented with predesigned DNA array probes (577 control probes, 225 replicates, and 11,620 backbone) specific for the reagents and the assay performance. These are used specifically for image orientation, to assess whether the samples are labeled, for the orientation of the platform during the scan process, and for measuring on element background. These probes form a hairpin and do not hybridize well with labeled sample of any species. In addition, 312 probes specific for three human housekeeping genes (ACTB, ARL1, CCDN1) and 109 probes specific for one Mosquito-specific virus and two plant viruses (*Aedes albopictus* densovirus 2, Maize streak virus, Tomato pseudo-curly top virus) were added to the design.

(115) The microarray includes probes for cytomegalovirus (CMV; also known as HHV-5), Epstein Barr virus (EBV; also known as HHV-4), human herpesvirus 8 (HHV-8), human papilloma virus (HPV) type 6b HPV6, HPV11, HPV 16, HPV 17, hepatitis B virus (HBV) subtype adw, HBV subtype ayw, HBV subtype adr, HBV subtype ayr, and human parvovirus B19. Exemplary probes provided in Table 8 and include SEQ ID NOs: 1770-1852 (CMV), SEQ ID NOs: 1853-1917 (EBV B95-8), SEQ ID NOs: 1918-2023 (EBV AG876), SEQ ID NOs: 2024-2108 (HHV-8), SEQ ID NOs: 2109-2192 (HPV 6b), SEQ ID NOs: 2193-2271 (HPV 11), SEQ ID NOs: 2272-2342 (HPV 16), SEQ ID NOs: 2343-2419 (HPV 18), SEQ ID NOs: 2420-2470 (HBV subtype adw), SEQ ID

NOs: 2471-2520 (HBV subtype ayw), SEQ ID NOs: 2521-2556 (HBV subtype adr), SEQ ID NOs: 2557-2602 (HBV subtype ayr), and SEQ ID NOs: 2603-2647 (human parvovirus B19).

(116) The microarray also includes probes for *Treponema pallidum*, *Ehrlichia chaffeensis*, *Ehrlichia ewingii*, *Ehrlichia muris*, *Borrelia burgdorferi*, *Coxiella burnetii*, *Trypanosoma brucei*, *Trypanosoma cruzi*, *Leishmania major*, *Babesia microti*, *Plasmodium falciparum*, and *Plasmodium vivax*. Exemplary probes provided in Table 9 and include SEQ ID NOs: 2648-2751 (*Treponema pallidum*), SEQ ID NOs: 2752-2852 (*Ehrlichia chaffeensis*), SEQ ID NOs: 2853-2861 (*Ehrlichia ewingii*), SEQ ID NOs: 2862-2922 (*Ehrlichia muris*), SEQ ID NOs: 2923-3001 (*Borrelia burgdorferi*), SEQ ID NOs: 3002-3085 (*Coxiella burnetii*), SEQ ID NOs: 3086-3097 (*Trypanosoma brucei*), SEQ ID NO: 3098 (*Trypanosoma cruzi*), SEQ ID NOs: 3099-3113 (*Leishmania major*), SEQ ID NOs: 3114-3154 (*Babesia microti*), SEQ ID NOs: 3155-3185 (*Plasmodium falciparum*), and SEQ ID NOs: 3186-3207 (*Plasmodium vivax*).

(117) Finally, the microarray includes housekeeping and negative control probes (Table 10). Exemplary probes include SEQ ID NOs: 3208-3301 (housekeeping gene ACTB), SEQ ID NOs: 3302-3385 (housekeeping gene ARL1), SEQ ID NOs: 3386-3519 (housekeeping gene CCDN1), SEQ ID NOs: 3520-3557 (*Aedes albopictus* densovirus 2), SEQ ID NO: 3558-3598 (Maize streak virus), and SEQ ID NOs: 3599-3628 (Tomato pseudo-curly top virus).

(118) For sample analysis, viral DNA from plasma specimens was extracted with the Invitrogen Dynabeads™ SILANE viral NA kit. The kit is designed for highly predictable and consistent isolation of viral nucleic acids. Beads and buffers are optimized for sensitive isolation of viral DNA. DNA from bacteria and protozoans was extracted from whole blood with the QIAamp® DNA Blood Mini kit (Qiagen) according to the manufacturer's protocol.

(119) SureTag® Labeling Kit (Agilent technology) was used to enzymatically label DNA from plasma and blood. A modified protocol was developed and optimized for efficient sample fragmentation, enzymatic labeling, and clean up. A master mix containing 10×aCGH blocking agent and 2×HI-RPM hybridization buffer, was added to 2.5-3 µg of labeled DNA, denatured, and hybridized to arrays under 8-chamber gasket slides at 67° C. with 20-rpm rotation for 24 hours in an Agilent hybridization oven. Arrays were processed using wash procedure A and scanned on an Agilent SureScan® G4900DA microarray scanner using 5-µm resolution.

(120) CMV, *Trypanosoma*, Parvovirus B19, HBV, EBV (HHV-4), *Treponema*, *Babesia*, *Leishmania*, *Coxiella*, *Borrelia*, Papilloma Virus (HPV 6, 11, 16, 18), and *P. falciparum* had 10.sup.4-10.sup.3 copies/mL limits of detection. There were no false negatives or false positives when testing the positive plasma. All the results were confirmed by RT-qPCR of the DNA aliquots used for testing. All positive results were confirmed and the copy numbers for each pathogen were calculated to define the limit of the detection for each species on the array.

(121) TABLE-US-00008 TABLE 8 Exemplary DNA virus probes SEQ ID Genomic ProbeID Start End NO: Virus Region Product CUST_P10000630 45812 45860 1770 CMV (HHV-5)

Cytomegalovirus UL34 protein CUST_P10000631 45872 45916 1771 CMV (HHV-5)

Cytomegalovirus UL34 protein CUST_P10000638 46336 46384 1772 CMV (HHV-5)

Cytomegalovirus UL34 protein CUST_P10001082 78547 78591 1773 CMV (HHV-5)

Cytomegalovirus UL54 DNA rep CUST_P10001099 79469 79514 1774 CMV (HHV-5)

Cytomegalovirus UL54 DNA rep CUST_P10001109 80183 80227 1775 CMV (HHV-5)

Cytomegalovirus UL54 DNA rep CUST_P10001111 80253 80301 1776 CMV (HHV-5)

Cytomegalovirus UL54 DNA rep CUST_P10001120 80933 80979 1777 CMV (HHV-5)

Cytomegalovirus UL54 DNA rep CUST_P10001123 81072 81116 1778 CMV (HHV-5)

Cytomegalovirus UL54 DNA rep CUST_P10001126 81245 81289 1779 CMV (HHV-5)

Cytomegalovirus UL54 DNA rep CUST_P10001131 81599 81643 1780 CMV (HHV-5)

Cytomegalovirus UL54 DNA rep CUST_P10001132 81738 81782 1781 CMV (HHV-5)

Cytomegalovirus UL54 DNA rep CUST_P10001140 82327 82375 1782 CMV (HHV-5)

Cytomegalovirus UL55 envelop CUST_P10001146 82856 82901 1783 CMV (HHV-5)

Cytomegalovirus UL55 envelop CUST_P10001152 83347 83399 1784 CMV (HHV-5)
Cytomegalovirus UL55 envelop CUST_P10001154 83475 83525 1785 CMV (HHV-5)
Cytomegalovirus UL55 envelop CUST_P10001155 83645 83701 1786 CMV (HHV-5)
Cytomegalovirus UL55 envelop CUST_P10001156 83677 83724 1787 CMV (HHV-5)
Cytomegalovirus UL55 envelop CUST_P10001158 83744 83797 1788 CMV (HHV-5)
Cytomegalovirus UL55 envelop CUST_P10001160 83961 84012 1789 CMV (HHV-5)
Cytomegalovirus UL55 envelop CUST_P10001161 83993 84052 1790 CMV (HHV-5)
Cytomegalovirus UL55 envelop CUST_P10001162 84021 84073 1791 CMV (HHV-5)
Cytomegalovirus UL55 envelop CUST_P10001164 84223 84275 1792 CMV (HHV-5)
Cytomegalovirus UL55 envelop CUST_P10001165 84398 84442 1793 CMV (HHV-5)
Cytomegalovirus UL55 envelop CUST_P10001169 84665 84716 1794 CMV (HHV-5)
Cytomegalovirus UL55 envelop CUST_P10001175 85014 85071 1795 CMV (HHV-5)
Cytomegalovirus UL56 encapsi CUST_P10001177 85085 85132 1796 CMV (HHV-5)
Cytomegalovirus UL56 encapsi CUST_P10001179 85221 85272 1797 CMV (HHV-5)
Cytomegalovirus UL56 encapsi CUST_P10001183 85645 85693 1798 CMV (HHV-5)
Cytomegalovirus UL56 encapsi CUST_P10001189 86404 86461 1799 CMV (HHV-5)
Cytomegalovirus UL56 encapsi CUST_P10001198 87249 87308 1800 CMV (HHV-5)
Cytomegalovirus UL56 encapsi CUST_P10001603 117602 117650 1801 CMV (HHV-5)
Cytomegalovirus UL80 capsid CUST_P10001618 118112 118156 1802 CMV (HHV-5)
Cytomegalovirus UL80 capsid CUST_P10001622 118567 118615 1803 CMV (HHV-5)
Cytomegalovirus UL80 capsid CUST_P10001664 121437 121483 1804 CMV (HHV-5)
Cytomegalovirus UL83 tegumen CUST_P10001665 121470 121522 1805 CMV (HHV-5)
Cytomegalovirus UL83 tegumen CUST_P10001666 121493 121543 1806 CMV (HHV-5)
Cytomegalovirus UL83 tegumen CUST_P10001675 122579 122623 1807 CMV (HHV-5)
Cytomegalovirus UL83 tegumen CUST_P10001945 141984 142028 1808 CMV (HHV-5)
Cytomegalovirus UL97 core CUST_P10001948 142100 142144 1809 CMV (HHV-5)
Cytomegalovirus UL97 core CUST_P10001960 142742 142789 1810 CMV (HHV-5)
Cytomegalovirus UL97 core CUST_P10001965 143122 143180 1811 CMV (HHV-5)
Cytomegalovirus UL97 core CUST_P10001966 143159 143203 1812 CMV (HHV-5)
Cytomegalovirus UL97 core CUST_P10002353 170852 170896 1813 CMV (HHV-5)
Cytomegalovirus UL122 Beta Ge CUST_P10002359 171207 171256 1814 CMV (HHV-5)
Cytomegalovirus UL122 Beta Ge CUST_P10002362 171393 171444 1815 CMV (HHV-5)
Cytomegalovirus UL122 Beta Ge CUST_P10002371 171811 171858 1816 CMV (HHV-5)
Cytomegalovirus UL122 Beta Ge CUST_P10002377 172241 172300 1817 CMV (HHV-5)
Cytomegalovirus UL122 Beta Ge CUST_P10002378 172286 172345 1818 CMV (HHV-5)
Cytomegalovirus UL122 Beta Ge CUST_P10002379 172307 172366 1819 CMV (HHV-5)
Cytomegalovirus UL122 Beta Ge CUST_P10002381 172506 172559 1820 CMV (HHV-5)
Cytomegalovirus UL122 Beta Ge CUST_P10002382 172633 172678 1821 CMV (HHV-5)
Cytomegalovirus UL122 Beta Ge CUST_P10002384 172987 173046 1822 CMV (HHV-5)
Cytomegalovirus UL122 Beta Ge CUST_P10002385 173023 173082 1823 CMV (HHV-5)
Cytomegalovirus UL122 Beta Ge CUST_P10002386 173044 173097 1824 CMV (HHV-5)
Cytomegalovirus UL122 Beta Ge CUST_P10002389 173134 173193 1825 CMV (HHV-5)
Cytomegalovirus UL122 Beta Ge CUST_P10002390 173156 173215 1826 CMV (HHV-5)
Cytomegalovirus UL122 Beta Ge CUST_P10002391 173191 173239 1827 CMV (HHV-5)
Cytomegalovirus UL122 Beta Ge CUST_P10002392 173374 173433 1828 CMV (HHV-5)
Cytomegalovirus UL122 Beta Ge CUST_P10002393 173395 173454 1829 CMV (HHV-5)
Cytomegalovirus UL122 Beta Ge CUST_P10002394 173432 173489 1830 CMV (HHV-5)
Cytomegalovirus UL122 Beta Ge CUST_P10002395 173456 173514 1831 CMV (HHV-5)
Cytomegalovirus UL122 Beta Ge CUST_P10002396 173503 173553 1832 CMV (HHV-5)
Cytomegalovirus UL122 Beta Ge CUST_P10002397 173520 173579 1833 CMV (HHV-5)

Cytomegalovirus UL122 Beta Ge CUST_P10002398 173555 173613 1835 CMV (HHV-5)
Cytomegalovirus UL122 Beta Ge CUST_P10002399 173596 173649 1835 CMV (HHV-5)
Cytomegalovirus UL122 Beta Ge CUST_P10002400 173621 173679 1836 CMV (HHV-5)
Cytomegalovirus UL122 Beta Ge CUST_P10002401 173674 173731 1837 CMV (HHV-5)
Cytomegalovirus UL122 Beta Ge CUST_P10002402 173693 173746 1838 CMV (HHV-5)
Cytomegalovirus UL122 Beta Ge CUST_P10002403 173735 173780 1839 CMV (HHV-5)
Cytomegalovirus UL122 Beta Ge CUST_P10002404 173766 173812 1840 CMV (HHV-5)
Cytomegalovirus UL122 Beta Ge CUST_P10002406 174051 174095 1841 CMV (HHV-5)
Cytomegalovirus UL122 Beta Ge CUST_P10002407 174065 174109 1842 CMV (HHV-5)
Cytomegalovirus UL123 Prot E CUST_P10002408 174215 174259 1843 CMV (HHV-5)
Cytomegalovirus UL123 Prot E CUST_P10002411 174671 174726 1844 CMV (HHV-5)
Cytomegalovirus UL124 Prot E CUST_P10002466 178473 178527 1845 CMV (HHV-5)
Cytomegalovirus UL132 Glyco CUST_P10002470 178801 178851 1846 CMV (HHV-5)
Cytomegalovirus UL132 Glyco CUST_P10002471 178825 178870 1847 CMV (HHV-5)
Cytomegalovirus UL132 Glyco CUST_P10002474 178926 178974 1848 CMV (HHV-5)
Cytomegalovirus UL132 Glyco CUST_P10002475 178954 179000 1849 CMV (HHV-5)
Cytomegalovirus UL132 Glyco CUST_P10002927 211545 211592 1850 CMV (HHV-5)
Cytomegalovirus US17 protein CUST_P10002930 211944 211988 1851 CMV (HHV-5)
Cytomegalovirus US17 protein CUST_P10002934 212083 212127 1852 CMV (HHV-5)
Cytomegalovirus US17 protein CUST_P10003273 1 60 1853 Human herpesvirus 4 (EBV), B95-8
LMP-2A transmembrane protein CUST_P10003278 444 503 1854 Human herpesvirus 4 (EBV),
B95-8 LMP-2A transmembrane protein CUST_P10003289 1435 1487 1855 Human herpesvirus 4
(EBV), B95-8 LMP-2A transmembrane protein CUST_P10003301 2062 2106 1856 Human
herpesvirus 4 (EBV), B95-8 BNFR1 tegument protein CUST_P10003303 2221 2265 1857 Human
herpesvirus 4 (EBV), B95-8 BNFR1 tegument protein CUST_P10003307 2620 2664 1858 Human
herpesvirus 4 (EBV), B95-8 BNFR1 tegument protein CUST_P10003313 3104 3148 1859 Human
herpesvirus 4 (EBV), B95-8 BNFR1 tegument protein CUST_P10003322 3928 3972 1860 Human
herpesvirus 4 (EBV), B95-8 BNFR1 tegument protein CUST_P10003342 5201 5245 1861 Human
herpesvirus 4 (EBV), B95-8 BNFR1 tegument protein CUST_P10003349 5834 5893 1862 Human
herpesvirus 4 (EBV), B95-8 BNFR1 tegument protein CUST_P10003351 5931 5985 1863 Human
herpesvirus 4 (EBV), B95-8 EBNA-1 Nuclear antigen CUST_P10003352 6010 6069 1864 Human
herpesvirus 4 (EBV), B95-8 EBNA-1 Nuclear antigen CUST_P10003367 7289 7347 1865 Human
herpesvirus 4 (EBV), B95-8 EBNA-1 Nuclear antigen CUST_P10003369 7409 7468 1866 Human
herpesvirus 4 (EBV), B95-8 EBNA-1 Nuclear antigen CUST_P10003372 7520 7579 1867 Human
herpesvirus 4 (EBV), B95-8 EBNA-1 Nuclear antigen CUST_P10003375 7605 7664 1868 Human
herpesvirus 4 (EBV), B95-8 EBNA-1 Nuclear antigen CUST_P10003376 7635 7694 1869 Human
herpesvirus 4 (EBV), B95-8 EBNA-1 Nuclear antigen CUST_P10003377 7706 7765 1870 Human
herpesvirus 4 (EBV), B95-8 EBNA-1 Nuclear antigen CUST_P10003378 7731 7790 1871 Human
herpesvirus 4 (EBV), B95-8 EBNA-1 Nuclear antigen CUST_P10003379 7786 7845 1872 Human
herpesvirus 4 (EBV), B95-8 EBNA-1 Nuclear antigen CUST_P10003380 7821 7880 1873 Human
herpesvirus 4 (EBV), B95-8 EBNA-1 Nuclear antigen CUST_P10003381 7871 7930 1874 Human
herpesvirus 4 (EBV), B95-8 EBNA-1 Nuclear antigen CUST_P10003382 7901 7960 1875 Human
herpesvirus 4 (EBV), B95-8 EBNA-1 Nuclear antigen CUST_P10003383 7941 8000 1876 Human
herpesvirus 4 (EBV), B95-8 EBNA-1 Nuclear antigen CUST_P10003384 8000 8051 1877 Human
herpesvirus 4 (EBV), B95-8 EBNA-1 Nuclear antigen CUST_P10003390 8236 8291 1878 Human
herpesvirus 4 (EBV), B95-8 EBNA-1 Nuclear antigen CUST_P10003398 8631 8690 1879 Human
herpesvirus 4 (EBV), B95-8 EBNA-1 Nuclear antigen CUST_P10003399 8659 8716 1880 Human
herpesvirus 4 (EBV), B95-8 EBNA-1 Nuclear antigen CUST_P10003400 8690 8737 1881 Human
herpesvirus 4 (EBV), B95-8 EBNA-1 Nuclear antigen CUST_P10003404 9006 9063 1882 Human
herpesvirus 4 (EBV), B95-8 EBNA-1 Nuclear antigen CUST_P10003405 9052 9111 1883 Human

herpesvirus 4 (EBV), B95-8 EBNA-1 Nuclear antigen CUST_P10003406 9110 9166 1884 Human
herpesvirus 4 (EBV), B95-8 EBNA-1 Nuclear antigen CUST_P10003408 9301 9350 1885 Human
herpesvirus 4 (EBV), B95-8 EBNA-1 Nuclear antigen CUST_P10003409 9349 9399 1886 Human
herpesvirus 4 (EBV), B95-8 EBNA-1 Nuclear antigen CUST_P10003412 9759 9811 1887 Human
herpesvirus 4 (EBV), B95-8 EBNA-1 Nuclear antigen CUST_P10003413 9788 9841 1888 Human
herpesvirus 4 (EBV), B95-8 EBNA-1 Nuclear antigen CUST_P10003418 9961 10012 1889
Human herpesvirus 4 (EBV), B95-8 EBNA-1 Nuclear antigen CUST_P10003427 10915 10966
1890 Human herpesvirus 4 (EBV), B95-8 EBNA-1 Nuclear antigen CUST_P10003429 10994
11046 1891 Human herpesvirus 4 (EBV), B95-8 EBNA-1 Nuclear antigen CUST_P10003432
11304 11362 1892 Human herpesvirus 4 (EBV), B95-8 EBNA-1 Nuclear antigen
CUST_P10003721 35383 35442 1893 Human herpesvirus 4 (EBV), B95-8 EBNA-1 Nuclear
antigen CUST_P10003725 35654 35708 1894 Human herpesvirus 4 (EBV), B95-8 EBNA-1
Nuclear antigen CUST_P10005257 152676 152720 1895 Human herpesvirus 4 (EBV), B95-8
BALF5 Binding Protein CUST_P10005266 153637 153681 1896 Human herpesvirus 4 (EBV),
B95-8 BALF5 Binding Protein CUST_P10005267 153658 153702 1897 Human herpesvirus 4
(EBV), B95-8 BALF5 Binding Protein CUST_P10005275 154346 154393 1898 Human
herpesvirus 4 (EBV), B95-8 BALF5 Binding Protein CUST_P10005279 154462 154516 1899
Human herpesvirus 4 (EBV), B95-8 BALF5 Binding Protein CUST_P10005291 155318 155362
1900 Human herpesvirus 4 (EBV), B95-8 BALF5 Binding Protein CUST_P10005293 155583
155627 1901 Human herpesvirus 4 (EBV), B95-8 BALF5 Binding Protein CUST_P10005300
155914 155958 1902 Human herpesvirus 4 (EBV), B95-8 BALF5 Binding Protein
CUST_P10005304 156254 156306 1903 Human herpesvirus 4 (EBV), B95-8 BALF5 Binding
Protein CUST_P10005325 157914 157963 1904 Human herpesvirus 4 (EBV), B95-8 BALF4
Binding Protein CUST_P10005327 158072 158121 1905 Human herpesvirus 4 (EBV), B95-8
BALF4 Binding Protein CUST_P10005329 158133 158183 1906 Human herpesvirus 4 (EBV),
B95-8 BALF4 Binding Protein CUST_P10005332 158230 158286 1907 Human herpesvirus 4
(EBV), B95-8 BALF4 Binding Protein CUST_P10005334 158408 158453 1908 Human
herpesvirus 4 (EBV), B95-8 BALF4 Binding Protein CUST_P10005335 158572 158625 1909
Human herpesvirus 4 (EBV), B95-8 BALF4 Binding Protein CUST_P10005336 158595 158644
1910 Human herpesvirus 4 (EBV), B95-8 BALF4 Binding Protein CUST_P10005340 158865
158916 1911 Human herpesvirus 4 (EBV), B95-8 BALF4 Binding Protein CUST_P10005355
160486 160545 1912 Human herpesvirus 4 (EBV), B95-8 BALF4 Binding Protein
CUST_P10005356 160515 160574 1913 Human herpesvirus 4 (EBV), B95-8 BALF3 Binding
Protein CUST_P10005367 161267 161318 1914 Human herpesvirus 4 (EBV), B95-8 BALF3
Binding Protein CUST_P10005380 162040 162086 1915 Human herpesvirus 4 (EBV), B95-8
BALF3 Binding Protein CUST_P10005381 162117 162161 1916 Human herpesvirus 4 (EBV),
B95-8 BALF3 Binding Protein CUST_P10005382 162322 162366 1917 Human herpesvirus 4
(EBV), B95-8 BALF3 Binding Protein CUST_P10005496 1 60 1918 Human herpesvirus 4 (EBV),
AG876 LMP-2B latency and B cell survival CUST_P10005503 903 961 1919 Human herpesvirus
4 (EBV), AG876 LMP-2B latency and B cell survival CUST_P10005506 1162 1221 1920 Human
herpesvirus 4 (EBV), AG876 LMP-2B latency and B cell survival CUST_P10005507 1290 1345
1921 Human herpesvirus 4 (EBV), AG876 LMP-2B latency and B cell survival CUST_P10005515
1658 1717 1922 Human herpesvirus 4 (EBV), AG876 LMP-2B latency and B cell survival
CUST_P10005567 5833 5892 1923 Human herpesvirus 4 (EBV), AG876 EBER-1 DNA
replication CUST_P10005568 5869 5928 1924 Human herpesvirus 4 (EBV), AG876 EBER-1
DNA replication CUST_P10005571 6014 6073 1925 Human herpesvirus 4 (EBV), AG876 EBER-
1 DNA replication CUST_P10005581 6584 6635 1926 Human herpesvirus 4 (EBV), AG876
EBER-1 DNA replication CUST_P10005585 6912 6967 1927 Human herpesvirus 4 (EBV),
AG876 EBER-1 DNA replication CUST_P10005589 7293 7348 1928 Human herpesvirus 4
(EBV), AG876 EBER-1 DNA replication CUST_P10005592 7417 7476 1929 Human herpesvirus

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CUST_P10009756 980 1039 2205 Human papillomavirus type 6b E7 Regulatory (HPV 11) Protein
CUST_P10009758 1159 1212 2206 Human papillomavirus type 6b E1 Regulatory (HPV 11)
Protein CUST_P10009759 1195 1242 2207 Human papillomavirus type 6b E1 Regulatory (HPV
11) Protein CUST_P10009760 1393 1452 2208 Human papillomavirus type 6b E1 Regulatory
(HPV 11) Protein CUST_P10009761 1419 1478 2209 Human papillomavirus type 6b E1
Regulatory (HPV 11) Protein CUST_P10009762 1450 1509 2210 Human papillomavirus type 6b
E1 Regulatory (HPV 11) Protein CUST_P10009763 1476 1535 2211 Human papillomavirus type
6b E1 Regulatory (HPV 11) Protein CUST_P10009765 1847 1899 2212 Human papillomavirus
type 6b E1 Regulatory (HPV 11) Protein CUST_P10009766 1864 1923 2213 Human
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Human papillomavirus type 6b E1 Regulatory (HPV 11) Protein CUST_P10009768 1962 2017
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2062 2216 Human papillomavirus type 6b E1 Regulatory (HPV 11) Protein CUST_P10009770
2127 2186 2217 Human papillomavirus type 6b E2 Regulatory (HPV 11) Protein
CUST_P10009771 2265 2316 2218 Human papillomavirus type 6b E2 Regulatory (HPV 11)
Protein CUST_P10009772 2303 2362 2219 Human papillomavirus type 6b E2 Regulatory (HPV
11) Protein CUST_P10009773 2327 2384 2220 Human papillomavirus type 6b E2 Regulatory
(HPV 11) Protein CUST_P10009774 2370 2420 2221 Human papillomavirus type 6b E2
Regulatory (HPV 11) Protein CUST_P10009775 2422 2481 2222 Human papillomavirus type 6b

E2 Regulatory (HPV 11) Protein CUST_P10009776 2458 2517 2223 Human papillomavirus type 6b E2 Regulatory (HPV 11) Protein CUST_P10009777 2590 2649 2224 Human papillomavirus type 6b E2 Regulatory (HPV 11) Protein CUST_P10009778 2625 2684 2225 Human papillomavirus type 6b E2 Regulatory (HPV 11) Protein CUST_P10009779 2655 2708 2226 Human papillomavirus type 6b E2 Regulatory (HPV 11) Protein CUST_P10009780 2692 2740 2227 Human papillomavirus type 6b E2 Regulatory (HPV 11) Protein CUST_P10009781 2920 2979 2228 Human papillomavirus type 6b E2 Regulatory (HPV 11) Protein CUST_P10009782 2985 3035 2229 Human papillomavirus type 6b E2 Regulatory (HPV 11) Protein CUST_P10009783 3047 3106 2230 Human papillomavirus type 6b E2 Regulatory (HPV 11) Protein CUST_P10009784 3081 3140 2231 Human papillomavirus type 6b E2 Regulatory (HPV 11) Protein CUST_P10009785 3120 3176 2232 Human papillomavirus type 6b E2 Regulatory (HPV 11) Protein CUST_P10009786 3265 3324 2233 Human papillomavirus type 6b E2 Regulatory (HPV 11) Protein CUST_P10009787 3380 3424 2234 Human papillomavirus type 6b E2 Regulatory (HPV 11) Protein CUST_P10009788 3432 3476 2235 Human papillomavirus type 6b E2 Regulatory (HPV 11) Protein CUST_P10009789 3505 3560 2236 Human papillomavirus type 6b E2 Regulatory (HPV 11) Protein CUST_P10009790 3678 3735 2237 Human papillomavirus type 6b E2 Regulatory (HPV 11) Protein CUST_P10009791 3709 3768 2238 Human papillomavirus type 6b E2 Regulatory (HPV 11) Protein CUST_P10009792 3901 3960 2239 Human papillomavirus type 6b E5 Regulatory (HPV 11) Protein CUST_P10009793 4136 4195 2240 Human papillomavirus type 6b E5 Regulatory (HPV 11) Protein CUST_P10009794 4283 4342 2241 Human papillomavirus type 6b E5 Regulatory (HPV 11) Protein CUST_P10009795 4408 4458 2242 Human papillomavirus type 6b E5 Regulatory (HPV 11) Protein CUST_P10009796 4497 4556 2243 Human papillomavirus type 6b L2 Capsid (HPV 11) CUST_P10009798 4792 4851 2244 Human papillomavirus type 6b L2 Capsid (HPV 11) CUST_P10009799 4818 4875 2245 Human papillomavirus type 6b L2 Capsid (HPV 11) CUST_P10009800 4855 4912 2246 Human papillomavirus type 6b L2 Capsid (HPV 11) CUST_P10009802 4936 4993 2247 Human papillomavirus type 6b L2 Capsid (HPV 11) CUST_P10009804 5164 5223 2248 Human papillomavirus type 6b L2 Capsid (HPV 11) CUST_P10009805 5341 5397 2249 Human papillomavirus type 6b L2 Capsid (HPV 11) CUST_P10009806 5423 5480 2250 Human papillomavirus type 6b L2 Capsid (HPV 11) CUST_P10009808 5491 5548 2251 Human papillomavirus type 6b L2 Capsid (HPV 11) CUST_P10009809 5514 5566 2252 Human papillomavirus type 6b L2 Capsid (HPV 11) CUST_P10009810 5539 5586 2253 Human papillomavirus type 6b L2 Capsid (HPV 11) CUST_P10009811 5576 5622 2254 Human papillomavirus type 6b L2 Capsid (HPV 11) CUST_P10009812 5603 5651 2255 Human papillomavirus type 6b L2 Capsid (HPV 11) CUST_P10009813 5618 5670 2256 Human papillomavirus type 6b L2 Capsid (HPV 11) CUST_P10009814 5745 5800 2257 Human papillomavirus type 6b L2 Capsid (HPV 11) CUST_P10009815 5819 5875 2258 Human papillomavirus type 6b L1 PolyA (HPV 11) CUST_P10009816 5978 6034 2259 Human papillomavirus type 6b L1 PolyA (HPV 11) CUST_P10009817 6008 6059 2260 Human papillomavirus type 6b L1 PolyA (HPV 11) CUST_P10009818 6031 6075 2261 Human papillomavirus type 6b L1 PolyA (HPV 11) CUST_P10009819 6069 6113 2262 Human papillomavirus type 6b L1 PolyA (HPV 11) CUST_P10009820 6089 6139 2263 Human papillomavirus type 6b L1 PolyA (HPV 11) CUST_P10009821 6118 6177 2264 Human papillomavirus type 6b L1 PolyA (HPV 11) CUST_P10009822 6338 6390 2265 Human papillomavirus type 6b L1 PolyA (HPV 11) CUST_P10009823 6356 6415 2266 Human papillomavirus type 6b L1 PolyA (HPV 11) CUST_P10009824 6383 6442 2267 Human papillomavirus type 6b L1 PolyA (HPV 11) CUST_P10009825 6408 6467 2268 Human papillomavirus type 6b L1 PolyA (HPV 11) CUST_P10009826 6707 6757 2269 Human papillomavirus type 6b L1 PolyA (HPV 11) CUST_P10009827 6796 6855 2270 Human papillomavirus type 6b L1 PolyA (HPV 11)

CUST_P10009828 7130 7187 2271 Human papillomavirus type 6b L1 PolyA (HPV 11)
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CUST_P10009833 221 280 2274 Human papillomavirus type 6b E6 Regulatory (HPV 16) Protein
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Protein CUST_P10009842 1131 1190 2282 Human papillomavirus type 6b E1 Regulatory (HPV
16) Protein CUST_P10009843 1269 1316 2283 Human papillomavirus type 6b E1 Regulatory
(HPV 16) Protein CUST_P10009844 1296 1346 2284 Human papillomavirus type 6b E1
Regulatory (HPV 16) Protein CUST_P10009845 1461 1520 2285 Human papillomavirus type 6b
E1 Regulatory (HPV 16) Protein CUST_P10009846 1734 1793 2286 Human papillomavirus type
6b E1 Regulatory (HPV 16) Protein CUST_P10009847 1841 1900 2287 Human papillomavirus
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Human papillomavirus type 6b E1 Regulatory (HPV 16) Protein CUST_P10009850 2326 2385
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2431 2291 Human papillomavirus type 6b E1 Regulatory (HPV 16) Protein CUST_P10009852
2583 2642 2292 Human papillomavirus type 6b E1 Regulatory (HPV 16) Protein
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Protein CUST_P10009854 2745 2803 2294 Human papillomavirus type 6b E1 Regulatory (HPV
16) Protein CUST_P10009855 2862 2921 2295 Human papillomavirus type 6b E2 Regulatory
(HPV 16) Protein CUST_P10009856 2895 2951 2296 Human papillomavirus type 6b E2
Regulatory (HPV 16) Protein CUST_P10009857 2932 2991 2297 Human papillomavirus type 6b
E2 Regulatory (HPV 16) Protein CUST_P10009858 2974 3033 2298 Human papillomavirus type
6b E2 Regulatory (HPV 16) Protein CUST_P10009859 2995 3054 2299 Human papillomavirus
type 6b E2 Regulatory (HPV 16) Protein CUST_P10009860 3091 3150 2300 Human
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Human papillomavirus type 6b E2 Regulatory (HPV 16) Protein CUST_P10009862 3325 3384
2302 Human papillomavirus type 6b E2 Regulatory (HPV 16) Protein CUST_P10009864 3437
3481 2303 Human papillomavirus type 6b E2 Regulatory (HPV 16) Protein CUST_P10009867
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Protein CUST_P10009869 3949 4008 2306 Human papillomavirus type 6b E5 Regulatory (HPV
16) Protein CUST_P10009870 4245 4297 2307 Human papillomavirus type 6b E5 Regulatory
(HPV 16) Protein CUST_P10009871 4323 4382 2308 Human papillomavirus type 6b L2 Capsid
(HPV 16) CUST_P10009872 4347 4406 2309 Human papillomavirus type 6b L2 Capsid (HPV 16)
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Protein CUST_P10009926 1134 1187 2358 Human papillomavirus type 6b E1 Regulatory (HPV
18) Protein CUST_P10009927 1302 1357 2359 Human papillomavirus type 6b E1 Regulatory
(HPV 18) Protein CUST_P10009928 1663 1722 2360 Human papillomavirus type 6b E1
Regulatory (HPV 18) Protein CUST_P10009929 1691 1750 2361 Human papillomavirus type 6b
E1 Regulatory (HPV 18) Protein CUST_P10009930 1719 1778 2362 Human papillomavirus type
6b E1 Regulatory (HPV 18) Protein CUST_P10009931 1883 1942 2363 Human papillomavirus
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Human papillomavirus type 6b E1 Regulatory (HPV 18) Protein CUST_P10009934 2210 2269
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2451 2367 Human papillomavirus type 6b E1 Regulatory (HPV 18) Protein CUST_P10009936

2430 2489 2368 Human papillomavirus type 6b E1 Regulatory (HPV 18) Protein
CUST_P10009937 2471 2520 2369 Human papillomavirus type 6b E1 Regulatory (HPV 18)
Protein CUST_P10009938 2512 2571 2370 Human papillomavirus type 6b E1 Regulatory (HPV
18) Protein CUST_P10009939 2558 2617 2371 Human papillomavirus type 6b E1 Regulatory
(HPV 18) Protein CUST_P10009940 2606 2665 2372 Human papillomavirus type 6b E1
Regulatory (HPV 18) Protein CUST_P10009941 2642 2701 2373 Human papillomavirus type 6b
E1 Regulatory (HPV 18) Protein CUST_P10009942 2808 2852 2374 Human papillomavirus type
6b E1 Regulatory (HPV 18) Protein CUST_P10009943 2837 2891 2375 Human papillomavirus
type 6b E2 Regulatory (HPV 18) Protein CUST_P10009944 2858 2917 2376 Human
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Human papillomavirus type 6b E2 Regulatory (HPV 18) Protein CUST_P10009946 2914 2973
2378 Human papillomavirus type 6b E2 Regulatory (HPV 18) Protein CUST_P10009947 3085
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Protein CUST_P10009952 3727 3781 2382 Human papillomavirus type 6b E2 Regulatory (HPV
18) Protein CUST_P10009953 3753 3807 2383 Human papillomavirus type 6b E4 Regulatory
(HPV 18) Protein CUST_P10009954 4025 4080 2384 Human papillomavirus type 6b E5
Regulatory (HPV 18) Protein CUST_P10009955 4262 4316 2385 Human papillomavirus type 6b
L2 Capsid (HPV 18) CUST_P10009956 4357 4414 2386 Human papillomavirus type 6b L2
Capsid (HPV 18) CUST_P10009959 4544 4599 2387 Human papillomavirus type 6b L2 Capsid
(HPV 18) CUST_P10009960 4678 4735 2388 Human papillomavirus type 6b L2 Capsid (HPV 18)
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CUST_P10010004 44 88 2420 Hepatitis B virus subtype adw S surface protein CUST_P10010005
78 126 2421 Hepatitis B virus subtype adw S surface protein CUST_P10010006 158 202 2422
Hepatitis B virus subtype adw S surface protein CUST_P10010008 230 281 2423 Hepatitis B virus
subtype adw S surface protein CUST_P10010009 267 311 2424 Hepatitis B virus subtype adw S
surface protein CUST_P10010010 328 372 2425 Hepatitis B virus subtype adw S surface protein
CUST_P10010011 394 453 2426 Hepatitis B virus subtype adw S surface protein
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Core CUST_P10010022 1170 1214 2436 Hepatitis B virus subtype adw C Core CUST_P10010023
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2223 2451 Hepatitis B virus subtype adw C Core CUST_P10010038 2208 2267 2452 Hepatitis B
virus subtype adw C Core CUST_P10010039 2238 2285 2453 Hepatitis B virus subtype adw C
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CUST_P10010065 234 283 2477 Hepatitis B virus subtype ayw S surface protein
CUST_P10010066 264 313 2478 Hepatitis B virus subtype ayw S surface protein
CUST_P10010067 295 339 2479 Hepatitis B virus subtype ayw S surface protein
CUST_P10010068 330 374 2480 Hepatitis B virus subtype ayw S surface protein
CUST_P10010069 369 416 2481 Hepatitis B virus subtype ayw S surface protein
CUST_P10010070 392 447 2482 Hepatitis B virus subtype ayw S surface protein
CUST_P10010071 443 493 2483 Hepatitis B virus subtype ayw S surface protein
CUST_P10010072 473 517 2484 Hepatitis B virus subtype ayw S surface protein
CUST_P10010073 600 645 2485 Hepatitis B virus subtype ayw S surface protein
CUST_P10010076 851 907 2486 Hepatitis B virus subtype ayw X x-protein CUST_P10010077
983 1033 2487 Hepatitis B virus subtype ayw X x-protein CUST_P10010078 1028 1087 2488
Hepatitis B virus subtype ayw X x-protein CUST_P10010079 1091 1147 2489 Hepatitis B virus
subtype ayw X x-protein CUST_P10010080 1161 1205 2490 Hepatitis B virus subtype ayw X x-
protein CUST_P10010081 1218 1262 2491 Hepatitis B virus subtype ayw X x-protein
CUST_P10010082 1265 1309 2492 Hepatitis B virus subtype ayw X x-protein CUST_P10010083
1319 1374 2493 Hepatitis B virus subtype ayw X x-protein CUST_P10010084 1384 1429 2494
Hepatitis B virus subtype ayw X x-protein CUST_P10010085 1446 1490 2495 Hepatitis B virus
subtype ayw X x-protein CUST_P10010086 1476 1520 2496 Hepatitis B virus subtype ayw X x-
protein CUST_P10010087 1528 1572 2497 Hepatitis B virus subtype ayw X x-protein
CUST_P10010088 1552 1596 2498 Hepatitis B virus subtype ayw X x-protein CUST_P10010089
1610 1654 2499 Hepatitis B virus subtype ayw X x-protein CUST_P10010090 1636 1685 2500
Hepatitis B virus subtype ayw X x-protein CUST_P10010092 1815 1870 2501 Hepatitis B virus
subtype ayw C Core CUST_P10010093 2042 2088 2502 Hepatitis B virus subtype ayw C Core
CUST_P10010094 2133 2192 2503 Hepatitis B virus subtype ayw C Core CUST_P10010095 2162
2221 2504 Hepatitis B virus subtype ayw C Core CUST_P10010096 2210 2269 2505 Hepatitis B
virus subtype ayw C Core CUST_P10010097 2343 2387 2506 Hepatitis B virus subtype ayw C
Core CUST_P10010098 2390 2434 2507 Hepatitis B virus subtype ayw C Core CUST_P10010099
2427 2477 2508 Hepatitis B virus subtype ayw C Core CUST_P10010100 2535 2594 2509
Hepatitis B virus subtype ayw C Core CUST_P10010101 2577 2636 2510 Hepatitis B virus
subtype ayw P Pol CUST_P10010102 2641 2700 2511 Hepatitis B virus subtype ayw P Pol
CUST_P10010103 2698 2757 2512 Hepatitis B virus subtype ayw P Pol CUST_P10010104 2756
2815 2513 Hepatitis B virus subtype ayw P Pol CUST_P10010105 2794 2845 2514 Hepatitis B
virus subtype ayw P Pol CUST_P10010106 2824 2872 2515 Hepatitis B virus subtype ayw P Pol
CUST_P10010107 2856 2900 2516 Hepatitis B virus subtype ayw P Pol CUST_P10010108 2885
2929 2517 Hepatitis B virus subtype ayw P Pol CUST_P10010109 2915 2963 2518 Hepatitis B
virus subtype ayw P Pol CUST_P10010110 3046 3090 2519 Hepatitis B virus subtype ayw P Pol
CUST_P10010111 3121 3165 2520 Hepatitis B virus subtype ayw P Pol CUST_P10010112 8 55
2521 Hepatitis B virus, subtype adr P Pol CUST_P10010113 79 125 2522 Hepatitis B virus,
subtype adr P Pol CUST_P10010114 112 156 2523 Hepatitis B virus, subtype adr P Pol
CUST_P10010119 325 370 2524 Hepatitis B virus, subtype adr P Pol CUST_P10010122 451 497
2525 Hepatitis B virus, subtype adr P Pol CUST_P10010123 477 521 2526 Hepatitis B virus,
subtype adr P Pol CUST_P10010124 638 682 2527 Hepatitis B virus, subtype adr P Pol
CUST_P10010125 667 714 2528 Hepatitis B virus, subtype adr P Pol CUST_P10010126 705 754
2529 Hepatitis B virus, subtype adr P Pol CUST_P10010127 839 894 2530 Hepatitis B virus,
subtype adr P Pol CUST_P10010128 971 1022 2531 Hepatitis B virus, subtype adr P Pol
CUST_P10010129 1088 1145 2532 Hepatitis B virus, subtype adr P Pol CUST_P10010131 1160
1204 2533 Hepatitis B virus, subtype adr P Pol CUST_P10010132 1203 1247 2534 Hepatitis B
virus, subtype adr P Pol CUST_P10010133 1315 1361 2535 Hepatitis B virus, subtype adr P Pol
CUST_P10010134 1458 1502 2536 Hepatitis B virus, subtype adr P Pol CUST_P10010136 1556
1600 2537 Hepatitis B virus, subtype adr P Pol CUST_P10010137 1635 1683 2538 Hepatitis B

virus, subtype adr P Pol CUST_P10010138 1679 1729 2539 Hepatitis B virus, subtype adr C Core CUST_P10010139 1810 1864 2540 Hepatitis B virus, subtype adr C Core CUST_P10010140 1999 2043 2541 Hepatitis B virus, subtype adr C Core CUST_P10010141 2109 2153 2542 Hepatitis B virus, subtype adr C Core CUST_P10010142 2220 2279 2543 Hepatitis B virus, subtype adr C Core CUST_P10010143 2247 2294 2544 Hepatitis B virus, subtype adr C Core CUST_P10010144 2295 2346 2545 Hepatitis B virus, subtype adr C Core CUST_P10010145 2365 2409 2546 Hepatitis B virus, subtype adr C Core CUST_P10010148 2545 2604 2547 Hepatitis B virus, subtype adr P Pol CUST_P10010149 2575 2634 2548 Hepatitis B virus, subtype adr P Pol CUST_P10010150 2618 2677 2549 Hepatitis B virus, subtype adr P Pol CUST_P10010151 2646 2705 2550 Hepatitis B virus, subtype adr P Pol CUST_P10010152 2672 2731 2551 Hepatitis B virus, subtype adr P Pol CUST_P10010153 2699 2758 2552 Hepatitis B virus, subtype adr P Pol CUST_P10010154 2735 2794 2553 Hepatitis B virus, subtype adr P Pol CUST_P10010156 2928 2972 2554 Hepatitis B virus, subtype adr S1 surface protein CUST_P10010157 2976 3020 2555 Hepatitis B virus, subtype adr S1 surface protein CUST_P10010158 3063 3107 2556 Hepatitis B virus, subtype adr S1 surface protein CUST_P10010159 42 86 2557 Hepatitis B virus, subtype ayr P Pol CUST_P10010160 79 126 2558 Hepatitis B virus, subtype ayr P Pol CUST_P10010161 133 177 2559 Hepatitis B virus, subtype ayr S surface protein CUST_P10010162 158 202 2560 Hepatitis B virus, subtype ayr S surface protein CUST_P10010163 190 238 2561 Hepatitis B virus, subtype ayr S surface protein CUST_P10010164 227 280 2562 Hepatitis B virus, subtype ayr S surface protein CUST_P10010165 272 316 2563 Hepatitis B virus, subtype ayr S surface protein CUST_P10010166 328 372 2564 Hepatitis B virus, subtype ayr S surface protein CUST_P10010167 366 414 2565 Hepatitis B virus, subtype ayr S surface protein CUST_P10010168 390 447 2566 Hepatitis B virus, subtype ayr S surface protein CUST_P10010169 452 500 2567 Hepatitis B virus, subtype ayr S surface protein CUST_P10010170 477 521 2568 Hepatitis B virus, subtype ayr S surface protein CUST_P10010171 598 642 2569 Hepatitis B virus, subtype ayr S surface protein CUST_P10010172 659 707 2570 Hepatitis B virus, subtype ayr S surface protein CUST_P10010173 839 898 2571 Hepatitis B virus, subtype ayr P Pol CUST_P10010174 971 1023 2572 Hepatitis B virus, subtype ayr P Pol CUST_P10010175 1087 1146 2573 Hepatitis B virus, subtype ayr P Pol CUST_P10010176 1116 1163 2574 Hepatitis B virus, subtype ayr P Pol CUST_P10010177 1151 1195 2575 Hepatitis B virus, subtype ayr P Pol CUST_P10010179 1318 1366 2576 Hepatitis B virus, subtype ayr P Pol CUST_P10010180 1356 1400 2577 Hepatitis B virus, subtype ayr X x-protein CUST_P10010181 1388 1432 2578 Hepatitis B virus, subtype ayr X x-protein CUST_P10010182 1419 1463 2579 Hepatitis B virus, subtype ayr X x-protein CUST_P10010183 1472 1516 2580 Hepatitis B virus, subtype ayr X x-protein CUST_P10010185 1565 1609 2581 Hepatitis B virus, subtype ayr X x-protein CUST_P10010187 1643 1690 2582 Hepatitis B virus, subtype ayr X x-protein CUST_P10010188 1675 1725 2583 Hepatitis B virus, subtype ayr X x-protein CUST_P10010189 1808 1856 2584 Hepatitis B virus, subtype ayr X x-protein CUST_P10010190 1941 1996 2585 Hepatitis B virus, subtype ayr C Core CUST_P10010191 2001 2045 2586 Hepatitis B virus, subtype ayr C Core CUST_P10010192 2111 2155 2587 Hepatitis B virus, subtype ayr C Core CUST_P10010193 2210 2266 2588 Hepatitis B virus, subtype ayr C Core CUST_P10010194 2290 2335 2589 Hepatitis B virus, subtype ayr C Core CUST_P10010195 2365 2409 2590 Hepatitis B virus, subtype ayr C Core CUST_P10010196 2410 2456 2591 Hepatitis B virus, subtype ayr C Core CUST_P10010197 2430 2484 2592 Hepatitis B virus, subtype ayr C Core CUST_P10010198 2516 2571 2593 Hepatitis B virus, subtype ayr C Core CUST_P10010199 2577 2636 2594 Hepatitis B virus, subtype ayr C Core CUST_P10010200 2629 2688 2595 Hepatitis B virus, subtype ayr C Core CUST_P10010201 2675 2734 2596 Hepatitis B virus, subtype ayr C Core CUST_P10010202 2702 2761 2597 Hepatitis B virus, subtype ayr C Core CUST_P10010203 2742 2801 2598 Hepatitis B virus, subtype ayr C Core CUST_P10010204 2890 2936 2599 Hepatitis B virus, subtype ayr S surface protein

CUST_P10010205	2955	3001	2600	Hepatitis B virus, subtype ayr S surface protein
CUST_P10010206	2978	3022	2601	Hepatitis B virus, subtype ayr S surface protein
CUST_P10010207	3012	3056	2602	Hepatitis B virus, subtype ayr S surface protein
CUST_P10010208	57	104	2603	Human parvovirus B19 NS non-structural protein NS1
CUST_P10010209	227	271	2604	Human parvovirus B19 NS non-structural protein NS2
CUST_P10010211	637	693	2605	Human parvovirus B19 NS non-structural protein NS1
CUST_P10010212	685	744	2606	Human parvovirus B19 NS non-structural protein NS1
CUST_P10010214	973	1032	2607	Human parvovirus B19 NS non-structural protein NS1
CUST_P10010215	1065	1124	2608	Human parvovirus B19 NS non-structural protein NS1
CUST_P10010216	1165	1210	2609	Human parvovirus B19 NS non-structural protein NS1
CUST_P10010217	1220	1279	2610	Human parvovirus B19 NS non-structural protein NS1
CUST_P10010218	1258	1317	2611	Human parvovirus B19 NS non-structural protein NS1
CUST_P10010219	1357	1416	2612	Human parvovirus B19 NS non-structural protein NS1
CUST_P10010220	1408	1467	2613	Human parvovirus B19 NS non-structural protein NS1
CUST_P10010221	1542	1601	2614	Human parvovirus B19 NS non-structural protein NS1
CUST_P10010222	1628	1687	2615	Human parvovirus B19 NS non-structural protein NS1
CUST_P10010223	1767	1812	2616	Human parvovirus B19 NS non-structural protein NS1
CUST_P10010224	1845	1902	2617	Human parvovirus B19 NS non-structural protein NS1
CUST_P10010225	1911	1970	2618	Human parvovirus B19 NS non-structural protein NS1
CUST_P10010226	2041	2096	2619	Human parvovirus B19 NS non-structural protein NS1
CUST_P10010227	2257	2301	2620	Human parvovirus B19 NS non-structural protein NS1
CUST_P10010228	2351	2399	2621	Human parvovirus B19 NS non-structural protein NS1
CUST_P10010229	2395	2446	2622	Human parvovirus B19 NS non-structural protein NS1
CUST_P10010230	2426	2478	2623	Human parvovirus B19 NS non-structural protein NS1
CUST_P10010231	2552	2607	2624	Human parvovirus B19 NS non-structural protein NS1
CUST_P10010232	2787	2846	2625	Human parvovirus B19 VP1 minor capsid protein
CUST_P10010233	2836	2895	2626	Human parvovirus B19 VP1 minor capsid protein
CUST_P10010234	2868	2915	2627	Human parvovirus B19 VP1 minor capsid protein
CUST_P10010235	2914	2973	2628	Human parvovirus B19 VP1 minor capsid protein
CUST_P10010236	3081	3140	2629	Human parvovirus B19 VP1 minor capsid protein
CUST_P10010237	3252	3297	2630	Human parvovirus B19 VP1 minor capsid protein
CUST_P10010238	3276	3330	2631	Human parvovirus B19 VP1 minor capsid protein
CUST_P10010239	3422	3481	2632	Human parvovirus B19 VP1 minor capsid protein
CUST_P10010240	3524	3583	2633	Human parvovirus B19 VP1 minor capsid protein
CUST_P10010241	3652	3711	2634	Human parvovirus B19 VP1 minor capsid protein
CUST_P10010242	3801	3856	2635	Human parvovirus B19 VP1 minor capsid protein
CUST_P10010243	3826	3885	2636	Human parvovirus B19 VP1 minor capsid protein
CUST_P10010244	3864	3923	2637	Human parvovirus B19 VP1 minor capsid protein
CUST_P10010245	3996	4042	2638	Human parvovirus B19 VP1 minor capsid protein
CUST_P10010246	4097	4156	2639	Human parvovirus B19 VP1 minor capsid protein
CUST_P10010247	4334	4393	2640	Human parvovirus B19 VP1 minor capsid protein
CUST_P10010248	4463	4522	2641	Human parvovirus B19 VP1 minor capsid protein
CUST_P10010249	4587	4646	2642	Human parvovirus B19 VP1 minor capsid protein
CUST_P10010250	4723	4782	2643	Human parvovirus B19 VP1 minor capsid protein
CUST_P10010251	4820	4864	2644	Human parvovirus B19 VP1 minor capsid protein
CUST_P10010252	4915	4960	2645	Human parvovirus B19 VP1 minor capsid protein
CUST_P10010254	5087	5146	2646	Human parvovirus B19 VP2 major capsid protein
CUST_P10010257	5492	5539	2647	Human parvovirus B19 VP2 major capsid protein

(122) TABLE-US-00009 TABLE 9 Exemplary bacterial and protozoan probes SEQ ID Genomic
NO: ProbeID Start End Pathogen Region Product 2648 CUST_P10011833 115100 115153

Treponema pallidum polA Polymerase 2649 CUST_P10011835 115167 115219 *Treponema pallidum* polA Polymerase 2650 CUST_P10011836 115187 115237 *Treponema pallidum* polA Polymerase 2651 CUST_P10011838 115259 115314 *Treponema pallidum* polA Polymerase 2652 CUST_P10011840 115554 115613 *Treponema pallidum* polA Polymerase 2653 CUST_P10011841 115579 115638 *Treponema pallidum* polA Polymerase 2654 CUST_P10011842 115605 115664 *Treponema pallidum* polA Polymerase 2655 CUST_P10011843 115667 115716 *Treponema pallidum* polA Polymerase 2656 CUST_P10011844 115696 115740 *Treponema pallidum* polA Polymerase 2657 CUST_P10011845 115755 115807 *Treponema pallidum* polA Polymerase 2658 CUST_P10011847 116076 116126 *Treponema pallidum* polA Polymerase 2659 CUST_P10011848 116171 116223 *Treponema pallidum* polA Polymerase 2660 CUST_P10011849 116242 116294 *Treponema pallidum* polA Polymerase 2661 CUST_P10011850 116332 116384 *Treponema pallidum* polA Polymerase 2662 CUST_P10011851 116352 116396 *Treponema pallidum* polA Polymerase 2663 CUST_P10011852 116408 116459 *Treponema pallidum* polA Polymerase 2664 CUST_P10011853 116430 116488 *Treponema pallidum* polA Polymerase 2665 CUST_P10011854 116601 116649 *Treponema pallidum* polA Polymerase 2666 CUST_P10011855 116623 116674 *Treponema pallidum* polA Polymerase 2667 CUST_P10011856 116654 116713 *Treponema pallidum* polA Polymerase 2668 CUST_P10011857 116677 116736 *Treponema pallidum* polA Polymerase 2669 CUST_P10011858 116707 116760 *Treponema pallidum* polA Polymerase 2670 CUST_P10011860 116852 116906 *Treponema pallidum* polA Polymerase 2671 CUST_P10011862 116903 116962 *Treponema pallidum* polA Polymerase 2672 CUST_P10011863 116925 116978 *Treponema pallidum* polA Polymerase 2673 CUST_P10011864 116987 117032 *Treponema pallidum* polA Polymerase 2674 CUST_P10011865 117028 117077 *Treponema pallidum* polA Polymerase 2675 CUST_P10011866 117128 117176 *Treponema pallidum* polA Polymerase 2676 CUST_P10011867 117270 117329 *Treponema pallidum* polA Polymerase 2677 CUST_P10011868 117441 117488 *Treponema pallidum* polA Polymerase 2678 CUST_P10011870 117516 117575 *Treponema pallidum* polA Polymerase 2679 CUST_P10011871 117570 117620 *Treponema pallidum* polA Polymerase 2680 CUST_P10011872 117777 117836 *Treponema pallidum* polA Polymerase 2681 CUST_P10011873 117806 117865 *Treponema pallidum* polA Polymerase 2682 CUST_P10011874 117873 117932 *Treponema pallidum* polA Polymerase 2683 CUST_P10011875 118152 118211 *Treponema pallidum* polA Polymerase 2684 CUST_P10011876 118181 118240 *Treponema pallidum* polA Polymerase 2685 CUST_P10011877 118281 118340 *Treponema pallidum* polA Polymerase 2686 CUST_P10011878 118302 118361 *Treponema pallidum* polA Polymerase 2687 CUST_P10011880 118541 118600 *Treponema pallidum* polA Polymerase 2688 CUST_P10011882 118756 118810 *Treponema pallidum* polA Polymerase 2689 CUST_P10018873 622214 622258 *Treponema pallidum* TP0576 protein 2690 CUST_P10018877 622589 622637 *Treponema pallidum* TP0576 protein 2691 CUST_P10018878 622725 622775 *Treponema pallidum* TP0576 protein 2692 CUST_P10018881 622908 622961 *Treponema pallidum* TP0576 protein 2693 CUST_P10018882 622953 622997 *Treponema pallidum* TP0576 protein 2694 CUST_P10018883 622978 623030 *Treponema pallidum* TP0576 protein 2695 CUST_P10018884 623033 623087 *Treponema pallidum* TP0576 protein 2696 CUST_P10018887 623258 623302 *Treponema pallidum* TP0576 protein 2697 CUST_P10024966 1067780 1067830 *Treponema pallidum* aspS aspartate-tRNA ligase 2698 CUST_P10024967 1067910 1067970 *Treponema pallidum* aspS aspartate-tRNA ligase 2699 CUST_P10024968 1067940 1067990 *Treponema pallidum* aspS aspartate-tRNA ligase 2700 CUST_P10024969 1068070 1068120 *Treponema pallidum* aspS aspartate-tRNA ligase 2701 CUST_P10024970 1068180 1068230 *Treponema pallidum* aspS aspartate-tRNA ligase 2702 CUST_P10024971 1068330 1068380 *Treponema pallidum* aspS aspartate-tRNA ligase 2703 CUST_P10024972 1068410 1068470 *Treponema pallidum* aspS aspartate-tRNA ligase 2704 CUST_P10024976 1068840 1068900 *Treponema pallidum* aspS aspartate-tRNA ligase 2705 CUST_P10024981 1069090 1069140 *Treponema pallidum* aspS aspartate-tRNA ligase 2706 CUST_P10024984 1069200 1069260 *Treponema*

pallidum aspS aspartate-tRNA ligase 2707 CUST_P10024986 1069380 1069380 *Treponema pallidum*
pallidum aspS aspartate-tRNA ligase 2708 CUST_P10024991 1069740 1069790 *Treponema pallidum*
pallidum aspS aspartate-tRNA ligase 2709 CUST_P10024994 1069890 1069940 *Treponema pallidum*
pallidum aspS aspartate-tRNA ligase 2710 CUST_P10024996 1070010 1070060 *Treponema pallidum*
pallidum aspS aspartate-tRNA ligase 2711 CUST_P10024997 1070040 1070080 *Treponema pallidum*
pallidum aspS aspartate-tRNA ligase 2712 CUST_P10024998 1070090 1070130 *Treponema pallidum*
pallidum aspS aspartate-tRNA ligase 2713 CUST_P10024999 1070140 1070190 *Treponema pallidum*
pallidum aspS aspartate-tRNA ligase 2714 CUST_P10025000 1070250 1070300 *Treponema pallidum*
pallidum aspS aspartate-tRNA ligase 2715 CUST_P10025003 1070380 1070430 *Treponema pallidum*
pallidum aspS aspartate-tRNA ligase 2716 CUST_P10025005 1070670 1070710 *Treponema pallidum*
pallidum aspS aspartate-tRNA ligase 2717 CUST_P10025006 1070800 1070850 *Treponema pallidum*
pallidum TP0986 protein 2718 CUST_P10025009 1071210 1071260 *Treponema pallidum* TP0986
protein 2719 CUST_P10025010 1071230 1071290 *Treponema pallidum* TP0986 protein 2720
CUST_P10025018 1071990 1072040 *Treponema pallidum* TP0986 protein 2721
CUST_P10025019 1072210 1072260 *Treponema pallidum* TP0989 protein 2722
CUST_P10025024 1072490 1072540 *Treponema pallidum* TP0989 protein 2723
CUST_P10025027 1072660 1072710 *Treponema pallidum* TP0989 protein 2724
CUST_P10025028 1072870 1072910 *Treponema pallidum* TP0989 protein 2725
CUST_P10025029 1072990 1073040 *Treponema pallidum* TP0989 protein 2726
CUST_P10025030 1073010 1073060 *Treponema pallidum* TP0989 protein 2727
CUST_P10025044 1074070 1074120 *Treponema pallidum* TP0990 protein 2728
CUST_P10025045 1074230 1074270 *Treponema pallidum* TP0991 protein 2729
CUST_P10025047 1074310 1074360 *Treponema pallidum* TP0992 protein 2730
CUST_P10025050 1074480 1074520 *Treponema pallidum* TP0993 protein 2731
CUST_P10025051 1074520 1074560 *Treponema pallidum* TP0994 protein 2732
CUST_P10025052 1074560 1074610 *Treponema pallidum* TP0995 protein 2733
CUST_P10025054 1074620 1074670 *Treponema pallidum* TP0996 protein 2734
CUST_P10025055 1074740 1074780 *Treponema pallidum* TP0997 protein 2735
CUST_P10025056 1074790 1074830 *Treponema pallidum* TP0998 protein 2736
CUST_P10025058 1075040 1075090 *Treponema pallidum* TP0999 protein 2737
CUST_P10025059 1075150 1075210 *Treponema pallidum* TP1000 protein 2738
CUST_P10025061 1075310 1075360 *Treponema pallidum* TP1001 protein 2739
CUST_P10025062 1075380 1075430 *Treponema pallidum* TP1002 protein 2740
CUST_P10025063 1075540 1075600 *Treponema pallidum* TP1003 protein 2741
CUST_P10025064 1075600 1075640 *Treponema pallidum* TP1004 protein 2742
CUST_P10025065 1075670 1075720 *Treponema pallidum* TP1005 protein 2743
CUST_P10025067 1075970 1076030 *Treponema pallidum* TP1006 protein 2744
CUST_P10025068 1076040 1076100 *Treponema pallidum* TP1007 protein 2745
CUST_P10025069 1076170 1076220 *Treponema pallidum* TP0990 protein 2746
CUST_P10025073 1076400 1076450 *Treponema pallidum* TP0991 protein 2747
CUST_P10025084 1077230 1077280 *Treponema pallidum* TP0992 protein 2748
CUST_P10025095 1077820 1077870 *Treponema pallidum* TP0992 protein 2749
CUST_P10025097 1077980 1078030 *Treponema pallidum* TP0993 protein 2750
CUST_P10025098 1078120 1078170 *Treponema pallidum* TP0993 protein 2751
CUST_P10025104 1078320 1078370 *Treponema pallidum* TP0993 protein 2752
CUST_P10025934 479 538 *Ehrlichia chaffeensis* ECH_RS00020 protein 2753 CUST_P10026529
69254 69313 *Ehrlichia chaffeensis* argF protein 2754 CUST_P10026957 118031 118090 *Ehrlichia chaffeensis*
ECH_RS00525 protein 2755 CUST_P10027106 133725 133784 *Ehrlichia chaffeensis*
ECH_RS00595 protein 2756 CUST_P10027296 155652 155711 *Ehrlichia chaffeensis*
ECH_RS00695 protein 2757 CUST_P10027314 158001 158060 *Ehrlichia chaffeensis*

ECH_RS00710 protein 2758 CUST_P10027972 235733 235792 *Ehrlichia chaffeensis*
ECH_RS01035 protein 2759 CUST_P10027973 235768 235827 *Ehrlichia chaffeensis*
ECH_RS01035 protein 2760 CUST_P10028360 279872 279931 *Ehrlichia chaffeensis*
ECH_RS01185 protein 2761 CUST_P10028636 313330 313389 *Ehrlichia chaffeensis*
ECH_RS01325 protein 2762 CUST_P10028976 353675 353734 *Ehrlichia chaffeensis* groL
chaperonin GroEL 2763 CUST_P10028977 353852 353911 *Ehrlichia chaffeensis* groL chaperonin
GroEL 2764 CUST_P10028978 354008 354067 *Ehrlichia chaffeensis* groL chaperonin GroEL
2765 CUST_P10028979 354044 354103 *Ehrlichia chaffeensis* groL chaperonin GroEL 2766
CUST_P10028980 354207 354266 *Ehrlichia chaffeensis* groL chaperonin GroEL 2767
CUST_P10028981 354405 354464 *Ehrlichia chaffeensis* groL chaperonin GroEL 2768
CUST_P10028982 354433 354492 *Ehrlichia chaffeensis* groL chaperonin GroEL 2769
CUST_P10028983 354474 354533 *Ehrlichia chaffeensis* groL chaperonin GroEL 2770
CUST_P10028984 354504 354563 *Ehrlichia chaffeensis* groL chaperonin GroEL 2771
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transposase 3023 CUST_P10056628 345483 345529 *Coxiella burnetii* IS1111A transposase
IS1111A transposase 3024 CUST_P10056629 345606 345656 *Coxiella burnetii* IS1111A
transposase IS1111A transposase 3025 CUST_P10056630 345637 345686 *Coxiella burnetii*
IS1111A transposase IS1111A transposase 3026 CUST_P10056631 345756 345802 *Coxiella*
burnetii IS1111A transposase IS1111A transposase 3027 CUST_P10056632 345778 345825
Coxiella burnetii IS1111A transposase IS1111A transposase 3028 CUST_P10056633 345818
345863 *Coxiella burnetii* IS1111A transposase IS1111A transposase 3029 CUST_P10056634
345835 345884 *Coxiella burnetii* IS1111A transposase IS1111A transposase 3030
CUST_P10056635 345870 345922 *Coxiella burnetii* IS1111A transposase IS1111A transposase
3031 CUST_P10056636 345979 346023 *Coxiella burnetii* IS1111A transposase IS1111A
transposase 3032 CUST_P10056637 346031 346082 *Coxiella burnetii* IS1111A transposase
IS1111A transposase 3033 CUST_P10056638 346136 346186 *Coxiella burnetii* IS1111A
transposase IS1111A transposase 3034 CUST_P10058309 465895 465941 *Coxiella burnetii*
IS1111A transposase IS1111A transposase 3035 CUST_P10058310 465926 465970 *Coxiella*
burnetii IS1111A transposase IS1111A transposase 3036 CUST_P10058311 466027 466080
Coxiella burnetii IS1111A transposase IS1111A transposase 3037 CUST_P10058312 466064
466113 *Coxiella burnetii* IS1111A transposase IS1111A transposase 3038 CUST_P10058313
466123 466170 *Coxiella burnetii* IS1111A transposase IS1111A transposase 3039
CUST_P10058314 466146 466192 *Coxiella burnetii* IS1111A transposase IS1111A transposase
3040 CUST_P10058315 466263 466311 *Coxiella burnetii* IS1111A transposase IS1111A
transposase 3041 CUST_P10058316 466301 466354 *Coxiella burnetii* IS1111A transposase
IS1111A transposase 3042 CUST_P10058317 466422 466466 *Coxiella burnetii* IS1111A
transposase IS1111A transposase 3043 CUST_P10058318 466457 466504 *Coxiella burnetii*
IS1111A transposase IS1111A transposase 3044 CUST_P10058319 466479 466530 *Coxiella*
burnetii IS1111A transposase IS1111A transposase 3045 CUST_P10058320 466511 466555
Coxiella burnetii IS1111A transposase IS1111A transposase 3046 CUST_P10058321 466595
466648 *Coxiella burnetii* IS1111A transposase IS1111A transposase 3047 CUST_P10058322
466644 466688 *Coxiella burnetii* IS1111A transposase IS1111A transposase 3048
CUST_P10058323 466684 466728 *Coxiella burnetii* IS1111A transposase IS1111A transposase
3049 CUST_P10058324 466749 466798 *Coxiella burnetii* IS1111A transposase IS1111A
transposase 3050 CUST_P10058325 466778 466835 *Coxiella burnetii* IS1111A transposase
IS1111A transposase 3051 CUST_P10058326 466943 466999 *Coxiella burnetii* IS1111A
transposase IS1111A transposase 3052 CUST_P10058816 502390 502443 *Coxiella burnetii*
IS1111A transposase IS1111A transposase 3053 CUST_P10058817 502499 502548 *Coxiella*
burnetii IS1111A transposase IS1111A transposase 3054 CUST_P10058818 502510 502563

Coxiella burnetii IS1111A transposase IS1111A transposase 3055 CUST_P10058819 502573
502619 *Coxiella burnetii* IS1111A transposase IS1111A transposase 3056 CUST_P10058820
502696 502746 *Coxiella burnetii* IS1111A transposase IS1111A transposase 3057
CUST_P10058821 502727 502776 *Coxiella burnetii* IS1111A transposase IS1111A transposase
3058 CUST_P10058822 502846 502892 *Coxiella burnetii* IS1111A transposase IS1111A
transposase 3059 CUST_P10058823 502868 502915 *Coxiella burnetii* IS1111A transposase
IS1111A transposase 3060 CUST_P10058824 502908 502953 *Coxiella burnetii* IS1111A
transposase IS1111A transposase 3061 CUST_P10058825 502925 502974 *Coxiella burnetii*
IS1111A transposase IS1111A transposase 3062 CUST_P10058826 502960 503012 *Coxiella*
burnetii IS1111A transposase IS1111A transposase 3063 CUST_P10058827 503069 503113
Coxiella burnetii IS1111A transposase IS1111A transposase 3064 CUST_P10058828 503121
503172 *Coxiella burnetii* IS1111A transposase IS1111A transposase 3065 CUST_P10058829
503226 503276 *Coxiella burnetii* IS1111A transposase IS1111A transposase 3066
CUST_P10058830 503308 503358 *Coxiella burnetii* IS1111A transposase IS1111A transposase
3067 CUST_P10058831 503329 503384 *Coxiella burnetii* IS1111A transposase IS1111A
transposase 3068 CUST_P10058832 503374 503433 *Coxiella burnetii* IS1111A transposase
IS1111A transposase 3069 CUST_P10058833 503541 503597 *Coxiella burnetii* IS1111A
transposase IS1111A transposase 3070 CUST_P10058834 503609 503658 *Coxiella burnetii*
IS1111A transposase IS1111A transposase 3071 CUST_P10058835 503681 503725 *Coxiella*
burnetii IS1111A transposase IS1111A transposase 3072 CUST_P10058836 503718 503765
Coxiella burnetii IS1111A transposase IS1111A transposase 3073 CUST_P10058837 503745
503791 *Coxiella burnetii* IS1111A transposase IS1111A transposase 3074 CUST_P10058838
503852 503911 *Coxiella burnetii* IS1111A transposase IS1111A transposase 3075
CUST_P10065644 1021442 1021488 *Coxiella burnetii* IS1111A transposase IS1111A transposase
3076 CUST_P10065645 1021559 1021607 *Coxiella burnetii* IS1111A transposase IS1111A
transposase 3077 CUST_P10065646 1021597 1021650 *Coxiella burnetii* IS1111A transposase
IS1111A transposase 3078 CUST_P10065647 1021718 1021762 *Coxiella burnetii* IS1111A
transposase IS1111A transposase 3079 CUST_P10065648 1021753 1021800 *Coxiella burnetii*
IS1111A transposase IS1111A transposase 3080 CUST_P10065649 1021775 1021826 *Coxiella*
burnetii IS1111A transposase IS1111A transposase 3081 CUST_P10065650 1021807 1021851
Coxiella burnetii IS1111A transposase IS1111A transposase 3082 CUST_P10065651 1021891
1021944 *Coxiella burnetii* IS1111A transposase IS1111A transposase 3083 CUST_P10065652
1021940 1021984 *Coxiella burnetii* IS1111A transposase IS1111A transposase 3084
CUST_P10065653 1021980 1022024 *Coxiella burnetii* IS1111A transposase IS1111A transposase
3085 CUST_P10065654 1022045 1022094 *Coxiella burnetii* IS1111A transposase IS1111A
transposase 3086 CUST_P10079271 1019 1078 *Trypanosoma brucei* kinetoplast apocy kinetoplast
apocy 3087 CUST_P10079269 471 530 *Trypanosoma brucei* kinetoplast apocy kinetoplast apocy
3088 CUST_P10079274 1363 1414 *Trypanosoma brucei* kinetoplast apocy kinetoplast apocy 3089
CUST_P10079272 1279 1334 *Trypanosoma brucei* kinetoplast apocy kinetoplast apocy 3090
CUST_P10079270 628 687 *Trypanosoma brucei* kinetoplast apocy kinetoplast apocy 3091
CUST_P10079273 1329 1373 *Trypanosoma brucei* kinetoplast apocy kinetoplast apocy 3092
CUST_P10079280 687 746 *Trypanosoma brucei* kinetoplast DNA m kinetoplast DNA m 3093
CUST_P10079276 478 526 *Trypanosoma brucei* kinetoplast DNA m kinetoplast DNA m 3094
CUST_P10079277 547 601 *Trypanosoma brucei* kinetoplast DNA m kinetoplast DNA m 3095
CUST_P10079275 1 60 *Trypanosoma brucei* kinetoplast DNA m kinetoplast DNA m 3096
CUST_P10079279 677 736 *Trypanosoma brucei* kinetoplast DNA m kinetoplast DNA m 3097
CUST_P10079278 554 610 *Trypanosoma brucei* kinetoplast DNA m kinetoplast DNA m 3098
CUST_P10079281 23 67 *Trypanosoma Cruzi* Mini satellite Mini satellite 3099 CUST_P10079284
267 312 *Leishmania major* kinetoplast DNA kinetoplast DNA 3100 CUST_P10079285 319 365
Leishmania major kinetoplast DNA kinetoplast DNA 3101 CUST_P10079286 360 404 *Leishmania*

major kinetoplast DNA kinetoplast DNA 3102 CUST_P10079283 146 205 *Leishmania major* kinetoplast DNA kinetoplast DNA 3103 CUST_P10079287 382 426 *Leishmania major* kinetoplast DNA kinetoplast DNA 3104 CUST_P10079282 1 60 *Leishmania major* kinetoplast DNA kinetoplast DNA 3105 CUST_P10079294 504 548 *Leishmania major* kinetoplast DNA kinetoplast DNA 3106 CUST_P10079292 333 377 *Leishmania major* kinetoplast DNA kinetoplast DNA 3107 CUST_P10079296 588 632 *Leishmania major* kinetoplast DNA kinetoplast DNA 3108 CUST_P10079291 298 342 *Leishmania major* kinetoplast DNA kinetoplast DNA 3109 CUST_P10079290 257 303 *Leishmania major* kinetoplast DNA kinetoplast DNA 3110 CUST_P10079295 551 595 *Leishmania major* kinetoplast DNA kinetoplast DNA 3111 CUST_P10079289 148 200 *Leishmania major* kinetoplast DNA kinetoplast DNA 3112 CUST_P10079288 116 168 *Leishmania major* kinetoplast DNA kinetoplast DNA 3113 CUST_P10079293 462 507 *Leishmania major* kinetoplast DNA kinetoplast DNA 3114 CUST_P10079326 82 141 *Babesia microti* 18S ribosomal RNA 18S ribosomal RNA 3115 CUST_P10079327 140 189 *Babesia microti* 18S ribosomal RNA 18S ribosomal RNA 3116 CUST_P10079328 168 227 *Babesia microti* 18S ribosomal RNA 18S ribosomal RNA 3117 CUST_P10079329 320 364 *Babesia microti* 18S ribosomal RNA 18S ribosomal RNA 3118 CUST_P10079330 336 380 *Babesia microti* 18S ribosomal RNA 18S ribosomal RNA 3119 CUST_P10079331 368 412 *Babesia microti* 18S ribosomal RNA 18S ribosomal RNA 3120 CUST_P10079332 398 450 *Babesia microti* 18S ribosomal RNA 18S ribosomal RNA 3121 CUST_P10079333 434 493 *Babesia microti* 18S ribosomal RNA 18S ribosomal RNA 3122 CUST_P10079334 699 758 *Babesia microti* 18S ribosomal RNA 18S ribosomal RNA 3123 CUST_P10079335 722 781 *Babesia microti* 18S ribosomal RNA 18S ribosomal RNA 3124 CUST_P10079336 771 825 *Babesia microti* 18S ribosomal RNA 18S ribosomal RNA 3125 CUST_P10079337 801 860 *Babesia microti* 18S ribosomal RNA 18S ribosomal RNA 3126 CUST_P10079338 826 885 *Babesia microti* 18S ribosomal RNA 18S ribosomal RNA 3127 CUST_P10079339 891 947 *Babesia microti* 18S ribosomal RNA 18S ribosomal RNA 3128 CUST_P10079340 933 987 *Babesia microti* 18S ribosomal RNA 18S ribosomal RNA 3129 CUST_P10079341 952 1006 *Babesia microti* 18S ribosomal RNA 18S ribosomal RNA 3130 CUST_P10079342 990 1042 *Babesia microti* 18S ribosomal RNA 18S ribosomal RNA 3131 CUST_P10079343 1016 1060 *Babesia microti* 18S ribosomal RNA 18S ribosomal RNA 3132 CUST_P10079344 1055 1101 *Babesia microti* 18S ribosomal RNA 18S ribosomal RNA 3133 CUST_P10079345 1108 1152 *Babesia microti* 18S ribosomal RNA 18S ribosomal RNA 3134 CUST_P10079346 1239 1298 *Babesia microti* 18S ribosomal RNA 18S ribosomal RNA 3135 CUST_P10079347 1357 1405 *Babesia microti* 18S ribosomal RNA 18S ribosomal RNA 3136 CUST_P10079348 1435 1479 *Babesia microti* 18S ribosomal RNA 18S ribosomal RNA 3137 CUST_P10079349 1458 1517 *Babesia microti* 18S ribosomal RNA 18S ribosomal RNA 3138 CUST_P10079350 1620 1675 *Babesia microti* 18S ribosomal RNA 18S ribosomal RNA 3139 CUST_P10079351 1659 1718 *Babesia microti* 18S ribosomal RNA 18S ribosomal RNA 3140 CUST_P10079352 1736 1786 *Babesia microti* 18S ribosomal RNA 18S ribosomal RNA 3141 CUST_P10079353 1758 1802 *Babesia microti* 18S ribosomal RNA 18S ribosomal RNA 3142 CUST_P10079354 1811 1855 *Babesia microti* 18S ribosomal RNA 18S ribosomal RNA 3143 CUST_P10079355 1845 1889 *Babesia microti* 18S ribosomal RNA 18S ribosomal RNA 3144 CUST_P10079356 1894 1938 *Babesia microti* 18S ribosomal RNA 18S ribosomal RNA 3145 CUST_P10079357 1914 1962 *Babesia microti* 18S ribosomal RNA 18S ribosomal RNA 3146 CUST_P10079358 1947 1997 *Babesia microti* 18S ribosomal RNA 18S ribosomal RNA 3147 CUST_P10079359 1975 2019 *Babesia microti* 18S ribosomal RNA 18S ribosomal RNA 3148 CUST_P10079360 2034 2093 *Babesia microti* 18S ribosomal RNA 18S ribosomal RNA 3149 CUST_P10079361 2056 2115 *Babesia microti* 18S ribosomal RNA 18S ribosomal RNA 3150 CUST_P10079362 2125 2174 *Babesia microti* 18S ribosomal RNA 18S ribosomal RNA 3151 CUST_P10079363 2266 2322 *Babesia microti* 18S ribosomal RNA 18S ribosomal RNA 3152

CUST_P10079364 2342 2392 *Babesia microti* 18S ribosomal RNA 18S ribosomal RNA 3153
CUST_P10079365 2363 2418 *Babesia microti* 18S ribosomal RNA 18S ribosomal RNA 3154
CUST_P10079366 2389 2446 *Babesia microti* 18S ribosomal RNA 18S ribosomal RNA 3155
CUST_P10079478 328 375 *Plasmodium falciparum* 18S ribosomal RNA 18S ribosomal RNA
3156 CUST_P10079479 433 492 *Plasmodium falciparum* 18S ribosomal RNA 18S ribosomal
RNA 3157 CUST_P10079480 517 566 *Plasmodium falciparum* 18S ribosomal RNA 18S
ribosomal RNA 3158 CUST_P10079481 612 671 *Plasmodium falciparum* 18S ribosomal RNA
18S ribosomal RNA 3159 CUST_P10079482 820 879 *Plasmodium falciparum* 18S ribosomal
RNA 18S ribosomal RNA 3160 CUST_P10079483 945 1003 *Plasmodium falciparum* 18S
ribosomal RNA 18S ribosomal RNA 3161 CUST_P10079484 971 1030 *Plasmodium falciparum*
18S ribosomal RNA 18S ribosomal RNA 3162 CUST_P10079485 1018 1077 *Plasmodium
falciparum* 18S ribosomal RNA 18S ribosomal RNA 3163 CUST_P10079486 1211 1257
Plasmodium falciparum 18S ribosomal RNA 18S ribosomal RNA 3164 CUST_P10079487 1284
1343 *Plasmodium falciparum* 18S ribosomal RNA 18S ribosomal RNA 3165 CUST_P10079488
1386 1445 *Plasmodium falciparum* 18S ribosomal RNA 18S ribosomal RNA 3166
CUST_P10079489 1673 1727 *Plasmodium falciparum* 18S ribosomal RNA 18S ribosomal RNA
3167 CUST_P10079490 1817 1876 *Plasmodium falciparum* 18S ribosomal RNA 18S ribosomal
RNA 3168 CUST_P10079491 1852 1911 *Plasmodium falciparum* 18S ribosomal RNA 18S
ribosomal RNA 3169 CUST_P10079492 70 129 *Plasmodium falciparum* 18S ribosomal RNA 18S
ribosomal RNA 3170 CUST_P10079493 321 373 *Plasmodium falciparum* 18S ribosomal RNA
18S ribosomal RNA 3171 CUST_P10079494 460 519 *Plasmodium falciparum* 18S ribosomal
RNA 18S ribosomal RNA 3172 CUST_P10079495 489 548 *Plasmodium falciparum* 18S
ribosomal RNA 18S ribosomal RNA 3173 CUST_P10079496 519 569 *Plasmodium falciparum*
18S ribosomal RNA 18S ribosomal RNA 3174 CUST_P10079497 599 658 *Plasmodium
falciparum* 18S ribosomal RNA 18S ribosomal RNA 3175 CUST_P10079498 800 859
Plasmodium falciparum 18S ribosomal RNA 18S ribosomal RNA 3176 CUST_P10079499 826
885 *Plasmodium falciparum* 18S ribosomal RNA 18S ribosomal RNA 3177 CUST_P10079500
990 1046 *Plasmodium falciparum* 18S ribosomal RNA 18S ribosomal RNA 3178
CUST_P10079501 1014 1073 *Plasmodium falciparum* 18S ribosomal RNA 18S ribosomal RNA
3179 CUST_P10079502 1053 1112 *Plasmodium falciparum* 18S ribosomal RNA 18S ribosomal
RNA 3180 CUST_P10079503 1264 1310 *Plasmodium falciparum* 18S ribosomal RNA 18S
ribosomal RNA 3181 CUST_P10079504 1332 1391 *Plasmodium falciparum* 18S ribosomal RNA
18S ribosomal RNA 3182 CUST_P10079505 1450 1509 *Plasmodium falciparum* 18S ribosomal
RNA 18S ribosomal RNA 3183 CUST_P10079506 1721 1770 *Plasmodium falciparum* 18S
ribosomal RNA 18S ribosomal RNA 3184 CUST_P10079507 1742 1801 *Plasmodium falciparum*
18S ribosomal RNA 18S ribosomal RNA 3185 CUST_P10079508 1890 1949 *Plasmodium
falciparum* 18S ribosomal RNA 18S ribosomal RNA 3186 CUST_P10079572 2 61 *Plasmodium
vivax* SSU rRNA external transcribed spacer 3187 CUST_P10079573 38 89 *Plasmodium vivax*
SSU rRNA external transcribed spacer 3188 CUST_P10079574 68 127 *Plasmodium vivax* SSU
rRNA external transcribed spacer 3189 CUST_P10079575 112 171 *Plasmodium vivax* SSU rRNA
external transcribed spacer 3190 CUST_P10079576 249 308 *Plasmodium vivax* SSU rRNA
external transcribed spacer 3191 CUST_P10079577 279 338 *Plasmodium vivax* SSU rRNA
external transcribed spacer 3192 CUST_P10079578 319 372 *Plasmodium vivax* SSU rRNA
external transcribed spacer 3193 CUST_P10079579 421 480 *Plasmodium vivax* SSU rRNA
external transcribed spacer 3194 CUST_P10079580 590 645 *Plasmodium vivax* SSU rRNA
external transcribed spacer 3195 CUST_P10079581 671 719 *Plasmodium vivax* SSU rRNA
external transcribed spacer 3196 CUST_P10079582 783 842 *Plasmodium vivax* SSU rRNA
external transcribed spacer 3197 CUST_P10079583 803 862 *Plasmodium vivax* SSU rRNA
external transcribed spacer 3198 CUST_P10079584 829 888 *Plasmodium vivax* SSU rRNA
external transcribed spacer 3199 CUST_P10079585 849 899 *Plasmodium vivax* SSU rRNA

external transcribed spacer 3200 CUST_P10079586 946 1005 *Plasmodium vivax* SSU rRNA
external transcribed spacer 3201 CUST_P10079587 1153 1212 *Plasmodium vivax* SSU rRNA
external transcribed spacer 3202 CUST_P10079588 1314 1373 *Plasmodium vivax* SSU rRNA
external transcribed spacer 3203 CUST_P10079589 1346 1405 *Plasmodium vivax* SSU rRNA
external transcribed spacer 3204 CUST_P10079590 1545 1591 *Plasmodium vivax* SSU rRNA
external transcribed spacer 3205 CUST_P10079591 1610 1669 *Plasmodium vivax* SSU rRNA
external transcribed spacer 3206 CUST_P10079592 1717 1776 *Plasmodium vivax* SSU rRNA
external transcribed spacer 3207 CUST_P10079593 1757 1816 *Plasmodium vivax* SSU rRNA
external transcribed spacer

(123) TABLE-US-00010 TABLE 10 Exemplary control probes SEQ ID Genomic NO: ProbeID
Start End Type Region 3208 CUST_P10079594 1561 1605 Housekeeping Gene ACTB 3209
CUST_P10079595 1703 1750 Housekeeping Gene ACTB 3210 CUST_P10079596 2220 2264
Housekeeping Gene ACTB 3211 CUST_P10079597 2242 2286 Housekeeping Gene ACTB 3212
CUST_P10079598 2276 2320 Housekeeping Gene ACTB 3213 CUST_P10079599 2402 2446
Housekeeping Gene ACTB 3214 CUST_P10079600 2489 2533 Housekeeping Gene ACTB 3215
CUST_P10079601 2659 2703 Housekeeping Gene ACTB 3216 CUST_P10079602 2696 2740
Housekeeping Gene ACTB 3217 CUST_P10079603 2823 2867 Housekeeping Gene ACTB 3218
CUST_P10079604 2847 2891 Housekeeping Gene ACTB 3219 CUST_P10079605 2874 2918
Housekeeping Gene ACTB 3220 CUST_P10079606 3005 3049 Housekeeping Gene ACTB 3221
CUST_P10079607 3046 3090 Housekeeping Gene ACTB 3222 CUST_P10079608 3213 3257
Housekeeping Gene ACTB 3223 CUST_P10079609 3338 3382 Housekeeping Gene ACTB 3224
CUST_P10079610 3376 3420 Housekeeping Gene ACTB 3225 CUST_P10079611 3393 3437
Housekeeping Gene ACTB 3226 CUST_P10079612 3438 3482 Housekeeping Gene ACTB 3227
CUST_P10079613 3545 3593 Housekeeping Gene ACTB 3228 CUST_P10079614 3568 3622
Housekeeping Gene ACTB 3229 CUST_P10079615 3601 3645 Housekeeping Gene ACTB 3230
CUST_P10079616 3744 3788 Housekeeping Gene ACTB 3231 CUST_P10079617 3858 3902
Housekeeping Gene ACTB 3232 CUST_P10079618 3973 4017 Housekeeping Gene ACTB 3233
CUST_P10079619 4130 4177 Housekeeping Gene ACTB 3234 CUST_P10079620 4223 4267
Housekeeping Gene ACTB 3235 CUST_P10079621 4280 4324 Housekeeping Gene ACTB 3236
CUST_P10079622 4304 4348 Housekeeping Gene ACTB 3237 CUST_P10079623 4315 4359
Housekeeping Gene ACTB 3238 CUST_P10079624 4459 4503 Housekeeping Gene ACTB 3239
CUST_P10079625 4592 4636 Housekeeping Gene ACTB 3240 CUST_P10079626 4906 4950
Housekeeping Gene ACTB 3241 CUST_P10079627 4932 4978 Housekeeping Gene ACTB 3242
CUST_P10079628 4972 5016 Housekeeping Gene ACTB 3243 CUST_P10079629 5024 5068
Housekeeping Gene ACTB 3244 CUST_P10079630 5040 5084 Housekeeping Gene ACTB 3245
CUST_P10079631 5076 5120 Housekeeping Gene ACTB 3246 CUST_P10079632 5110 5154
Housekeeping Gene ACTB 3247 CUST_P10079633 5200 5244 Housekeeping Gene ACTB 3248
CUST_P10079634 5357 5401 Housekeeping Gene ACTB 3249 CUST_P10079635 5390 5434
Housekeeping Gene ACTB 3250 CUST_P10079636 5415 5459 Housekeeping Gene ACTB 3251
CUST_P10079637 5453 5497 Housekeeping Gene ACTB 3252 CUST_P10079638 5474 5518
Housekeeping Gene ACTB 3253 CUST_P10079639 5622 5666 Housekeeping Gene ACTB 3254
CUST_P10079640 5662 5706 Housekeeping Gene ACTB 3255 CUST_P10079641 5691 5736
Housekeeping Gene ACTB 3256 CUST_P10079642 5712 5756 Housekeeping Gene ACTB 3257
CUST_P10079643 5760 5804 Housekeeping Gene ACTB 3258 CUST_P10079644 5783 5827
Housekeeping Gene ACTB 3259 CUST_P10079645 5817 5861 Housekeeping Gene ACTB 3260
CUST_P10079646 5963 6007 Housekeeping Gene ACTB 3261 CUST_P10079647 6004 6048
Housekeeping Gene ACTB 3262 CUST_P10079648 6106 6150 Housekeeping Gene ACTB 3263
CUST_P10079649 6310 6354 Housekeeping Gene ACTB 3264 CUST_P10079650 6421 6465
Housekeeping Gene ACTB 3265 CUST_P10079651 6507 6553 Housekeeping Gene ACTB 3266
CUST_P10079652 6696 6740 Housekeeping Gene ACTB 3267 CUST_P10079653 6722 6769

Housekeeping Gene ACTB 3268 CUST_P10079654 6745 6789 Housekeeping Gene ACTB 3269
CUST_P10079655 6772 6816 Housekeeping Gene ACTB 3270 CUST_P10079656 6793 6837
Housekeeping Gene ACTB 3271 CUST_P10079657 6844 6888 Housekeeping Gene ACTB 3272
CUST_P10079658 7000 7044 Housekeeping Gene ACTB 3273 CUST_P10079659 7321 7365
Housekeeping Gene ACTB 3274 CUST_P10079660 7418 7462 Housekeeping Gene ACTB 3275
CUST_P10079661 7554 7598 Housekeeping Gene ACTB 3276 CUST_P10079662 7683 7727
Housekeeping Gene ACTB 3277 CUST_P10079663 7777 7821 Housekeeping Gene ACTB 3278
CUST_P10079664 7825 7869 Housekeeping Gene ACTB 3279 CUST_P10079665 7861 7917
Housekeeping Gene ACTB 3280 CUST_P10079666 8033 8077 Housekeeping Gene ACTB 3281
CUST_P10079667 8138 8182 Housekeeping Gene ACTB 3282 CUST_P10079668 8344 8388
Housekeeping Gene ACTB 3283 CUST_P10079669 8459 8503 Housekeeping Gene ACTB 3284
CUST_P10079670 8519 8563 Housekeeping Gene ACTB 3285 CUST_P10079671 8543 8587
Housekeeping Gene ACTB 3286 CUST_P10079672 8574 8618 Housekeeping Gene ACTB 3287
CUST_P10079673 8628 8672 Housekeeping Gene ACTB 3288 CUST_P10079674 8732 8776
Housekeeping Gene ACTB 3289 CUST_P10079675 8941 8985 Housekeeping Gene ACTB 3290
CUST_P10079676 8962 9006 Housekeeping Gene ACTB 3291 CUST_P10079677 8987 9031
Housekeeping Gene ACTB 3292 CUST_P10079678 9103 9147 Housekeeping Gene ACTB 3293
CUST_P10079679 9155 9199 Housekeeping Gene ACTB 3294 CUST_P10079680 9429 9474
Housekeeping Gene ACTB 3295 CUST_P10079681 9554 9600 Housekeeping Gene ACTB 3296
CUST_P10079682 9739 9785 Housekeeping Gene ACTB 3297 CUST_P10079683 9774 9820
Housekeeping Gene ACTB 3298 CUST_P10079684 9794 9838 Housekeeping Gene ACTB 3299
CUST_P10079685 10015 10062 Housekeeping Gene ACTB 3300 CUST_P10079686 10064 10108
Housekeeping Gene ACTB 3301 CUST_P10079687 10193 10245 Housekeeping Gene ACTB
3302 CUST_P10079715 3325 3369 Housekeeping Gene ARL1 3303 CUST_P10079713 3255
3314 Housekeeping Gene ARL1 3304 CUST_P10079727 5957 6016 Housekeeping Gene ARL1
3305 CUST_P10079741 10779 10838 Housekeeping Gene ARL1 3306 CUST_P10079759 13558
13604 Housekeeping Gene ARL1 3307 CUST_P10079701 934 979 Housekeeping Gene ARL1
3308 CUST_P10079692 526 570 Housekeeping Gene ARL1 3309 CUST_P10079750 11884 11939
Housekeeping Gene ARL1 3310 CUST_P10079730 6728 6787 Housekeeping Gene ARL1 3311
CUST_P10079746 11349 11397 Housekeeping Gene ARL1 3312 CUST_P10079744 11307 11354
Housekeeping Gene ARL1 3313 CUST_P10079732 7235 7286 Housekeeping Gene ARL1 3314
CUST_P10079733 7270 7314 Housekeeping Gene ARL1 3315 CUST_P10079760 13597 13641
Housekeeping Gene ARL1 3316 CUST_P10079762 13648 13693 Housekeeping Gene ARL1 3317
CUST_P10079758 13449 13508 Housekeeping Gene ARL1 3318 CUST_P10079728 6518 6577
Housekeeping Gene ARL1 3319 CUST_P10079738 9891 9950 Housekeeping Gene ARL1 3320
CUST_P10079742 10939 10998 Housekeeping Gene ARL1 3321 CUST_P10079696 746 796
Housekeeping Gene ARL1 3322 CUST_P10079703 1025 1079 Housekeeping Gene ARL1 3323
CUST_P10079688 137 181 Housekeeping Gene ARL1 3324 CUST_P10079755 12703 12762
Housekeeping Gene ARL1 3325 CUST_P10079748 11728 11787 Housekeeping Gene ARL1 3326
CUST_P10079722 5055 5114 Housekeeping Gene ARL1 3327 CUST_P10079724 5213 5268
Housekeeping Gene ARL1 3328 CUST_P10079736 8842 8901 Housekeeping Gene ARL1 3329
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Housekeeping Gene ARL1 3334 CUST_P10079753 12428 12487 Housekeeping Gene ARL1 3335
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CUST_P10079745 11327 11379 Housekeeping Gene ARL1 3342 CUST_P10079712 3205 3264

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12412 Housekeeping Gene CCDN1 3519 CUST_P10079905 12402 12446 Housekeeping Gene
CCDN1 3520 CUST_P10081361 16 63 Negative Control *Aedes albopictus* densovirus 2 3521
CUST_P10081362 47 101 Negative Control *Aedes albopictus* densovirus 2 3522
CUST_P10081363 79 131 Negative Control *Aedes albopictus* densovirus 2 3523
CUST_P10081364 204 248 Negative Control *Aedes albopictus* densovirus 2 3524
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CUST_P10081366 330 385 Negative Control *Aedes albopictus* densovirus 2 3526
CUST_P10081367 390 435 Negative Control *Aedes albopictus* densovirus 2 3527
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CUST_P10081384 1836 1893 Negative Control *Aedes albopictus* densovirus 2 3544
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CUST_P10081396 2766 2825 Negative Control *Aedes albopictus* densovirus 2 3556
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CUST_P10081399 411 457 Negative Control Maize streak virus 3559 CUST_P10081400 572 630
Negative Control Maize streak virus 3560 CUST_P10081401 727 786 Negative Control Maize
streak virus 3561 CUST_P10081402 760 819 Negative Control Maize streak virus 3562
CUST_P10081403 799 852 Negative Control Maize streak virus 3563 CUST_P10081404 839 891
Negative Control Maize streak virus 3564 CUST_P10081405 871 919 Negative Control Maize
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Maize streak virus 3593 CUST_P10081434 3586 3645 Negative Control Maize streak virus 3594
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3923 Negative Control Maize streak virus 3596 CUST_P10081437 3967 4011 Negative Control
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Negative Control Tomato pseudo- curly top virus 3600 CUST_P10081441 341 387 Negative
Control Tomato pseudo- curly top virus 3601 CUST_P10081442 436 480 Negative Control Tomato
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top virus 3603 CUST_P10081444 490 534 Negative Control Tomato pseudo- curly top virus 3604
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CUST_P10081453 1166 1214 Negative Control Tomato pseudo- curly top virus 3611

CUST_P10081454 1187 1246 Negative Control Tomato pseudo- curly top virus 3612
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 CUST_P10081461 1615 1669 Negative Control Tomato pseudo- curly top virus 3619
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 CUST_P10081464 2072 2128 Negative Control Tomato pseudo- curly top virus 3622
 CUST_P10081465 2114 2173 Negative Control Tomato pseudo- curly top virus 3623
 CUST_P10081466 2158 2217 Negative Control Tomato pseudo- curly top virus 3624
 CUST_P10081467 2296 2346 Negative Control Tomato pseudo- curly top virus 3625
 CUST_P10081468 2446 2490 Negative Control Tomato pseudo- curly top virus 3626
 CUST_P10081470 2548 2592 Negative Control Tomato pseudo- curly top virus 3627
 CUST_P10081471 2596 2650 Negative Control Tomato pseudo- curly top virus 3628
 CUST_P10081472 2633 2683 Negative Control Tomato pseudo- curly top virus

(124) In view of the many possible embodiments to which the principles of the disclosure may be applied, it should be recognized that the illustrated embodiments are only examples and should not be taken as limiting the scope of the invention. Rather, the scope of the invention is defined by the following claims. We therefore claim as our invention all that comes within the scope and spirit of these claims.

Claims

1. A probe set comprising: (a) probes having at least 90% identity with the nucleic acid sequences of SEQ ID NOs: 1-1300, 1391-1570, and 1691-1769; (b) probes having at least 95% identity with the nucleic acid sequences of SEQ ID NOs: 1-1300, 1391-1570, and 1691-1769; (c) probes comprising the nucleic acid sequence of SEQ ID NOs: 1-1300, 1391-1570, and 1691-1769; or (d) probes comprising the nucleic acid sequence of SEQ ID NOs: 1-1769, wherein each of the probes is covalently linked to a solid support and each of the probes is 60 nucleotides in length.
2. The probe set of claim 1, wherein the probe set comprises probes for each of Chikungunya virus, Dengue virus type 1, Dengue virus type 2, Dengue virus type 3, Dengue virus type 4, Hepatitis A virus, Hepatitis C virus type 1, Hepatitis C virus type 2, Hepatitis C virus type 3, Hepatitis E virus, Human immunodeficiency virus type 1, Human immunodeficiency virus type 2, Human T-lymphotropic virus type I, Human T-lymphotropic virus type II, West Nile virus, and Zika virus.
3. The probe set of claim 1, further comprising at least one negative control probe and/or further comprising at least one positive control probe.
4. The probe set of claim 3, wherein the at least one negative control probe comprises a set of probes with at least 90% identity with each of the nucleic acid sequences of SEQ ID NOs: 1571-1690 and wherein each of the probes is 60 nucleotides in length.
5. The probe set of claim 1, further comprising: (a) probes having at least 90% identity with the nucleic acid sequences of SEQ ID NOs: 1770-2647; (b) probes having at least 95% identity with the nucleic acid sequences of SEQ ID NOS: 1770-2647; or (c) probes comprising the nucleic acid sequence of SEQ ID NOs: 1770-2647; wherein each of the probes is covalently linked to a solid support and each of the probes is 40-60 nucleotides in length.
6. The probe set of claim 5, wherein the probe set comprises probes for each of cytomegalovirus, Epstein Barr virus subtype B95-8, Epstein Barr virus subtype AG876, human herpes virus 8, Hepatitis B virus subtype adw, Hepatitis B virus subtype ayw, Hepatitis B virus subtype adr,

- Hepatitis B virus subtype ayr, human parvovirus B19, human papillomavirus type 6, human papillomavirus type 11, human papillomavirus type 16, and human papillomavirus type 18.
7. The probe set of claim 5, further comprising at least one negative control probe and/or further comprising at least one positive control probe.
8. The probe set of claim 7, wherein the at least one negative control probe comprises a set of probes with at least 90% identity with the nucleic acid sequences of each of SEQ ID NOs: 3520-3628 and each of the probes is 45-60 nucleotides in length.
9. The probe set of claim 1, further comprising: (a) probes having at least 90% identity with the nucleic acid sequences of SEQ ID NOs: 2648-3207; (b) probes having at least 95% identity with the nucleic acid sequences of SEQ ID NOs: 2648-3207; or (c) probes comprising the nucleic acid sequence of SEQ ID NOs: 2648-3207; wherein each of the probes is covalently linked to a solid support and each of the probes is 45-60 nucleotides in length.
10. The probe set of claim 9, wherein the probe set comprises at least one probe for each of *Treponema pallidum*, *Ehrlichia chaffeensis*, *Ehrlichia ewingii*, *Ehrlichia muris*, *Borrelia burgdorferi*, *Coxiella burnetii*, *Trypanosoma brucei*, *Trypanosoma cruzi*, *Leishmania major*, *Babesia microti*, *Plasmodium falciparum*, and *Plasmodium vivax*.
11. The probe set of claim 9, further comprising at least one negative control probe and/or further comprising at least one positive control probe.
12. The probe set of claim 11, wherein the at least one positive control probe comprises a set of probes with at least 90% identity with the nucleic acid sequences of each of SEQ ID NOs: 3208-3519 and each of the probes is 45-60 nucleotides in length.
13. A microarray comprising the probe set of claim 1.
14. The microarray of claim 13, wherein the probe set comprises probes comprising the nucleic acid sequence of each of SEQ ID NOs: 1-1769.
15. A method of detecting one or more pathogen nucleic acids in a sample, comprising: contacting the sample with the probe set of claim 1 under conditions sufficient to allow hybridization of pathogen nucleic acids present in the sample to the probes of the probe set; and measuring hybridization of the sample to one or more of the probes, thereby detecting one or more pathogen nucleic acids in the sample.
16. The method of claim 15, wherein the sample comprises a blood, serum, or plasma sample or nucleic acids isolated from a blood, serum, or plasma sample.
17. The method of claim 16, further comprising isolating nucleic acids from the sample prior to contacting the sample with the probe set.
18. The method of claim 17, further comprising labeling the isolated nucleic acids from the sample.
19. The method of claim 18, wherein the isolated nucleic acids are isolated DNA, isolated RNA, isolated cDNA, or a combination of two or more thereof.
20. The method of claim 19, wherein labeling the isolated nucleic acids comprises labeling the nucleic acids with one or more fluorescent labels.
21. The method of claim 17, wherein the isolated nucleic acids are cDNA.
22. The method of claim 17, wherein isolating the nucleic acids does not comprise amplifying total RNA from the sample prior to preparing cDNA.
23. The method of claim 15, wherein measuring hybridization comprises detecting $\geq 50\%$ of the probes for the virus have a log ratio of >1.5 and/or the log ratio between the signal intensity mean for the probe set and the mean of a control group probe set is ≥ 1.5 .
24. The method of claim 15, wherein the pathogen nucleic acids comprise nucleic acids from one or more of Chikungunya virus, Dengue virus types 1, 2, 3, or 4, Hepatitis A virus, Hepatitis C virus types 1, 2, or 3, Hepatitis E virus, Human immunodeficiency virus types 1 or 2, Human T-lymphotropic virus types I or II, West Nile virus, and Zika virus.
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